

## Local Coverage Determination (LCD)

# Botulinum Toxin Injections

L35172

Links in PDF documents are not guaranteed to work. To follow a web link, please use the MCD Website.

## Contractor Information

| Contractor Name                                    | Contract Type  | Contract Number  | Jurisdiction | States     |
|----------------------------------------------------|----------------|------------------|--------------|------------|
| <a href="#">Noridian Healthcare Solutions, LLC</a> | A and B<br>MAC | 02101 -<br>MAC A | J - F        | Alaska     |
| <a href="#">Noridian Healthcare Solutions, LLC</a> | A and B<br>MAC | 02102 -<br>MAC B | J - F        | Alaska     |
| <a href="#">Noridian Healthcare Solutions, LLC</a> | A and B<br>MAC | 02201 -<br>MAC A | J - F        | Idaho      |
| <a href="#">Noridian Healthcare Solutions, LLC</a> | A and B<br>MAC | 02202 -<br>MAC B | J - F        | Idaho      |
| <a href="#">Noridian Healthcare Solutions, LLC</a> | A and B<br>MAC | 02301 -<br>MAC A | J - F        | Oregon     |
| <a href="#">Noridian Healthcare Solutions, LLC</a> | A and B<br>MAC | 02302 -<br>MAC B | J - F        | Oregon     |
| <a href="#">Noridian Healthcare Solutions, LLC</a> | A and B<br>MAC | 02401 -<br>MAC A | J - F        | Washington |

| Contractor Name                    | Contract Type  | Contract Number  | Jurisdiction | States          |
|------------------------------------|----------------|------------------|--------------|-----------------|
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 02402 -<br>MAC B | J - F        | Washington      |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03101 -<br>MAC A | J - F        | Arizona         |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03102 -<br>MAC B | J - F        | Arizona         |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03201 -<br>MAC A | J - F        | Montana         |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03202 -<br>MAC B | J - F        | Montana         |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03301 -<br>MAC A | J - F        | North<br>Dakota |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03302 -<br>MAC B | J - F        | North<br>Dakota |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03401 -<br>MAC A | J - F        | South<br>Dakota |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03402 -<br>MAC B | J - F        | South<br>Dakota |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03501 -<br>MAC A | J - F        | Utah            |
| Noridian Healthcare Solutions, LLC | A and B<br>MAC | 03502 -<br>MAC B | J - F        | Utah            |

| Contractor Name                    | Contract Type | Contract Number | Jurisdiction | States  |
|------------------------------------|---------------|-----------------|--------------|---------|
| Noridian Healthcare Solutions, LLC | A and B MAC   | 03601 - MAC A   | J - F        | Wyoming |
| Noridian Healthcare Solutions, LLC | A and B MAC   | 03602 - MAC B   | J - F        | Wyoming |

## LCD Information

### Document Information

**LCD ID**

L35172

**LCD Title**

Botulinum Toxin Injections

**Proposed LCD in Comment Period**

N/A

**Source Proposed LCD**[DL35172](#) **Original Effective Date**

For services performed on or after 10/01/2015

**Revision Effective Date**

For services performed on or after 02/22/2026

**Revision Ending Date**

N/A

**Retirement Date**

N/A

**Notice Period Start Date**

01/08/2026

**Notice Period End Date**

02/21/2026

CPT codes, descriptions, and other data only are copyright 2025 American Medical Association. All Rights Reserved. Fee schedules, relative value units, conversion factors and/or related components are not assigned by the AMA, are not part of CPT, and the AMA is not recommending their use. The AMA does not directly or indirectly practice medicine or dispense medical services. The AMA assumes no liability for data contained or not contained herein. CPT is a registered trademark of the American Medical Association.

Current Dental Terminology © 2025 American Dental Association. All rights reserved.

Copyright © 2025, the American Hospital Association, Chicago, Illinois. Reproduced with permission. No portion of the AHA copyrighted materials contained within this publication may be copied without the express written consent of the AHA. AHA copyrighted materials including the UB-04 codes and descriptions may not be removed, copied, or utilized within any software, product, service, solution, or derivative work without the written consent of the AHA. If an entity wishes to utilize any AHA materials, please contact the AHA at [ub04@aha.org](mailto:ub04@aha.org) or 312-422-3366.

Making copies or utilizing the content of the UB-04 Manual, including the codes and/or descriptions, for internal purposes, resale and/or to be used in any product or publication; creating any modified or derivative work of the UB-04 Manual and/or codes and descriptions; and/or making any commercial use of UB-04 Manual or any portion thereof, including the codes and/or descriptions, is only authorized with an express license from the American Hospital Association. The American Hospital Association (the "AHA") has not reviewed, and is not responsible for, the completeness or accuracy of any information contained in this material, nor was the AHA or any of its affiliates, involved in the preparation of this material, or the analysis of information provided in the material. The views and/or positions presented in the material do not necessarily represent the views of the AHA. CMS and its products and services are not endorsed by the AHA or any of its affiliates.

## Issue

### Issue Description

This Local Coverage Determination (LCD) has been developed to create a policy consistent with current evidence for FDA approved and off-label uses of Botulinum Toxins in the Medicare population.

## CMS National Coverage Policy

Language quoted from Centers for Medicare and Medicaid Services (CMS), National Coverage Determinations (NCDs) and coverage provisions in interpretive manuals is italicized throughout the policy. NCDs and coverage provisions in interpretive manuals are not subject to the Local Coverage Determination (LCD) Review Process (42 CFR 405.860[b] and 42 CFR 426 [Subpart D]). In addition, an administrative law judge may not review an NCD. See §1869(f)(1)(A)(i) of the Social Security Act.

Unless otherwise specified, *italicized* text represents quotation from one or more of the following CMS sources:

### Title XVIII of the Social Security Act (SSA):

Section 1862(a)(1)(A) excludes expenses incurred for items or services which are not

reasonable and necessary for the diagnosis or treatment of illness or injury or to improve the functioning of a malformed body member.

Section 1862(a)(10) excludes coverage for cosmetic surgery.

Section 1833(e) prohibits Medicare payment for any claim which lacks the necessary information to process the claim.

### **Code of Federal Regulations:**

42 CFR Section 410.32 indicates that diagnostic tests may only be ordered by the treating physician (or other treating practitioner acting within the scope of his or her license and Medicare requirements) who furnishes a consultation or treats a beneficiary for a specific medical problem and who uses the results in the management of the beneficiary's specific medical problem. Tests not ordered by the physician (or other qualified non-physician provider) who is treating the beneficiary are not reasonable and necessary (see Sec. 411.15(k)(1) of this chapter).

### **CMS Publications:**

CMS Publication 100-02, *Medicare Benefit Policy Manual*, Chapter 8:

50.5 Drugs and Biologicals [Coverage of SNF services]

70 Medical and Other Health Services Furnished to SNF Patients.

CMS Publication 100-02, *Medicare Benefit Policy Manual*, Chapter 12:

40.10 Drugs and Biologicals [Coverage of Comprehensive Outpatient Rehabilitation Facility services]

CMS Publication 100-02, *Medicare Benefit Policy Manual*, Chapter 15:

50.1–50.5 Drugs and Biologicals 120 Ambulatory Surgical Center Services

CMS Publication 100-02, *Medicare Benefit Policy Manual*, Chapter 16:

260 Non-coverage for Cosmetic Procedures

CMS Publication 100-04; *Medicare Claims Processing Manual*, Chapter 17:

40 Discarded Drugs and Biologicals

CMS Publication 100-04; *Medicare Claims Processing Manual*, Chapter 30:

20.2.1 Categorical Denials

## Coverage Guidance

### Coverage Indications, Limitations, and/or Medical Necessity

#### General Indications and Limitations of Coverage

1. Botulinum toxin dosing must be used in accordance with the United States Food and Drug Administration (FDA) approved labeling.
2. For off-label botulinum toxin use without an FDA approved dosing indication, the qualified healthcare professional must provide robust published clinical evidence to support the dosing use.
3. For off-label botulinum toxin use without an FDA approved serotype indication, the qualified healthcare professional must provide robust published clinical evidence to support the serotype use.
4. For new FDA approved indications for botulinum toxin serotype or dosing recommendations after the effective date of this policy, the qualified healthcare professional must provide the updated published FDA prescribing information to support the use of the botulinum toxin serotype or dose.
5. Botulinum toxin administration must not be given more frequently than every 12 weeks, regardless of diagnosis, unless specifically addressed in the policy.
6. The use of botulinum toxin for all cosmetic procedures is not a covered benefit under Medicare; AND
7. When botulinum toxin is used for an approved diagnosis, but the botulinum is also being used with cosmetic intent, the entire claim for the botulinum is not considered reasonable and necessary, AND
8. The potency of each of the botulinum toxin serotypes are not interchangeable with other botulinum toxin serotypes and, therefore, units of biological activity of any 1 serotype cannot be compared to or converted into units of any other botulinum toxin serotype; AND
9. Botulinum toxin injections (BTIs) are not considered reasonable and necessary for patients with a contraindication to botulinum toxin; AND
10. BTIs are not considered reasonable and necessary for patients with existing medical conditions which could affect the neuromuscular function; AND
11. BTIs are not considered reasonable and necessary for patients with hypersensitivity to any botulinum toxin preparation or to any of the components in the formulation of the serotype; AND
12. BTIs are not considered reasonable and necessary for patients with severe clotting disorders; AND
13. BTIs are not considered reasonable and necessary for patients with an infection at the injection site; AND
14. Both conscious sedation and monitored anesthesia care (MAC) are not medically necessary for BTIs; AND

15. When conservative treatment needs to be provided prior to the administration of botulinum toxin, a vague or nonspecific physician statement that conservative treatment measures were completed is not sufficient. The documentation must provide specific information regarding the conservative treatments which were used and an objective assessment of the intolerance or inadequacy of the response to the conservative measures.
16. Image guidance is not considered reasonable and necessary for injection of botulinum toxin.
17. Medicare will allow payment for 1 injection per site regardless of the number of injections made into the site. A site is defined as 1 area.

## **Specific Indications and Limitations of Coverage by Diagnosis**

### **Achalasia**

#### **Definition:**

Achalasia is a relatively rare primary motor esophageal disorder, characterized by incomplete relaxation of the esophageal gastric junction (EGJ) coupled with the absence of organized peristalsis along the esophageal body. Three achalasia subtypes have been defined based on the high-resolution manometry (HRM) findings in the esophageal body: type I or classic achalasia with low intraesophageal pressure, type II with pan-esophageal pressurization, and type III with high-amplitude spastic contractions.

#### **Diagnosis:**

The diagnostic criterion for achalasia is evaluated by HRM, which is recognized as the gold standard for diagnosis.<sup>1,2</sup> HRM measures the integrated relaxation pressure (IRP) at the EGJ, with an IRP greater than 15 mmHg serving as a critical diagnostic marker for achalasia. The Chicago Classification system enables the categorization of achalasia into 3 distinct subtypes based on the IRP and esophageal pressurization patterns observed during swallowing. Each subtype has specific clinical implications and guides the management strategy, with type II achalasia generally associated with the most favorable treatment outcomes.<sup>1,2</sup>

#### **Treatment:**

BTI is an FDA off-label use consisting of endoscopic injections of the toxin into 4 quadrants of the lower esophageal sphincter (LES). The treatment options in achalasia patients aim to improve symptoms by reducing the functional obstruction at the level of the gastroesophageal junction. Injection of botulinum toxin reduces LES pressure by inhibiting release of acetylcholine from nerve endings.<sup>3-9</sup>

### **Indications of Coverage**

## Initial Botulinum Toxin Injections

In the management of achalasia, BTI is an FDA off-label use consisting of endoscopic injections of the toxin into 4 quadrants of the LES.<sup>3-9</sup> Initial BTIs for achalasia will be considered reasonable and necessary when the following requirements are met:

1. Objective documentation of the clinical features consistent with the diagnosis of achalasia; AND
2. Chronic achalasia measured on objective clinical scale\*; AND
3. Medically high-risk patients diagnosed with achalasia who cannot undergo other invasive treatments (peroral endoscopic myotomy (POEM), Heller myotomy, pneumatic dilation [PD]);<sup>1,2,8,10-15</sup> OR
4. As a bridge for those patients diagnosed with achalasia awaiting more effective treatments such as Heller myotomy, PDs or POEM;<sup>2</sup> OR
5. During work-up and treatment planning of definitive treatments for achalasia.<sup>16,17</sup>

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. There are several clinical scales to measure severity of achalasia, for example the Eckert Scale.

## Initial Dosing Guidelines

1. The initial dose of [onabotulinumtoxinA](#)  is 80 to 100 Units.

## Subsequent Botulinum Toxin Injections

Subsequent BTIs for achalasia will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decisions regarding repeat botulinum toxin injections; AND
2. Reassessment of the degree of persistent moderate to severe achalasia; AND
3. There is evidence of a significant beneficial symptomatic response to the initial dose.

## Subsequent Dosing Guidelines

1. The subsequent dose of up to 100 Units onabotulinumtoxinA may be given 30 days after the initial dose.

## Limitations of Coverage

BTIs are not considered reasonable and necessary for:



1. The patient who has achalasia symptoms but insufficient manometric criteria to make the diagnosis.
2. Patients with contraindications for upper endoscopy.<sup>18</sup>
3. Injection of botulinum toxin in the esophageal body.<sup>1,2,8,13</sup>
4. Initial or subsequent onabotulinumtoxinA doses above 100 Units.<sup>1,2</sup>

## **Anal Fissure**

### **Definition:**

A linear tear in the anal mucosa typically extending into the internal anal sphincter. Anal fissures that persist for more than 6 weeks are classified as chronic. The internal sphincter spasm is believed to be the main etiology for the pathogenesis of chronic anal fissure.<sup>19</sup>

### **Diagnosis:**

Anal fissures are characterized by a longitudinal linear tear primarily caused by trauma or irritation from constipation or diarrhea.<sup>20</sup> Fissures are commonly located in the posterior midline. Variations in location, such as anterior midline or atypical lateral positions, necessitate a thorough evaluation due to potential underlying conditions.<sup>20</sup>

Chronic fissures exhibit additional features such as a hypertrophied anal papilla, a sentinel tag, and possibly exposed internal anal sphincter muscle, indicative of underlying issues like sphincter hypertonicity and impaired wound healing. This distinction is crucial for diagnosis and treatment, as chronic fissures may display additional physical features such as a hypertrophied anal papilla at the proximal edge of the fissure, a sentinel tag (or skin tag) at the distal edge, and exposure of the internal anal sphincter muscle at the base of the fissure, indicating a more complex condition requiring specialized management strategies.<sup>20</sup>

## **Indications of Coverage**

### **Initial Botulinum Toxin Injections**

Initial BTIs for anal fissure will be considered reasonable and necessary when the following requirements are met:

1. The anal fissure has been present for more than 6 weeks; AND
2. Conservative treatment has been tried for eliminating constipation and reducing anal sphincter spasm.

### **Initial Dosing Guidelines**

1. The initial treatment with onabotulinumtoxinA dosing for anal fissure is 20 Units (10 Units on each side of the fissure).<sup>21,22</sup>

### Subsequent Botulinum Toxin Injections

Subsequent BTIs for anal fissure will be considered reasonable and necessary when the following requirements are met:

1. Documentation for informed clinical decision-making regarding repeat botulinum toxin injections and surgical treatment; AND
2. Reassessment of the symptoms and degree of persistent anal fissure.

### Subsequent Dosing Guidelines

1. A subsequent dose of onabotulinumtoxinA for anal fissure is up to 60 Units with the clinical trials demonstrating greater efficacy with lower doses.<sup>20</sup>

## Blepharospasm

### Definition:

A focal dystonia involving the involuntary closure of the eyelids and spasms of the orbicularis oculi muscles characterized by sustained or intermittent muscle contractions resulting in mild blinking or sustained forced closure of the eyes.

### Diagnosis:

To diagnose a patient with blepharospasm, clinicians should observe for involuntary contraction of the orbicularis oculi and other muscles involved in eyelid closure, which can range from sporadic and mildly irritating to functionally blinding.<sup>23</sup> The disorder often begins with infrequent bilateral eyelid twitching and may progress to forceful and frequent spasms, including symptoms like eyelid apraxia. Sensory stimulation such as touching the eyelids or certain activities may temporarily reduce the contractions.

Diagnosis is primarily clinical and considered a diagnosis of exclusion, with imaging or laboratory studies typically not indicated. The diagnostic criteria for blepharospasm have been proposed, with key features being bilateral spasms of the orbicularis oculi and nearby muscles of the upper face often with excessive blinking.<sup>23</sup> Clinicians may encounter patients presenting with associated symptoms such as anxiety, depression, or ocular surface diseases like blepharitis.<sup>4</sup> Differential diagnosis includes distinguishing from conditions like Meige syndrome, myokymia, hemifacial spasm, and others based on the symmetry, distribution, and accompanying symptoms. Photophobia and response to sensory triggers are common, and the patient's history of triggers and symptom progression is essential for

accurate diagnosis. To accurately diagnose blepharospasm without inadvertently diagnosing Meige Syndrome, it's essential to focus on the primary symptom of involuntary eyelid closure, ensuring it is bilateral, synchronous, and stereotyped, without the presence of significant lower facial or neck muscle involvement.<sup>23</sup>

Despite this focal blepharospasm being the second most common type of dystonia, a high percentage of individuals given this diagnosis had dystonia outside of the eye/upper face region<sup>5</sup> which makes the diagnosis inconsistent with existing guidelines for the diagnosis and classification of focal blepharospasm. These authors proposed that the diagnosis of focal blepharospasm be based on dystonic spasms of muscles in the upper face around the eyes only.

The most sensitive findings to diagnose blepharospasm is a stereotypical bilateral and synchronous orbicularis oculi muscle spasms inducing eyelid narrowing closure, the presence of a sensory trick (the sensory trick was defined as any kind of maneuver performed by the patient that led to a transient reduction in spasm severity in the period of time immediately after its execution<sup>24</sup>), and increased blinking.<sup>4</sup> Blepharospasm can be classified into idiopathic, acquired, and inherited subtypes.<sup>25</sup>

The rating instruments in the medical literature have coalesced into several main clinical scales, including the Jankovic Rating Scale (JRS) and Blepharospasm Disability Index (BSDI).<sup>26</sup>

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial BTIs for blepharospasm will be considered reasonable and necessary when the following requirements are met:

1. Objective documentation of the clinical features consistent with the diagnosis of blepharospasm; AND
2. Chronic blepharospasm of at least 30 days duration measured using an objective clinical scale\*; AND
3. BTI therapy is accepted first line treatment for patients with blepharospasm.

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. For example, there are several clinical scales to measure severity of blepharospasm, including the JRS and BSDI.

### Initial Dosing Guidelines

1. The initial treatment with onabotulinumtoxinA dosing for blepharospasm associated with dystonia is 1.25 Units-2.5 Units into each of 3 sites per affected facial or ocular muscle.<sup>27</sup>

### Subsequent Botulinum Toxin Injections

Subsequent BTIs for blepharospasm will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decisions regarding repeat botulinum toxin injections; AND
2. Reassessment of the severity and frequency of persistent blepharospasm; AND
3. Persistence or reoccurrence of blepharospasm; AND
4. The initial treatment is considered sufficient, and administration of same dose is recommended; OR
5. The initial treatment is considered insufficient (defined as an effect which does not last longer than 2 months) and administration of an increased dose is recommended. Based upon the initial response, the dosage may be increased up to 5 units per site.

### Subsequent Dosing Guidelines

1. The subsequent treatment with [incobotulinumtoxinA](#)  dosing is based on the previous dosing with onabotulinumtoxinA. If previously treated with onabotulinumtoxinA, if not known or not previously treated with onabotulinumtoxinA, the incobotulinumtoxinA starting dose is 1.25 Units-2.5 Units per injection site.<sup>28</sup>
2. For onabotulinumtoxinA in blepharospasm, subsequent doses may be increased up to two-fold if the initial response is insufficient. However, injecting more than 5 units per facial or ocular muscle site provides little additional benefit.
3. Clinical circumstances that necessitate an exception of OnabotulinumtoxinA dose up to 10 units per site may be considered on an individual basis and the rationale for the higher dose must be clearly documented in the medical record.

### Blepharospasm Associated with Orofacial Dystonia

#### Definition

A rare neurological movement disorder characterized by involuntary and often forceful contractions of the muscles of the jaw and tongue (oromandibular dystonia) and involuntary muscle spasms and contractions of the muscles around the eyes (blepharospasm) is known as Meige Syndrome.<sup>29</sup> Overall, half of all blepharospasm subjects experience spread over a period of 5 years.<sup>30</sup>

## Diagnosis

The syndrome is clinically suspected by the clinical symptoms of eyelid spasms accompanied by jaw clenching or mouth opening, grimacing, and/or tongue movement. This condition must be objectively diagnosed by a clinical scale such as the Craniocervical Dystonia Questionnaire.

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial BTIs for blepharospasm associated with orofacial dystonia will be considered reasonable and necessary when the following requirements are met:

1. Objective documentation of the clinical features consistent with the diagnosis of blepharospasm associated with orofacial dystonia; AND
2. Moderate to severe chronic blepharospasm associated with orofacial dystonia measured on objective clinical scale\*, AND
3. OnabotulinumtoxinA injection therapy is accepted first line treatment for patients with blepharospasm associated with orofacial dystonia.

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. There are several clinical scales to measure severity of blepharospasm associated with orofacial dystonia, including the Burke-Fahn-Marsden scale (BFMS), the Global Dystonia Severity Rating scale (GDRS), Craniocervical Dystonia Questionnaire, and the JRS.

### Initial Dosing Guidelines

1. There are no standard dosing recommendations of onabotulinumtoxinA for orofacial dystonia which is typically administered into masseters, temporalis, medial pterygoids, submental, lateral pterygoids, platysma, genioglossus, and hyoglossus.<sup>31</sup>

### Subsequent Botulinum Toxin Injections

Subsequent BTIs for associated with orofacial dystonia will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum injections; AND
2. Reassessment of persistent blepharospasm associated with orofacial dystonia.

### Subsequent Dosing Guidelines

1. The number of injections sites and muscles injected is determined by the response to initial injections.
2. When the initial treatment is considered sufficient, and administration of same dose is recommended, OR
3. When the initial treatment or subsequent treatments are considered insufficient (defined as an effect which does not last longer than 2-months) administration of an increased dose is recommended.

### Cervical Dystonia (CD).

#### **Definition:**

Cervical dystonia (CD), also known as spasmodic torticollis, is a rare neurological disorder caused by an impairment of the central nervous system<sup>32</sup> with an estimated prevalence range from 5 to 30 cases per 100,000 individuals.<sup>33,34</sup> The disorder is characterized by upper motor neuron velocity-dependent hypertonic skeletal muscles, the inability to relax the muscles and characteristic intermittent or sustained muscle contractions causing abnormal and frequent repetitive movements and posture<sup>32</sup> of the cervical and/or shoulder muscles.<sup>35</sup>

There are 2 groups of CD. The first type of CD is idiopathic or primary CD which is believed to be due to genetic or sporadic onset. The patients with primary CD have no evidence by history, physical examination or laboratory studies (except primary dystonia gene) of any secondary cause for the dystonic symptoms. CD is a part of either generalized or focal dystonic syndrome which may have a genetic basis, with an identifiable genetic association.

The second type of CD is acquired or secondary CD (or sometimes described as "symptomatic" CD). Secondary CD may be caused by central or peripheral trauma, exposure to dopamine receptor antagonists (tardive CD), neurodegenerative disease, and other conditions associated with abnormal functioning of the basal ganglia.<sup>36</sup>

There is controversy as to the development of post-traumatic CD<sup>37</sup> and the contribution of peripheral acute trauma to causing idiopathic dystonia is negligible. CD is not the only cause of neck rotation and torticollis may be caused by orthopedic, musculofibrotic, infectious and other neurological conditions that affect the anatomy of the neck, and structural causes need to be assessed.<sup>36</sup> The accuracy of the diagnosis is critical. There was a pivotal trial for the FDA approval of botulinum toxin for CD.<sup>36</sup>

#### **Diagnosis**

CD is clinically suspected but the condition is not simply a manifestation of focal muscle pain, muscle spasm, cervical movements and cervical posture.<sup>38</sup> This clinical diagnosis must be associated with excessive pulling of the muscles of the neck and shoulder. These dystonic movements, which may be twisting, sustained, jerking, or tremulous, are the

hallmark symptoms. The abnormal postures or movements, often exacerbated by voluntary actions, can lead to significant neck pain or discomfort. The presence of such symptoms, alongside the use of specific maneuvers like *gestes antagonistes*, which can temporarily alleviate dystonic postures confirms the clinical diagnosis. The diagnostic criteria emphasize the importance of these involuntary movements' characteristics and their impact on the patient's quality of life.<sup>38</sup>

This condition must be objectively measured by a clinical scale. Several scales have been recommended by medical literature. Scales that are used to evaluate CD include but are not limited to the Fahn-Marsden Scale, Unified Dystonia Rating Scale, Columbia Torticollis Rating Scale, Tsui Scale, Toronto Western Spasmodic Torticollis Rating Scale (TWSTRS), and Cervical Dystonia Rating Scale. The TWSTRS scale is the most widely utilized rating scale for CD.<sup>39</sup> Some authors have indicated that the rating scales (such as the Tsui score and the TWSTRS) used at present have not been rigorously tested for responsiveness to detect significant changes in clinical status after therapeutic interventions.<sup>39</sup>

Botulinum toxin is used to reduce the severity of the abnormal head positions and neck pain associated with CD.

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial BTIs for CD will be considered reasonable and necessary when the following requirements are met:

1. The documentation supports a diagnosis of CD; AND
2. The etiology of the central nervous system impairment which is causing the CD is documented, AND
3. Beneficiary has a history of recurrent clonic or tonic involuntary contractions of 1 or more of the following muscles: sternocleidomastoid, splenius, trapezius and/or posterior cervical muscles; AND
4. There is moderate to severe CD assessed by an objective scale\*; AND
5. There are objective measurements of abnormal posturing, with limited range of motion in the neck, or sustained head tilt; AND
6. The duration of the CD is greater than 6 months; AND
7. The initial onabotulinumtoxinA dose is based on the patient's head and neck position, localization of pain, muscle hypertrophy, patient response, and adverse event history; use lower initial dose in botulinum toxin naïve patients.<sup>27</sup>

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. Some clinical scales include Tsui score and the TWSTRS.

## Initial Dosing Guidelines

1. The initial dose of [abobotulinumtoxinA](#) is up to 500 Units given intramuscularly as a divided dose among the affected cervical muscles<sup>40</sup>;
2. The initial dose of [rimabotulinumtoxinB](#) is up to a total dosage of 2,500 Units to 5,000 Units divided among affected cervical muscles.<sup>41</sup>
3. The initial dose of incobotulinumtoxinA is up to 120 Units divided among affected cervical muscles per treatment session.<sup>28</sup>
4. The initial dose of daxibotulinumtoxinA-lanm is 125 Units to 250 Units given intramuscularly as a divided dose among affected muscles.<sup>330</sup>

## Subsequent Botulinum Toxin Injections

Subsequent BTIs for CD will be considered reasonable and necessary when the following requirements are met:

1. Beneficiary is requesting subsequent injections; AND
2. Response to initial treatment is documented in the medical records; AND
3. Reassessment of clinical utility of the injection with documentation of informed clinical decision regarding repeat botulinum injections.

## Subsequent Dosing Guidelines

1. The subsequent onabotulinumtoxinA dosing for CD is based on the clinical response, head and neck position, localization of pain, and muscle hypertrophy.
2. The subsequent abobotulinumtoxinA doses are administered between 250 and 1000 Units to optimize clinical benefit and titrated in 250 Units in incremental steps according to patient's response.<sup>40</sup>
3. The subsequent dose of rimabotulinumtoxinB is up to a total dosage of 2,500 Units to 5,000 Units divided among effected cervical muscles.<sup>41</sup>
4. The subsequent dose of IncobotulinumtoxinA is 120 Units per treatment session.<sup>28</sup>
5. The subsequent doses of daxibotulinumtoxinA-lanm can be adjusted in 50 to 75 Unit increments according to the individual patient's response. The total dose administered in a single treatment is between 125 Units and 250 Units.<sup>330</sup>

## Limitations of Coverage:

1. It is not reasonable and necessary for multiple procedures (e.g., epidural injections, sympathetic blocks, trigger point injections, etc.) to be provided to a beneficiary on the same day as the BTI.
2. BTIs are not medically necessary for generalized pain conditions (such as fibromyalgia), chronic nonspecific neck pain with or without cervical rotation, or chronic centralized pain syndromes, temporomandibular disorders with or without neck pain, severe



bruxism with or without neck pain, myofascial pain syndrome, and cervical spondylosis with or without neck pain.

## **Chronic Migraine**

### **Definition:**

Chronic migraine is a recurrent chronic headache disorder with 2 major types described as either episodic migraines or chronic migraines based on the frequency of the headaches. Patients presenting with headaches should be evaluated to determine whether their headache is a primary or a secondary headache disorder. Migraine headaches can present with or without aura.

### **Diagnosis:**<sup>42,43</sup>

Migraine without aura is a clinical syndrome characterized by the following diagnostic criteria:

1. At least 5 attacks fulfilling criteria b.-d.
2. Headache attacks lasting 4-72 hours (untreated or unsuccessfully treated)
3. Headache has at least 2 of the following 4 characteristics:
  - a. Unilateral location
  - b. Pulsating quality
  - c. Moderate or severe pain intensity
  - d. Aggravation by or causing avoidance of routine physical activity (e.g., walking or climbing stairs)
4. During headache at least 1 of the following:
  - a. Nausea and/or vomiting
  - b. Photophobia and Phonophobia
5. Not better accounted for by another International Classification of Headache Disorders, Third Edition (ICHD-3) diagnosis.

Migraine with aura is a clinical syndrome characterized by the following diagnostic criteria:

1. At least 2 attacks fulfilling criteria b. and c.
2. One or more of the following fully reversible aura symptoms:
  - a. Visual
  - b. Sensory
  - c. Speech and/or language
  - d. Motor

e. Brainstem

f. Retinal

3. At least 3 of the following 6 characteristics:

a. At least 1 aura symptom spreads gradually over  $\geq 5$  minutes

b. Two or more aura symptoms occur in succession.

c. Each individual aura symptom lasts 5-60 minutes.

d. At least 1 aura symptom is unilateral.

e. At least 1 aura symptom is positive.

f. The aura is accompanied, or followed within 60 minutes, by headache.

4. Not better accounted for by another ICHD-3 diagnosis.

### **Migraine Headache Prophylaxis:**

The botulinum toxin, onabotulinumtoxinA, is beneficial for the prophylaxis of chronic migraine headaches based upon FDA approval, a few randomized controlled clinical trials, published practice guidelines, and professional society evidence reviews.

The reporting of headaches is often clinically based on the patient's historical recall of the past headache frequency, intensity and duration. There is a potential for recall bias when patients retrospectively report the past headaches and some studies have shown the recall has been "reasonably accurate"<sup>44</sup> with the headache intensity appearing to be more difficult to remember and report than headache frequency and duration. Additional studies have indicated that recall questionnaires administered on a monthly basis may underestimate headache frequency compared to headache diaries.<sup>45</sup> Daily prospective headache diaries may reduce recall bias and increase the reliability of patient's descriptions of migraine headache characteristics.<sup>46</sup> The use of the standardized questionnaires has shown moderately high test-retest reliability between the questionnaire and the headache diaries.<sup>46</sup>

van der Meer endorsed that more research is needed to enhance the level of evidence for existing measurement instruments for multiple headaches.<sup>47</sup>

### **Indications of Coverage**

#### **Initial Botulinum Toxin Injections**

Initial BTIs for chronic migraine prophylaxis will be considered reasonable and necessary when the following requirements are met:

1. The BTI is being used only for the indication of chronic migraine prophylaxis, AND

2. The monthly headache days have occurred  $\geq 15$  headache days per month, AND

3. The monthly migraine headache days have occurred  $\geq 8$  migraine headache days per month, AND
4. The migraine headaches are documented to be lasting for a duration  $\geq 4$  hours on a migraine day, AND
5. The chronic headaches have been present for a period of at least 3 months, AND
6. The beneficiary has had a trial of and inadequate response to a 2-month trial of at least 1 agent in any 2 of the following classes or has contraindication to the following medications; AND
  - a. Antidepressant class: amitriptyline, venlafaxine, nortriptyline, duloxetine; OR
  - b. Beta blocker class: metoprolol, propranolol, timolol (oral), nadolol, atenolol, nebivolol; OR
  - c. Calcium channel blocker class: verapamil; OR
  - d. Antiepileptic class: valproate sodium, divalproex sodium, topiramate, gabapentin.
7. The 2-month trial of the oral pharmacologic classes above for chronic migraine prophylaxis must have been at the target dose of the usual effective dose or an intolerance to the 2 agents in each class, AND
8. If the beneficiary also currently using a calcitonin gene-related peptide (CGRP) agent for chronic migraine prophylaxis and is going to be using CGRP and botulinum toxin together the following must apply:
  - a. The beneficiary had a reduction in the overall number of migraine days or reduction in number of severe migraine days per month with CGRP use, but still has chronic migraines requiring additional therapy for chronic migraine prevention; AND
9. The headaches are causing an objective significant functional disability, AND
10. The headaches are moderate to severe intensity with typical migraine headache characteristics; AND
11. The botulinum toxin serotype is approved by the FDA for chronic migraine prophylaxis.

The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment.

### Initial Dosing Guidelines

1. The onabotulinumtoxinA dose ranges between 155-195 Units. The initial onabotulinumtoxinA dose is 155 Units given as 5 Units per each site divided across 7 head/neck muscles (frontalis, corrugator, procerus, occipitalis, temporalis, trapezius, and cervical paraspinal muscle group).<sup>27</sup>

### Subsequent Botulinum Toxin Injections

Clinicians must be aware and address the importance of cognitive-affective processes in headache disorders and address the potential for hypervigilance to bodily sensations and anxiety regarding the symptoms of headache pain when managing chronic migraine headaches. Subsequent BTIs for migraine prophylaxis will be considered reasonable and necessary when the following requirements are met:

1. The number of the total chronic migraine headache days have demonstrated  $\geq 50\%$  reduction in migraine headache days per month; AND
2. The frequency of the total chronic migraine headache episodes has demonstrated  $\geq 50\%$  reduction in migraine headache episodes per month; AND
3. There is a minimal important change (MIC) and significant reduction of headache-related disability and objective improvement in functioning; AND
4. Biobehavioral therapy (cognitive behavioral therapy, biofeedback, relaxation therapies, mindfulness-based therapies, acceptance and commitment therapy) has been assessed and implemented as appropriate for preventive and acute headache treatment; AND
5. If the beneficiary is using concurrently with a CGRP agent for migraine prophylaxis, the beneficiary has had further reduction in the overall number of migraine days or reduction in number of severe migraine days per month compared to monotherapy with the initial agent (either botulinum toxin or the CGRP agent); AND
6. The botulinum toxin serotype is approved by the FDA for chronic migraine prophylaxis.

### Subsequent Dosing Guidelines

1. The subsequent onabotulinumtoxinA dose ranges between 155-195 Units to allow a discretionary 40 Units to be administered using a “follow-the-pain” strategy, resulting in 195 Units onabotulinumtoxinA over 39 sites.<sup>27</sup>

### Limitations of Coverage

1. It is not reasonable and necessary for multiple procedures (e.g., epidural injections, sympathetic blocks, trigger point injections, etc.) to be provided to a beneficiary on the same day as the BTI.
2. BTIs are not medically necessary for generalized pain conditions (such as fibromyalgia) or chronic centralized pain syndromes, episodic migraine prophylaxis, temporomandibular disorders with or without headaches, tension headaches, severe bruxism with or without headache, chronic daily headaches, myofascial pain syndrome with or without headaches and cervical spondylosis with or without headaches.
3. Initial or subsequent onabotulinumtoxinA doses above 195 Units.

### Focal Hand Dystonia(FHD).

**Definition:**

Focal hand dystonia (FHD) is a disabling task-specific movement disorder, which can generalize across tasks and be present at rest. It is also referred to as writer's cramp, musician's dystonia, and other occupational hand dystonias.

**Diagnosis:**

FHD is a rare primary hand dystonia. The diagnosis of FHDs includes neuropathies, myopathies, myotonias, radiculopathies, plexopathies, complex regional pain syndrome, repetitive stress injury, focal seizures, thoracic outlet syndrome, medications effects, and a psychogenic movement disorder. The patient's history and clinical examination are both critical for its diagnosis and for identification of dystonic muscles. Since FHDs are uncommon, the use of botulinum toxin for the treatment of a FHD should be infrequent. BTI therapy dosing in FHD is highly individualized due to different activity types, levels, multiple potential target muscles with varying BTI doses and intervals.<sup>2</sup> Currently, no effective alternative medical or proven surgical therapies have been established for FHD.<sup>42,43,48-52</sup>

**Indications of Coverage****Initial Botulinum Toxin Injections**

BTI therapy is an accepted FDA off-label use first line treatment for patients with FHDs. In the management of FHD, BTI is considered reasonable and necessary consisting of focal injections of the toxin into the muscles responsible for the abnormal postures, and initial BTIs for FHD will be considered reasonable and necessary when the following requirements are met:

1. Objective documentation of the clinical features consistent with the diagnosis of FHD;  
AND
2. Moderate to severe chronic FHDs measured on objective clinical scale\*; AND
3. The injections are used with guidance either by ultrasound or by electromyography (EMG) with or without electrostimulation.

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. For example, scales that could be used include the Fahn-Marsden rating scale, Unified Dystonia Rating Scale, Arm Dystonia Disability Scale (ADDS); Tubiana-Chamagne Scale; and Writer's Cramp Rating Scale.

**Initial Dosing Guidelines**

1. The appropriate dosing range of onabotulinumtoxinA depends on the site of muscle injection. Clinician must assess the specific pattern of dystonic postures and

movements.<sup>53</sup>

2. For flexor muscles (flexor digitorum superficialis: 25-50 Units, flexor pollicis longus: 5-20 Units, flexor digitorum profundus: 20-60 Units, flexor carpi radialis: 20-50 Units, flexor carpi ulnaris: 15-60 Units).
3. For extensor muscles (extensor digitorum communis: 10-25 Units, extensor pollicis longus: 5-20 Units, extensor indicis: 5-10 Units, extensor carpi radialis longus: 5-20 Units, extensor carpi ulnaris: 5-20 Units).

### **Subsequent Botulinum Toxin Injections**

Subsequent BTIs for FHD will be considered reasonable and necessary when all the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum injections; AND
2. Reassessment of the degree of persistent FHD; AND
3. The initial treatment is considered sufficient, and administration of same dose is recommended; OR
4. The initial treatment is considered insufficient (defined as an effect which does not last longer than 2 months) and administration of an increased dose is recommended.

### **Subsequent Dosing Guidelines**

1. Appropriate subsequent dosage of onabotulinumtoxinA range depends on the site of muscle injection. Clinician must assess the specific pattern of dystonic postures and movements.<sup>53</sup>
2. For flexor muscles (flexor digitorum superficialis: 25-50 Units, flexor pollicis longus: 5-20 Units, flexor digitorum profundus: 20-60 Units, flexor carpi radialis: 20-50 Units, flexor carpi ulnaris: 15-60 Units).
3. For extensor muscles (extensor digitorum communis: 10-25 Units, extensor pollicis longus: 5-20 Units, extensor indicis: 5-10 Units, extensor carpi radialis longus: 5-20 Units, extensor carpi ulnaris: 5-20 Units).

### **Hemifacial Spasm/Facial Dystonia**

#### **Definition:**

Hemifacial spasm (HFS) is a chronic neurological disorder characterized by usually unilateral hyperkinetic movements with short or persistent, intermittent synchronous twitching of the muscles innervated by the facial nerve and spontaneous recovery is rare. The etiology of the disorder is the seventh cranial nerve. HFS is classified as primary facial nerve damage (79%) or secondary facial nerve damage (21%).<sup>54</sup> Most cases of hemifacial spasm are idiopathic

and probably caused by vascular compression of the facial nerve, other etiologies should be considered in the differential diagnosis, particularly if there are atypical features.<sup>55</sup>

## Diagnosis:

HFS is characterized by involuntary contractions of the facial muscles innervated by the seventh cranial nerve. An accurate diagnosis is important as many conditions can mimic HFS. The diagnosis of this condition is made clinically based on the detailed history coupled with a neurological and local physical examination. Electrophysiological testing may not be required to make the diagnosis, but an electromyogram can be used in the early stages of the disease when it is difficult to distinguish clinically from facial myokymia, blepharospasm, complex partial motor seizures, or motor tics. The diagnostic finding on electrophysiological testing is lateral spread and variable synkinesis on blink reflex testing. Brain magnetic resonance imaging may be needed to determine if there is a suspected facial nerve compression at the brainstem nerve root exit zone or near the cerebellopontine angle.<sup>56</sup>

Similarly, facial dystonia is a movement disorder characterized by sustained or intermittent muscle contractions causing abnormal, often repetitive movements, postures, or both involving the facial muscle(s). Diagnosis is made clinically but will often require neuroimaging and for certain other etiologies, may require additional laboratory testing.

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial BTIs for HFS will be considered reasonable and necessary when all the following requirements are met:

1. BTI therapy is accepted as first line treatment for patients with primary or secondary HFS; AND
2. Objective documentation of the clinical features consistent with the diagnosis of primary or secondary HFS; AND
3. Moderate to severe primary or secondary HFS measured on objective clinical scale to measure severity, complexity and psychosocial aspects of HFS.\*

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. The selection of the clinical scale is not defined but the selected scale should provide a comprehensive and sensitive rating tool with both clinical and subjective parameters for HFS to enable a standardized assessment of HFS and treatment outcome.<sup>26,57</sup>

### Initial Dosing Guidelines

1. The administration of a total dose 25-30 Units of onabotulinumtoxinA into the orbicularis oculi, procerus, mentalis, platysma, orbicularis oris and depressor anguli oris on the side of the face affected by the HFS.<sup>27</sup>

### **Subsequent Botulinum Toxin Injections**

Subsequent BTIs for HFS will be considered reasonable and necessary when all the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum injections; AND
2. Reassessment of the degree of persistent moderate to severe HFS.

### **Subsequent Dosing Guidelines**

1. A gradual increase in the number of onabotulinumtoxinA Units (5-15 additional Units) after 1 year, may be required.

## **Hyperhidrosis**

### **Definition:**

Primary hyperhidrosis is defined as excessive, uncontrollable sweating without any discernible cause and commonly involves the axillae, palms, and soles. Secondary hyperhidrosis is caused by an underlying medical condition or due to taking certain medications, such as pain relievers, antidepressants, diabetes medications, and hormonal medications. Severely affected patients have skin maceration and secondary microbial infections.

### **Diagnosis:**

Hyperhidrosis is characterized by excessive sweating beyond thermoregulatory needs.<sup>58-60</sup> This condition is classified into mild, moderate and severe forms based on the quantity of sweating and its impact on patients' lives.<sup>58-60</sup> Those with mild hyperhidrosis experience uncomfortable yet tolerable sweating that does not require changes of clothing. Moderate hyperhidrosis involves visible sweat droplets but no need for multiple clothing changes per day. However, patients with severe hyperhidrosis undergo profuse, uncontrollable sweating necessitating multiple clothing changes daily, causing significant disruption of professional, social and personal activities.<sup>58-60</sup>

Accurately diagnosing hyperhidrosis requires qualitative and quantitative assessments. Qualitative methods like visual inspection and patient reporting assess localization and situational triggers of sweating through examining clothing sweat stains and patient narratives. More precise quantitative techniques like gravimetric testing directly measure



sweat production but can be impractical for routine diagnosis. Minor's starch-iodine test localizes affected areas but does not quantify sweat volumes.<sup>58-60</sup> Critically, quality-of-life questionnaires like the Hyperhidrosis Disease Severity Scale evaluate functional life impairment across professional, social and psychological domains. Comprehensive assessment combining patient-reported symptoms, visual observations, sweat quantification and impact evaluation is key for confirming hyperhidrosis and determining optimal, personalized treatment approaches tailored to the severity.<sup>58-60</sup>

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial BTIs for primary axillary hyperhidrosis will be considered reasonable and necessary when the following requirements are met:

1. Beneficiary has a diagnosis of primary or secondary axillary hyperhidrosis; AND
2. Excessive sweating in the axilla lasting 6 months or more; AND
3. Bilateral symmetric sweating in the axilla; AND
4. Cessation of focal sweating while asleep; AND
5. The failure of a 6-month trial to respond to other noninvasive conservative management for axillary hyperhidrosis (systemic anticholinergics, tranquilizers, topical dermatologics such as aluminum chloride, tannic acid, or glutaraldehyde, or non-steroid anti-inflammatory drugs); AND
6. Severe chronic hyperhidrosis measured on objective clinical scale\*; AND
7. There is severe chronic axillary hyperhidrosis manifested by medical complications or skin maceration with secondary infection; AND
8. There is severe chronic axillary hyperhidrosis associated with and impairment of daily activities; AND
9. Significant functional impairment due to the hyperhidrosis

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment, such as the Hyperhidrosis Disease Severity Scale (HDSS).<sup>61,62</sup>

### Initial Dosing Guidelines

1. The initial administration of onabotulinumtoxinA is up to a total 50 Units per axilla.<sup>27</sup>

### Subsequent Botulinum Toxin Injections

Subsequent BTIs for primary axillary hyperhidrosis will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum toxin injections;  
AND
2. Reassessment of the degree of persistent hyperhidrosis and assessment of previous response to botulinum toxin; AND
3. Severe chronic hyperhidrosis measured on objective clinical scale\*; AND
4. Retreatment no greater than once every 6 months.

### **Subsequent Dosing Guidelines**

1. The subsequent dose of onabotulinumtoxinA is 50-75 Units per axilla.

### **Limitations of Coverage**

1. The safety and effectiveness of onabotulinumtoxinA for hyperhidrosis in body areas other than axillary are not considered to be reasonable and necessary.
2. Secondary axillary hyperhidrosis due to endocrine, neurologic, medication adverse effects, infection or malignancy is not considered to be reasonable and necessary.
3. Previous surgical debulking of the axillary sweat glands make the administration of botulinum toxin not reasonable and necessary.
4. Initial administration of onabotulinumtoxinA over 50 Units or subsequent doses over 75 Units.

### **Laryngeal Dystonia (Spasmodic Dysphonia)**

#### **Definition:**

Laryngeal spasmodic dysphonia, a type of laryngeal dystonia (LD), is a task-specific focal dystonia involving the laryngeal muscles characterized by sustained or intermittent muscle contractions and due to abnormal neuronal function at several potential sites along the neuroaxis from the motor cortex, supplementary motor areas, brainstem, putamen, globus pallidus, cerebellum, and other potential locations in the basal ganglia.

#### **Diagnosis:**

The diagnostic criteria for laryngeal spasmodic dysphonia include the evaluation of voice quality changes such as alterations in pitch, loudness, or vocal effort that impair communication or reduce the quality of life.<sup>63</sup> A thorough history and physical examination are crucial to identify underlying causes of dysphonia and to differentiate laryngeal spasmodic dysphonia from other voice disorders. Laryngoscopy is recommended when dysphonia fails to resolve or improve within 4 weeks or when a serious underlying cause is suspected, irrespective of the duration of symptoms.<sup>63</sup>

Laryngeal dystonia (spasmodic dysphonia) typically presents as adductor type, Adductor Spasmodic Dysphonia (ADSD), 90% versus the less common abductor type, Abductor Spasmodic Dysphonia (ABSD). Studies show onabotulinum toxinA to be an effective treatment option for ADSD, however, there is currently inadequate evidence to support the effectiveness of botulinum toxin injections for the ABSD variant. Consequently, consistent with evidence-based guidelines, coverage has been extended for the off-label coverage for botulinum toxin for ADSD.

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial BTIs for LD will be considered reasonable and necessary when the following requirements are met:

1. Objective documentation of the clinical features consistent with the diagnosis of adductor type, ADSD; AND
2. Objective assessment and documentation to rule out non-organic voice disorders; AND
3. Moderate to severe chronic ADSD measured on objective clinical scale\*;<sup>64</sup> AND
4. BTI therapy is an accepted FDA off-label use first line treatment for patients with ADSD.

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. For example, the Voice Handicap Index (VHI) and the Vocal Performance Questionnaire (VPQ).

### Initial Dosing Guidelines

1. The initial injection of onabotulinumtoxinA takes place at two separate sites of the thyroarytenoid (TA) and/or lateral cricoarytenoid (LCA) muscle on one side to a total dose between 1.5 to 3.5 Units per site.<sup>65</sup>

### Subsequent Botulinum Toxin Injections

Subsequent BTIs for LD will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum toxin injections; AND
2. Reassessment of the degree of persistent LD and assessment of previous response to botulinum toxin.

### Subsequent Dosing Guidelines

1. A second injection of onabotulinumtoxinA in contralateral side at separate time (typically 2 to 4 weeks later); titrate to response for patients who require bilateral injections for ADSD.
2. Subsequent injections are usually given 12-week intervals (a maximum of 7 Units).

**Limitations of Coverage:**

1. Medicare will allow payment for one injection per site regardless of the number of injections made into the site. A site is defined as one area (including all the muscles around the vocal cords).
2. Bilateral TA and LCA injections for ADSD will not be considered reasonable and necessary when administered concurrently.

**Neurogenic Bladder****Definition:**

The term neurogenic lower urinary tract dysfunction (NLUTD) is usually referred to as a neurogenic bladder which is the loss of normal bladder function caused by damage to part of the nervous system. The neurogenic bladder can be classified as an uninhibited bladder, upper motor neuron bladder, or lower motor neuron bladder. The voiding dysfunction may cause urinary incontinence (UI) which may be due to neurogenic detrusor overactivity (and possibly associated with detrusor sphincter dyssynergia for neurologic lesions below the pontine micturition center). The uninhibited bladder and upper motor neuron bladder commonly occur after a neurologic injury to the central nervous system such as spinal cord injuries (SCI), cerebral vascular accident (CVA) or multiple sclerosis (MS).

**Diagnosis:**

The diagnostic criteria for NLUTD involve a detailed assessment of the patient's neurologic condition, urinary symptoms, and urodynamic findings.<sup>66</sup> A NLUTD is diagnosed and risk-stratified by objective testing such as post-void residual (PVR), urinary tract imaging, and urodynamics. Clinicians should first evaluate the patient's neurological status and history of neurological diseases, such as SCI, MS, or CVAs, which are commonly associated with NLUTD. The presence of lower urinary tract symptoms (LUTS), including UI, urinary retention, frequency, urgency, and nocturia, should be documented. Additionally, a history of recurrent urinary tract infections (UTIs) and autonomic dysreflexia in susceptible individuals could indicate NLUTD. Urodynamic studies (UDS) are crucial in the diagnostic process, providing objective data on bladder storage and voiding function, detrusor activity, bladder compliance, and the presence of detrusor-sphincter dyssynergia.<sup>66</sup>

**Indications of Coverage**

## Initial Botulinum Toxin Injections

Initial BTIs for neurogenic detrusor overactivity will be considered reasonable and necessary when the following requirements are met:

1. The documentation supports a diagnosis of neurogenic detrusor overactivity; AND
2. The etiology of the central nervous system impairment which is causing the neurogenic detrusor overactivity is documented; AND
3. There is moderate to severe neurogenic detrusor overactivity and/or detrusor sphincter dyssynergia assessed by an objective diagnostic testing; AND
4. The voiding dysfunction has failed or become refractory to conservative measures such as lifestyle interventions, bladder training, intermittent catheterizations, pelvic floor muscle exercises and anticholinergic drugs or beta-3 adrenergic agonists (in absence of absolute contraindication to the medications) for a minimum of 12 weeks.<sup>27,67</sup>

## Initial Dosing Guidelines

1. The detrusor overactivity or low bladder compliance may be treated with an initial dose of onabotulinumtoxinA between 100 Units to 200 Units,<sup>68</sup> OR
2. The sphincter dyssynergia or detrusor underactivity with nonrelaxing urethra may be treated with an initial dose of onabotulinumtoxinA between 100 Units to 200 Units.<sup>68</sup>
3. The recommended dose of onabotulinumtoxinA 200 Units per treatment must not be exceeded.<sup>27</sup>

An objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment.

## Subsequent Botulinum Toxin Injections

Subsequent BTIs for NLUTD will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decisions regarding repeat botulinum injections; AND
2. Reassessment of the degree of persistent neurogenic detrusor overactivity and assessment of previous response to botulinum toxin.

## Subsequent Dosing Guidelines

1. The detrusor overactivity or low bladder compliance may be treated with a subsequent dose of onabotulinumtoxinA between 100 Units to 200 Units.<sup>68</sup>
2. The sphincter dyssynergia or detrusor underactivity with nonrelaxing urethra may be treated with a subsequent dose of onabotulinumtoxinA between 100 Units to 200

Units.<sup>68</sup>

3. The dose of onabotulinumtoxinA 200 Units per treatment should not be exceeded.<sup>27</sup>

### **Overactive Bladder (OAB)/Urinary Incontinence (UI)**

#### **Definition:**

Overactive bladder (OAB) is not a disease; it is a symptom complex that generally is not a life-threatening condition characterized by the presence of bothersome urinary symptoms and not related to neurologic conditions.<sup>69</sup>

#### **Diagnosis Overactive Bladder:**

To diagnose OAB, clinicians should document symptoms of urinary urgency, usually accompanied by increased daytime urinary frequency and nocturia, with or without urgency UI.<sup>69,70</sup> These symptoms must be bothersome to the patient as measured with a careful history to characterize the symptoms and assess degree of bother.<sup>69,70</sup> A physical exam and urinalysis should be done to exclude UTIs, neurological disorders, medications, or other pathology that could cause similar symptoms. If diagnosis is uncertain, consider additional testing such as urine culture, post-void residual urine volume assessment, frequency-volume charts, or validated questionnaires. However, definitive testing like urodynamics, cystoscopy and imaging are not mandated upfront and should be reserved for recalcitrant cases.<sup>69,70</sup>

#### **Indications of Coverage**

##### **Initial Botulinum Toxin Injections**

Initial BTIs for OAB will be considered reasonable and necessary when the following requirements are met:

1. The documentation supports a diagnosis of refractory overactive bladder; AND
2. The OAB has been diagnosed by a history and physical exam and a urine analysis to rule out infection or blood in the urine; AND
3. There is moderate to severe OAB assessed by an objective scale\*; AND
4. Conservative treatment for OAB has been tried but the OAB symptoms are refractory to a minimum of 12 weeks of standard of care treatment.
  1. Conservative management which may consist of education of normal bladder function, self-care practices, behavioral modifications, stress management practices, manual physical therapy, and combination therapy; AND
  2. Pharmacological therapy: anticholinergic or beta-3 adrenergic agonists (in absence of absolute contraindication to the medications).

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each

assessment, such as the Overactive Bladder Symptom Score (OABSS), International Prostate Symptom Score–Storage Subscore (IPSS-S), the modified Urgency Severity Scale (USS), and hypersensitive bladder (HSB).

### Initial Dosing Guidelines

1. The initial dose of onabotulinumtoxinA is a total dose of 100 Units per treatment session.

### Subsequent Botulinum Toxin Injections

Subsequent BTIs for OAB will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum toxin injections; AND
2. Reassessment of the degree of persistent OAB and assessment of previous response to botulinum toxin; AND
3. Documentation of a positive response to the initial onabotulinumtoxinA injections.

### Subsequent Dosing Guidelines

1. The subsequent dose of onabotulinumtoxinA is a total dose of 100 Units per treatment session.

## Interstitial cystitis (IC)/Bladder Pain Syndrome (BPS)

### Definition

Interstitial cystitis/bladder pain syndrome (IC/BPS)<sup>71</sup> is an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the urinary bladder, associated with lower urinary tract symptoms of more than 6 weeks duration, in the absence of infection or other identifiable causes.

### Diagnosis

IC/BPS is diagnosed based on characteristic symptoms that persist over time without evidence of infection or other disorders that could explain the symptoms. To confirm a diagnosis of IC/BPS, the following criteria should be met:<sup>71,72</sup>

- The patient should have bothersome bladder sensations perceived to be related to the urinary bladder, typically described as pain, pressure or discomfort.

- These sensations should be associated with lower urinary tract symptoms such as urinary frequency and urgency that have lasted for at least 6 weeks.

The pain or discomfort may be localized to the bladder but can also be felt throughout the pelvis including the urethra, vulva, vagina or rectum, or in extragenital locations like the lower abdomen. The clinician should document the location, duration, and severity of the symptoms.<sup>71,72</sup> There should be no evidence of infection based on negative urine cultures. Other disorders that could potentially explain the symptoms, such as bladder cancer, urinary stones, vaginitis or chronic prostatitis, should be excluded through appropriate evaluation. While IC/BPS remains a clinical diagnosis without definitive objective testing, documentation of characteristic symptoms lasting over 6 weeks that persist despite a basic evaluation to rule out infections and alternate diagnoses supports an accurate IC/BPS diagnosis.<sup>71,72</sup> Baseline voiding symptoms and pain levels should be obtained in order to measure subsequent treatment effects. IC diagnosis often requires cystoscopy with bladder hydrodistension and/or some peculiar morphological findings in bladder biopsies. The findings of cystoscopy can be glomerulations or Hunner lesions. Histologic evaluation of mucosal biopsy can show inflammatory infiltrates, granulation tissue, detrusor mastocytosis, and/or interfascicular fibrosis.

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial BTIs for IC will be considered reasonable and necessary when the following requirements are met:

1. Diagnosis of IC/BPS established based on symptoms, frequency and cystoscopic findings consistent with IC/BPS.
2. IC/BPS are refractory or failed all other management options for a minimum of 6 months including:<sup>31</sup>
  - a. Conservative management/standard of care which may consist of education of normal bladder function, self-care practices, behavioral modifications, stress management practices, manual physical therapy, and combination therapy AND
  - b. Pharmacological therapy (at least 1 agent) AND
  - c. Bladder instillations AND
  - d. Hydrodistention

### Initial Dosing Guidelines

1. The initial dose of onabotulinumtoxinA is a total dose of 100 Units per treatment session.



\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment, such as the Pelvic Pain and Urgency/Frequency Patient Symptom Scale (PUF questionnaire), Global Response Assessment (GRA), O'Leary Sant Symptom and Problem Indexes, NIH-Chronic Prostatitis Symptom Index (NIH-CPSI).

### **Subsequent Botulinum Toxin Injections**

Subsequent BTIs for IC/BPS will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum injections; AND
2. Reassessment of the degree of persistent IC/BPS and assessment of previous response to botulinum toxin; AND
3. Repeat injections may be considered if the patient has a positive response to initial injections.

### **Subsequent Dosing Guidelines**

1. A subsequent dose of onabotulinumtoxinA is a total dose of 100 Units per treatment session.

### **Limitations of Coverage**

1. Botulinum toxin must be injected via cystoscopy and administration without injection into the bladder is non-covered.

### **Sialorrhea**

#### **Definition:**

An exocrine gland disorder due to hypersalivation (ptyalism) associated with neurological disorders, adverse effect of medications, or localized anatomical abnormalities in the oral cavity. Sialorrhea is classified as several types, anterior, posterior or both. Anterior sialorrhea is the flowing of saliva over the lower lip, known as drooling, salivary incontinence or involuntary spillage. Posterior sialorrhea is the flowing of saliva from the tongue to the pharynx. Sialorrhea can occur as a result of increased production of saliva, dental malocclusion, postural problems, cognitive dysfunction, the inability to swallow secretions because of tongue spasticity, weakness of face, mouth and pharyngeal muscles, and dysphagia. Sialorrhea due to psychotropic drugs has been reported repeatedly in the literature.

#### **Diagnosis:**

Sialorrhea is diagnosed clinically but the degree of sialorrhea should be assessed objectively by a clinical scale, such as the Sialorrhea Clinical Scale.

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial BTIs for sialorrhea will be considered reasonable and necessary when the following requirements are met:

1. The documentation supports a diagnosis of sialorrhea; AND
2. The etiology of the impairment which is causing the sialorrhea is documented; AND
3. There is moderate to severe sialorrhea assessed by an objective scale\*; AND

The condition has failed conservative measures such as observation, positioning, behavioral therapies, speech therapy and pharmacological therapy\*. The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. For example, clinical scales include Thomas-Stonell drooling severity scale and the Thomas-Stonell drooling frequency scale.

4. Ultrasound guidance is recommended to ensure delivery into the submandibular gland and is advisable for the parotids; however, experienced clinicians have relied on landmarks for the parotids.

### Initial Dosing Guidelines

1. The rimabotulinumtoxinB initial dosage is 1,500 Units to 3,500 Units; 500 Units to 1,500 Units per parotid gland and 250 Units per submandibular gland.<sup>41</sup>
2. The incobotulinumtoxinA initial total dose is 100 Units per treatment session consisting of 30 Units per parotid gland and 20 Units per submandibular gland.<sup>28</sup>
  - a. IncobotulinumtoxinA is intended for intramuscular or intraglandular injection in the parotid and submandibular glands only.
3. The abobotulinumA initial total dose is 15 per gland (submandibular, parotid or both) either unilaterally or bilaterally

### Subsequent Botulinum Toxin Injections

Subsequent BTIs for sialorrhea will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum injections; AND
2. Reassessment of the degree of persistent sialorrhea and assessment of previous response to botulinum toxin.

## Subsequent Dosing Guidelines

1. The subsequent dosage of rimabotulinumtoxinB is 1,500 Units to 3,500 Units; 500 Units to 1,500 Units per parotid gland and 250 Units per submandibular gland; no more frequent than every 12 weeks.<sup>41</sup>
2. The subsequent dosage of incobotulinumtoxinA is 100 Units per treatment session consisting of 30 Units per parotid gland and 20 Units per submandibular gland, no sooner than every 16 weeks.<sup>28</sup>
3. The subsequent dosage of abobotulinum toxinA is 15 to 75 units injected per gland (submandibular, parotid or both) either unilaterally or bilaterally with intervals of 4 to 6 months between treatments.

## Upper and Lower Spasticity

### Definition:

Spasticity is a subset of dystonia, describing a velocity-dependent increase in tonic-stretch reflexes with exaggerated tendon jerks as a result from an upper motor neuron lesion which disinhibits the tendon stretch reflex.<sup>73</sup>

### Diagnosis:

Spasticity is diagnosed clinically but the degree of spasticity should be assessed objectively by a clinical scale, such as the Ashworth Scale, Modified Ashworth Scale, Tardieu Scale, Modified Tardieu Scales, and the pendulum test.

## Indications of Coverage

### Initial Botulinum Toxin Injections

Initial botulinum toxin injections for spasticity will be considered reasonable and necessary when the following requirements are met:

1. Objective documentation of the clinical features consistent with the diagnosis of spasticity; AND
2. Moderate to severe chronic spasticity; AND
3. The clinical treatment goals are documented; AND
4. Electromyography, electrical stimulation, and/or ultrasonography, rather than site pain or tenderness, is required to determine injection site(s) for botulinum toxin especially for spastic conditions of the face, neck and upper extremity.

\*An objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment.

## Initial Dosing Guidelines

1. The initial dose of onabotulinumtoxinA is indicated for upper limb spasticity and lower limb spasticity in adults is based on the muscles affected and severity of muscle spasticity. The dose is divided among selected muscles at a given treatment session. The initial dose of onabotulinumtoxinA must not exceed a total dose of 400 Units for the lower limb, or a total of 600 units per treatment session of multiple limbs.<sup>27</sup>
2. The initial dose of onabotulinumtoxinA is indicated for upper limb spasticity and lower limb spasticity in a pediatric beneficiary is based on the muscles affected and severity of muscle spasticity. The dose is divided among selected muscles at a given treatment session. The total initial dose of onabotulinumtoxinA for upper limb spasticity must not exceed a dose of 3-6 Units/kg up to a maximum 200 Units per upper limb. The total initial dose of onabotulinumtoxinA for lower limb spasticity must not exceed a dose of 4-8 Units/kg up to a maximum 300 Units per lower limb.
3. The initial dose of abobotulinumtoxinA for the treatment of upper limb spasticity and lower limb spasticity in adults is based on muscles affected and severity of muscle spasticity.<sup>40</sup> The dose is divided among selected muscles at a given treatment session. The initial dose of abobotulinumtoxinA must not exceed a dose of 500-1000 Units for the upper limb, 1000 Units for the lower limb, or a total dose of 1500 Units per treatment session of multiple limbs.
4. The initial dose of abobotulinumtoxinA is indicated for upper limb spasticity and lower limb spasticity in a pediatric beneficiary is based on the muscles affected and severity of muscle spasticity. The dose is divided among selected muscles at a given treatment session. The total initial dose of abobotulinumtoxinA for upper limb spasticity must not exceed a dose of 8-15 Units/kg per upper limb up to a maximum 1500 Units per upper limb. The total initial dose of abobotulinumtoxinA for lower limb spasticity must not exceed a dose of 10-15 Units/kg per lower limb up to a maximum 1000 Units per lower limbs, or a total dose of 1500 Units per treatment session of multiple limbs.
5. The initial dose of incobotulinumtoxinA for the treatment of upper limb spasticity and lower limb spasticity in adults is based on muscles affected and severity of muscle spasticity. The dose is divided among selected muscles at a given treatment session. The total initial dose of incobotulinumtoxinA must not exceed a dose of 400 Units for the upper limb, 400 Units for the lower limb, or a total dose of 600 Units per treatment session of multiple limbs.

## Subsequent Botulinum Toxin Injections

Subsequent botulinum toxin injections for spasticity will be considered reasonable and necessary when the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum injections; AND

2. Reassessment of the degree of persistent spasticity and assessment of the previous response to botulinum toxin; AND
3. Electromyography, electrical stimulation, and/or ultrasonography rather than site pain or tenderness, is required to determine injection site(s) for botulinum toxin especially for spastic conditions of the face, neck and upper extremity.

### Subsequent Dosing Guidelines

1. The subsequent dose of onabotulinumtoxinA for both upper limb and lower limb spasticity in adults is based on the muscles affected, severity of muscle spasticity, prior response and adverse reaction history following treatment with botulinum toxins. The dose is divided among selected muscles at a given treatment session. The subsequent dose of onabotulinumtoxinA in adults must not exceed a dose of 400 Units for the upper limb, 400 Units for the lower limb, or a total dose of 600 Units per treatment session of multiple limbs per treatment session when clinically justified and supported by documentation in the medical record.
2. The subsequent dose of onabotulinumtoxinA is indicated for upper limb spasticity and lower limb spasticity in a pediatric beneficiary is based on the muscles affected and severity of muscle spasticity. The dose is divided among selected muscles at a given treatment session. The total subsequent dose of onabotulinumtoxinA for upper limb spasticity must not exceed a dose of 3-6 Units/kg up to a maximum 200 Units per upper limb. The total subsequent dose of onabotulinumtoxinA for lower limb spasticity must not exceed a dose of 4-8 Units/kg up to a maximum 300 Units per lower limb.
3. The subsequent dose of abobotulinumtoxinA for the treatment of upper limb spasticity and lower limb spasticity in adults is based on the muscles affected, severity of muscle spasticity, prior response and adverse reaction history following treatment with botulinum toxins. The dose is divided among selected muscles at a given treatment session. The subsequent dose of abobotulinumtoxinA in adults must not exceed for upper limb spasticity is between 500-1000 Units divided among selected muscles at a given treatment session.<sup>40</sup> The maximum total dose (upper and lower limb combined) of abobotulinumtoxinA for the treatment of spasticity in adults is abobotulinumtoxinA 1500 Units when clinically justified and supported by documentation in the medical record.
4. The subsequent dose of abobotulinumtoxinA for upper limb spasticity must not exceed a dose of 8-15 Units/kg per upper limb up to a maximum 1500 Units per upper limb. The total initial dose of abobotulinumtoxinA for lower limb spasticity must not exceed a dose of 10-15 Units/kg per lower limb up to a maximum 1000 Units per lower limbs. The maximum total dose (upper and lower limb combined) of abobotulinumtoxinA is 1500 Units per treatment session of multiple limbs when clinically justified and supported by documentation in the medical record.

5. The subsequent dose of incobotulinumtoxinA for both upper limb and lower limb spasticity in adults is based on the muscles affected, severity of muscle spasticity, prior response and adverse reaction history following treatment with botulinum toxins. The dose is divided among selected muscles at a given treatment session. The subsequent dose of onabotulinumtoxinA must not exceed a dose of 400 Units for the upper limb, 400 Units for the lower limb, or a total dose of 600 Units per treatment session of multiple limbs when clinically justified and supported by documentation in the medical record.

## **Strabismus**

### **Definition:**

Strabismus is the misalignment of the eyes not lining up in the same direction which may be congenital or acquired.<sup>74</sup> Strabismus is also known as hypertropia and crossed eyes. The condition may present with one eye deviating inward (esotropia), outward (exotropia), upward (hypertropia), or downward (hypotropia).

### **Diagnosis:**

Strabismus diagnosis involves a comprehensive assessment where the deviation of the eyes is quantified and characterized.<sup>74,75</sup> Diagnostic evaluations include measuring the angle of deviation, differentiating between accommodative and non-accommodative strabismus, assessing the frequency of the deviation (constant or intermittent), and determining its nature (concomitant or paralytic).<sup>74,75</sup>

The diagnostic process also includes examining binocular functions and the presence of amblyopia or suppression, where one eye's vision is favored over the other. This comprehensive approach ensures that the diagnosis of strabismus is accurate, facilitating targeted management and intervention strategies to correct eye alignment and improve visual function. Conditions such as convergence insufficiency, divergence insufficiency, and sagging eye syndrome are specifically mentioned, highlighting the need for diagnostic criteria that can categorize strabismus not only by its direction and magnitude but also by its underlying cause.<sup>74,75</sup>

### **Indications of Coverage**

#### **Initial Botulinum Toxin Injections**

Initial BTIs for strabismus will be considered reasonable and necessary when the following requirements are met:

1. Objective documentation of the clinical features consistent with the diagnosis of strabismus; AND
2. Moderate to severe strabismus\*; AND
3. The clinical objective treatment goals are documented.

\* The objective assessment must be performed and documented at baseline, after each diagnostic procedure, and at each follow-up assessment using the same scale during each assessment. For example, clinical scales to measure severity of strabismus include Ashworth Scale score and the Tardieu Scale.

### **Initial Dosing Guidelines**

1. The initial dose of onabotulinumtoxinA is 1.25 Units-2.5 Units in any one muscle.<sup>27</sup>

### **Subsequent Botulinum Toxin Injections**

Subsequent BTIs for strabismus will be considered reasonable and necessary when all the following requirements are met:

1. Documentation of informed clinical decision regarding repeat botulinum injections; AND
2. Reassessment of the degree of persistent strabismus and assessment of previous response to botulinum toxin.

### **Subsequent Dosing Guidelines**

1. The subsequent treatment dose of onabotulinumtoxinA is 1.25 Units-2.5 Units in any one muscle.<sup>27</sup>

### **Provider Qualifications**

The Medicare Program Integrity Manual states services will be considered reasonable and necessary only if performed by appropriately trained providers.

Patient safety and quality of care mandate that healthcare professionals who perform botulinum injections/procedures for chronic pain (not surgical anesthesia) are appropriately trained and/or credentialed by a formal residency/fellowship program and/or are certified by either an accredited and nationally recognized organization or by a post-graduate training course accredited by an established national accrediting body or accredited professional training program whose core curriculum includes the performance and management of the procedures addressed in this policy. Credentialing or privileges are required for procedures performed in inpatient and outpatient settings.<sup>76</sup>

All aspects of care must be within the provider's medical licensure and scope of practice. Reimbursement for procedures utilizing imaging techniques may be made to providers who meet training requirements for the procedures in this policy only if their respective state



allows such in their practice act and formally licenses or certifies the practitioner to use and interpret these imaging modalities (ionizing radiation and associated contrast material, magnetic resonance imaging, ultrasound). At a minimum, training must cover and develop an understanding of anatomy and drug pharmacodynamics and kinetics as well as proficiency in diagnosis and management of disease, the technical performance of the procedure, and utilization of the required associated imaging modalities.

## Summary of Evidence

### Achalasia

#### Society Guidelines

We found 3 guidelines issued by medical societies and 1 position statement: the American Society of Gastroenterology (ASGE)'s Guideline on the Management of Achalasia 2020<sup>13</sup>, the International Society for Diseases of the Esophagus (ISDE)'s 2018 Achalasia Guidelines<sup>2</sup>, the American College of Gastroenterology (ACG) Clinical Guidelines for the Diagnosis and Management of Achalasia, 2020<sup>8</sup>, and the European Society of Gastrointestinal Endoscopy (ESGE) Guideline 2020<sup>1</sup> support BTI into the distal esophageal sphincter as an effective short-term treatment for achalasia, in medically high-risk patients who are not candidates for other invasive therapies.

The ASGE guidelines<sup>13</sup> were based on a systematic review and meta-analysis by Khashab et al. (2020). The review included 22 uncontrolled studies (published from inception to October 2017) in 730 achalasia patients who were treated with BTI. Clinical success, defined by an Eckardt score of 3, was achieved in 77% (95% confidence interval [CI], 72%-81%; I<sup>2</sup> value 35; P= 0.04) over a follow-up period ranging from 1 to 6 months. There was a statistically significant decrease in average LES pressure from 38.23 mm Hg (range, 34.40-42.06) before the procedure to 23.30 mm Hg (range, 20.79-25.81) after BTI (P < 0.01). At 12 months, clinical success rates were 73.3% (55/75) and 37.5% (27/72), for pneumatic dilation (PD) and BTI respectively (Risk ratio, 1.88; 95% CI, 1.35-2.61; P= 0.0002). Serious adverse events were uncommon, with transient chest pain reported after 4.4% of injections. The authors recommended against the use of BTI as definitive therapy for achalasia patients (AMSTAR moderate quality of evidence, or moderately confident in the effect estimate: the true effect is likely to be close to the estimate of effect, but there is a possibility that it is substantially different).

The ESGE<sup>1</sup> performed a systematic review (discussed below under the systematic review by Weusten et al. (2020)) and Delphi consensus and aligned with previous recommendations. They additionally made specific dosing recommendations of 100 units onabotulinumtoxinA or equivalent of the toxin diluted in preservative-free saline that is injected in aliquots of 0.5–1mL using an injection needle in forward view just above the squamocolumnar junction in at least 4 quadrants. This was a strong recommendation, high quality of evidence, with a level



of agreement of 100%. A multicenter randomized trial with injections of 50, 100, and 200 Units of BTI resulted in similar short-term results in LES pressure 1 month after injection<sup>11</sup> and in an SR, a dose of 100 units of botulinum toxin was used most frequently<sup>77</sup>, supporting this dosing recommendation.

The 2018 ISDE achalasia guidelines were based on a systematic review by Zaninotto et al. (2018)<sup>2</sup> (discussed below). The strength of recommendation was graded according to Grading of Recommendations Assessment (GRADE). These guidelines recommend BTI for patients who are unfit for surgery or as a bridge to more effective therapies, such as surgery or endoscopic dilation (GRADE – moderate).

The ACG (2020)<sup>8</sup> performed a literature review and generated consensus-based recommendations using a modified Delphi process. They recommend BTI as first-line therapy for patients with achalasia that are unfit for definitive therapies compared with other less-effective pharmacological therapies.

## Systematic Reviews

Recent systematic reviews and meta-analyses comparing the use of BTI with alternative treatments have found that, while BTI may provide short-term symptom relief and are associated with low complication rates, its results are inferior in both medium- and long-term compared to all other non-pharmacological alternatives.

Gong et al. (2023)<sup>12</sup> conducted a systematic review comparing the efficacy of 9 interventional treatments (BTI, PD, BTI + PD, Laparoscopic Heller's myotomy (LHM) + Toupet, LHM + anterior wrap (D), anterior POEM (APOEM), traditional posterior POEM, double-scope POEM, and waterjet assisted JPOEM) for achalasia. The authors included 27 RCTs published from 1990 to December 2020 with 2,278 achalasia patients (BTI [n = 158], BTI + PD [n = 72]) in a meta-analysis. Primary outcome was comparisons of the induction of clinical remission after 1-year follow-up. Results indicated that APOEM was the most effective (rank 1, 50%) strategy to significantly relieve achalasia symptoms, while BTI ranked last (rank 9, 99%), although BTI had the highest safety profile. However, the recommendations provided by the authors were not supported by the reported data.

Shiu et al. (2022)<sup>78</sup> conducted a systematic review and meta-analysis of RCTs published between January 2000 and June 2021 to compare the efficacy of 8 treatments for achalasia. Twenty-four studies involved a total of 1,987 participants for analysis. LHM + Dor (OR 2.29, 95% CI: 1.50–3.49) all showed significantly greater efficacy and were at least twice as effective as the reference group with only BTI (OR 0.33, 95% CI: 0.17–0.63) showing significantly lower efficacy. The median interquartile ranges (IQRs) for short-term clinical successful rates (%) of Anterior POEM, Posterior POEM, LHM + Toupet, BTI + PD, LHM + Dor, PD, and BTI were 91.1 (IQR, 84.4–95.4), 92.1 (IQR, 87.0–93.0), 93.9 (IQR, 90.2–97.0), 85.4

(IQR, 84.4–86.5), 86.7 (IQR, 79.7–88.7), 66.35 (IQR, 56.0–77.7), and 53.3 (IQR, 37.5–60.0), respectively.

Weusten et al. (2020)<sup>1</sup> conducted a systematic review of the technical aspects of the endoscopic management of gastrointestinal motility disorders (including achalasia). Evidence levels and recommendation strengths were assessed using the GRADE system. The total number of studies included were unclear as was the methods for analysis and combination of study results. The authors concluded that BTI can be considered an effective and safe therapy for short-term symptom relief in esophageal achalasia (GRADE recommendation= Moderate). However, the authors' recommendations for policy and practice were not supported by the reported data.

Andolfi et al. (2019)<sup>10</sup> conducted a systematic review of articles published between 2008 and 2018 comparing the effect of different invasive treatments on symptomatic outcomes across achalasia subtypes. Their search strategy included PubMed® and MEDLINE only. They included 20 studies (1,575 participants) reporting clinical outcomes after BTI, PD, LHM and POEM based on manometric subtypes. The authors performed a meta-analysis. Three studies reported data on botulinum toxin (58 patients). BTI was the treatment modality with the worse outcomes (18% for type I, 59% for type II and 21% for type III). About half of patients needed further injections at intervals of 6–24 months. However, the authors did not assess the likelihood of publication bias.

Zaninotto et al. (2018)<sup>2</sup> conducted a systematic review with interdisciplinary and international authors to establish best evidence-based principles to the diagnosis and management of achalasia. The authors included 466 articles in their analysis. The authors found that BTI has a high safety profile with mild adverse events (heartburn or chest pain) observed in less than 10% of patients treated. At 2-year follow up, only 34% of BTI patients versus 87.5% of the Heller patients were asymptomatic. Similarly, 4 randomized trials and a Cochrane meta-analysis comparing BTI with PD consistently reported a higher cumulative rate of remission rate at 1 year after treatment after PD. The authors recommended against BTI as an effective therapy (control of symptoms) for achalasia in patients fit for surgery (LHM) or PD GRADE: moderate. Repeated BTI may be successful, if there are contraindications to invasive, but more durable treatments. The authors also found that BTI efficacy may decrease over time. However, it should be noted that the authors did not assess the likelihood of publication bias. Further, the authors' recommendations for policy and practice were not supported by the reported data.

Leyden et al. (2014)<sup>14</sup> conducted a Cochrane systematic review and meta-analysis comparing the efficacy and safety of 2 endoscopic treatments, PD and BTI. They included 7 randomized controlled trials (RCT) with 178 patients published from 1946 through March 2014. The authors examined symptom remission rates within the first month, at 6 months, and at 12 months. Based on the AMSTAR-2 critical appraisal tool, the overall confidence in

the result of this meta-analysis was deemed to be “high.” The overall methodological quality of the studies was good although the risk of bias was high. Only 1 of the included studies was double blinded. There was no significant difference between PD and BTI arms in clinical success rates or LES pressures within 4 weeks of the initial intervention (risk ratio of remission, 1.11; 95% CI, 0.97-1.27) with a weighted mean difference for PD of -0.77 (95% CI, -2.44 to 0.91;  $P = 0.37$ ). Clinical success rates beyond 4 weeks were available for 3 studies at 6 months and 4 studies at 12 months. At 6 months, clinical success was achieved in 80.7% of patients (46/57) who underwent PD as compared with 51.8% of patients (29/56) who underwent BTI (risk ratio, 1.57; 95% CI, 1.19-2.08;  $P = 0.0015$ ). At 12 months, clinical success rates were 73.3% (55/75) and 37.5% (27/72), respectively (risk ratio, 1.88; 95% CI, 1.35-2.61;  $P = 0.0002$ ). There were no adverse events with BTI (total of 151 injection procedures), whereas perforation occurred in 3 cases (total of 188 PD procedures) in the PD arm. These data demonstrate that PD is a more effective long-term (>6 months) endoscopic treatment option compared with BTI for patients with achalasia. However, the authors’ policy and practice recommendations were not supported by the reported data.

## Meta-Analyses

N/A

## Randomized Controlled Trials

N/A

## Anal Fissure

## Society Guidelines

We found 1 medical society guideline issued in 2022 by the American Society of Colon and Rectal Surgeons (ASCRS).<sup>79</sup> They state that botulinum toxin has similar results compared with topical therapies as first-line therapy for chronic anal fissures and modest improvement in healing rates as second-line therapy following failed treatment with topical therapies.<sup>79</sup> Grade of recommendation: strong recommendation based on moderate-quality evidence, 1B. The guideline was based on a systematic review (described below).<sup>79</sup>

## Systematic Reviews

### ASCRS’ Systematic Review

ASCRS’ systematic review (n = 1577) included 86 studies treated with botulinum toxin for chronic anal fissures. The review found no clear optimal dosing or injection protocol, with doses ranging from 20 to 100 units and injection sites varying in the literature. Pooled analysis showed botulinum toxin has a 33-96% fissure healing rate, with an overall 5% risk of transient fecal incontinence and no evidence of dose-dependent efficacy or complication

rates.<sup>79</sup> Compared to topical nitroglycerin or calcium channel blockers, botulinum toxin showed comparable efficacy of 67-71% healing but better side effect profile. As first-line therapy, prospective studies found botulinum toxin heals 67% of fissures, similar to 71% with topicals. Second-line after failed topical therapy, small studies show botulinum improves symptoms and healing without needing sphincterotomy surgery. Limitations include significant heterogeneity amongst studies in botulinum toxin protocol and comparisons.<sup>79</sup>

### Other Systematic Reviews

A large systematic analysis by Vitoopinyoparb et al. (2022) provides moderate quality evidence that low dose botulinum toxin injected out of the fissure (OOF) offers the best short-term healing, while low dose both sides of the fissure (BSF) optimizes longer term outcomes.<sup>80</sup> It reviewed 27 RCTs (n = 1,880 patients) to determine the optimal botulinum toxin dose and injection site for chronic anal fissures. Treatments were categorized as low dose ( $\leq 20$  units) or high dose ( $> 20$  units) botulinum toxin injected either OOF, on BSF, or at both sites (BS). For short-term healing (n = 1,720), high dose OOF had significantly better outcomes than high dose BSF, with similar efficacy to low dose OOF but higher incontinence risk (n = 1,458). Low dose BSF with added topical agents demonstrated significantly improved healing over low dose BSF alone.<sup>80</sup> For recurrence (n = 1,622), OOF sites had higher rates than BSF sites regardless of dosage.

Boland et al. (2020) found sphincterotomy is most effective for fissure healing, while medical therapies have higher risks of recurrence and variable side effect profiles during its systematic review of 9 RCTs (n = 775) comparing treatments for chronic anal fissure.<sup>19</sup> Lateral internal sphincterotomy (LIS) had the highest healing rate at 8 weeks at 95.1% (n = 349) but a 10% overall incontinence risk and 2.3% permanent incontinence rate. Botulinum toxin (n = 132) had a 66.7% healing rate at 8 weeks and a 41.7% recurrence rate but a 14.4% temporary incontinence rate. Topical nitrates (n = 206) had 63.6% healing and 11.2% recurrence but fewer side effects. Permanent incontinence is a concern with sphincterotomy.<sup>19</sup> Additional data is still needed comparing sphincter-preserving procedures.<sup>19</sup>

Sahebally et al. (2018) systematically reviewed 6 RCTs (n = 393) comparing BTI to topical nitrates (TN) for chronic anal fissures (n = 194 botox, n = 199 nitrates).<sup>81</sup> There was no significant difference in fissure healing or recurrence rates, but BTI had significantly fewer side effects overall (6.4% vs 33.1%) and less headache specifically (4.8% vs 27.9%) than TN. However, BTI was associated with higher transient anal incontinence (10.4% vs 4.4%,  $P = 0.06$ ). In summary, while equally effective for healing chronic anal fissures, BTI appears to have a better side effect profile than TN, albeit with a non-significantly higher risk of temporary incontinence. Further adequately powered, high quality head-to-head trials with uniform methodology are still needed to definitively compare these 2 modalities.<sup>81</sup>

Lin et al. (2016) performed a systematic review and meta-analysis of 18 studies ( $n = 1,158$ ) with ( $n = 661$ ) using Botulinum toxin to determine the optimal botulinum toxin dose and injection site for chronic anal fissures.<sup>82</sup> Meta-regression analysis found that higher doses were associated with slightly lower healing rates (0.34% reduction per unit increase,  $P = 0.048$ ) and higher risks of incontinence (1.02 times higher per unit,  $P = 0.048$ ) and recurrence (1.037 times higher per unit,  $P = 0.0002$ ).<sup>82</sup> There was significant heterogeneity amongst studies regarding injection locations, preventing definitive conclusions, but injecting laterally into the internal anal sphincter was most common. In summary, this well-conducted analysis provides evidence that lower botulinum toxin doses (10-20 units) have similar or better efficacy than higher doses for fissure healing, with less side effects, supporting use of the lowest effective dose.<sup>82</sup>

## Meta-Analyses

Bobkiewicz et al. (2016) performed a meta-analysis of 34 studies involving 1,577 patients to determine if there is a dose-dependent efficacy of BTIs for chronic anal fissures.<sup>83</sup> This large meta-analysis found no evidence that higher Botulinum toxin doses improve fissure healing, but similarly no increased side effects, providing support for using the minimum effective dose in treatment regimens. Across a wide 5-150-unit dose range, they found no correlation between dose and healing rates (33-96% range,  $P = 0.0708$ ), complication rates, or incontinence risks (occurring in 5.01%). There was also no difference between injection sites or number of injections per session. The analysis did reveal short follow-up times, generally under 6 months, which could underestimate recurrence rates. Longer-term data is still needed.<sup>83</sup>

## Randomized Controlled Trials (RCTs)

Pilkington et al. (2018) performed a randomized trial that demonstrated equivalent efficacy but some long-term advantages of unilateral over bilateral BTIs for chronic anal fissures.<sup>84</sup> Their study comparing bilateral ( $n = 49$ ) to unilateral ( $n = 51$ ) injection of 100 units botulinum toxin for chronic anal fissures. There was no difference in injection pain between groups.<sup>84</sup> Healing rates were similar at 53.5% (bilateral) and 52.2% (unilateral). While bilateral injection provided better pain relief initially, at 1-year unilateral injection resulted in significantly greater fissure pain improvement (-39.1 mm on VAS vs -22.3 mm) and higher quality of life, with no increased incontinence risk.<sup>84</sup>

Gandomkar et al. (2015) performed a randomized trial that provides evidence that combined botulinum toxin and diltiazem cream has similar efficacy to sphincterotomy surgery for short duration chronic anal fissures, with fewer risks. It compared combined botulinum toxin A injection and topical diltiazem ( $n = 49$ ) to partial lateral internal sphincterotomy (PLIS) ( $n = 50$ ) for chronic anal fissure.<sup>85</sup> After 1 year, PLIS had a significantly higher healing rate than the combined treatment (94% vs 65%,  $P < 0.001$ ).<sup>85</sup>

However, PLIS was also associated with higher rates of incontinence (16% vs 4%,  $P = 0.04$ ) and other complications. For patients with shorter duration fissures ( $\leq 12$  months), there was no difference in 1-year healing between groups (100% in both). PLIS remains more effective for longer duration lesions.<sup>85</sup>

Berkel et al. (2014) performed a randomized trial comparing botulinum toxin A ( $n = 27$ ) to topical isosorbide dinitrate (ISDN) ( $n = 33$ ) for primary treatment of chronic anal fissure.<sup>86</sup> The study provides evidence that botulinum toxin A injection is more effective and better tolerated than topical ISDN as initial therapy for chronic anal fissures. After a median 9 weeks, botulinum toxin A had a significantly higher healing rate than ISDN (67% vs 33%,  $P = 0.01$ ).<sup>86</sup> While pain improvement was similar between groups, botulinum toxin A was associated with significantly fewer side effects. However, fissure recurrence at 1 year was still substantial in both groups (28% botulinum toxin A, 50% ISDN,  $P = 0.29$ ).<sup>86</sup>

## Blepharospasm

### Society Guidelines

The American Academy of Neurology (AAN)<sup>5</sup> recommends onabotulinumtoxinA (OnaBoNT-A) and incobotulinumtoxinA (incoBoNT-A) as level B and abobotulinumtoxinA (AboBoNT-A) as level C treatment options. Overall, botulinum toxin A (BtA) should be considered the first line of treatment.

### Systematic Reviews

Duarte et al. (2020)<sup>87</sup> performed a systematic review on BtA therapy for patients with blepharospasm. Three RCTs were included with a total of 313 patients. Two of the studies included excluded patients with worse responses to BtA. The trials included and evaluated a single treatment session which showed a moderate to large improvement in blepharospasm severity and a reduction of 0.93 on the JRS at 4-6 weeks. A moderate to large improvement was also found in blepharospasm-specific disability and blepharospasm-specific involuntary movements at 4-6 weeks. The mean duration of effects was about 10.6 weeks. There was an increased risk of vision complaints and eyelid ptosis seen throughout the entire study group compared to the placebo.

### Meta-Analyses

N/A

### Randomized Controlled Trials (RCTs)

N/A

### Other Studies



Calace et al. (2003)<sup>88</sup> found that patients who had undergone treatments for over 10 years did not vary in the average duration of relief or response to treatment.

Engstrom et al. (1987)<sup>89</sup> found that the average response duration for repeat BTIs increased for the first 3 treatment sessions and the average response time found in this study was 6-12 weeks. The longer response time was reversed starting at the fourth session.

Shorr et al. (1985)<sup>90</sup> found that patients had an average of 6.1 weeks of relief after the initial injection of BtA and 8.44 weeks of relief after the second injection.

Scott et al. (1985)<sup>91</sup> found that the middle of the upper eyelid should be avoided during injection as this showed an increased risk of ptosis. The superior rectus muscle should also be avoided as this could cause hypotropia. This study also found that patients who had injections of BtA into the orbicularis oculi experienced some relief from their symptoms. It was also found that the orbicularis oculi muscle remained weakened after each subsequent injection, but spasms still reoccurred. This effect was prolonged in patients who had been operated on previously.<sup>91</sup>

## Cervical Dystonia

### Society Guidelines

We found 3 society guidelines founded on systematic review outcomes. The AAN issued a guideline in 2016 for treating CD, updating their 2008 guidelines.<sup>92</sup> They recommend abobotulinumtoxinA (aboBoNT-A) and rimabotulinumtoxinB (rimaBoNT-B) as highly effective (Level A) treatments, with onabotulinumtoxinA (onaBoNT-A) and incobotulinumtoxinA (incoBoNT-A) as also effective options (Level B).<sup>92</sup>

The European Federation of the Neurological Societies (EFNS) in 2011 advised using BtA, or botulinum toxin B (BtB) in case of resistance to type A, for CD.<sup>93</sup> The American Academy of Otolaryngology-Head and Neck Surgery (AAO-HNS) in 2021 declared botulinum toxin as a safe and effective treatment for CD in a position statement.<sup>94</sup>

Both AAN and EFNS guidelines are based on systematic reviews performed on their behalf by researchers (discussed below).<sup>92,93</sup> The AAO-HNS's stance was supported by a single systematic review published in 2013 found as part of a literature review.<sup>94,95</sup>

### Systematic Reviews

#### AAN's Systematic Review

AAN's systematic review focused on randomized, masked trials (RMTs) and the medical society found 23 that focused on CD.<sup>92</sup> Its review found that various botulinum toxin formulations have shown differing levels of efficacy and safety. AboBoNT-A and rimaBoNT-

B are established as effective, based on 2 and 3 Class I studies respectively. OnaBoNT-A and incoBoNT-A are probably effective, supported by Class I and II studies. A study on incoBoNT-A showed a significant improvement in the TWSTRS scores at 4 weeks (placebo: 22.2, 120 U: 29.9, 240 U: -10.9,  $p < 0.001$ ) with dysphagia in 23.4% (240 U) and 10.7% (120 U) of cases. OnaBoNT-A improved the CD Severity Scale by 21.81 points and the Global Assessment Scale by 61.7% compared to 20.31 points and 41.6% for placebo ( $p = 0.012$  and  $p = 0.022$ ). Comparative studies found no significant efficacy difference between onaBoNT-A, aboBoNT-A, and rimaBoNT-B. Long-term studies show onaBoNT-A's benefits persisting for up to 2 years.

### EFNS's Systematic Review

Method specifics for EFNS's guidelines were less specified than AAN's. Researchers noted that a total of 299 papers were found relevant to dystonia treatments broadly, with 191 being primary diagnostic studies and 108 efficacy studies.<sup>96</sup> But it was unclear how many of the studies reviewed were specific to examining the use of botulinum toxins as a treatment for CD. EFNS' researchers broadly noted that studies reviewed specific to Botox and CD included systematic reviews, long-term observational studies, RCTs (Class I and II), and comparative efficacy studies (Class III and IV).<sup>96</sup>

### Other Systematic Reviews

Rodrigues et al. (2020)<sup>97</sup> performed a systematic review on BtA therapy for CD. The analysis included 9 RCTs ( $n = 1,144$ ) comparing a single BtA treatment to placebo. Seven studies excluded patients with poorer responses to BtA treatment, therefore, the population had a higher probability of benefiting from BtA. BtA doses ranged from 150-1000 units depending on formulation. The RCTs evaluated patients for a single treatment session and a moderate to large improvement from the patient's baseline clinical status was found. At week 4, BtA resulted in a mean reduction of 8.09 points in the TWSTRS (95% CI 6.22 to 9.96;  $I^2 = 0\%$ ), which was an 18.4% improvement compared to placebo. The mean difference in TWSTRS pain sub-score was 2.11 (95% CI 1.38 to 2.83;  $I^2 = 0\%$ ). Patients and clinicians both reported improvement of the patients' subjective clinical status. Three of the studies included in the systematic review examined the impact of BtA on a patient's quality of life and it was shown that there was improvement when treated with BtA. A single treatment session of BtA has been found to be effective and well tolerated in the treatment of moderately impaired adults with CD. There has been moderate to large improvement in severity, disability, and pain at 4 weeks but the mean duration of effect for BtA has been inconclusive.<sup>97</sup> Adverse events are common but are not commonly associated with discontinuing treatment. It is uncertain whether the clinical effectiveness of botulinum toxin decreases with repeated treatments.<sup>97</sup> The systematic review could not evaluate long-term duration as all trials included but 2 evaluated a single treatment session. Overall, with moderate certainty, BtA increased the risk of any adverse event (RR 1.23; 95% CI 1.05 to



1.43; I<sup>2</sup> = 28%). The most common were neck weakness (14%; RR 3.40), dysphagia (11%; RR 3.19), and diffuse weakness/tiredness (8%; RR 1.80). Similarly, with moderate certainty, BtA resulted in clinically relevant reductions in dystonia impairment and pain compared to placebo. With high certainty, it was well tolerated.<sup>97</sup>

Marques et al. (2016)<sup>98</sup> performed a systematic review evaluating 4 RCTs (n = 441 patients) that compared botulinum toxin type B (BtB) to placebo for CD. These RCTs evaluated patients during a single treatment session. BtB doses ranged from 2,500-10,000 units. At week 4, BtB improved clinical status by 14.7% (95% CI 9.8–19.5%) and reduced TWSTRS total score by 6.8 points (95% CI 4.54-9.01) versus placebo. The mean difference in TWSTRS pain score was 2.20 (95% CI 1.25-3.15). Patients as well as clinicians reported improvement in subjective status. Withdrawal rates due to adverse events were similar between groups, but BtB increased risk of dry mouth (RR 7.65, 95% CI 2.75–21.32) and dysphagia (RR 6.78, 95% CI 2.42–19.05). There was a dose-dependent response for duration of effect and risk of adverse events (the average duration of benefit lasted from 12-16 weeks).<sup>98</sup> BtB was more effective in BtA non-responders, with no difference in effect size between responder subgroups. Overall, the authors concluded that BtB significantly reduced CD severity, disability and pain versus placebo, but increased dry mouth and dysphagia risks. However, data were lacking on repeated BtB cycles, optimal doses/intervals, injection techniques, and quality of life impact.<sup>98</sup>

Boyce et al. (2022) sought to examine satisfaction rates among CD patients using a botulinum toxin product.<sup>99</sup> Their systematic review of 8 studies (n = 1,764) found that the top barrier to patient satisfaction with botulinum toxin treatment was perceived ineffectiveness, with up to 66% reporting dissatisfaction.<sup>99</sup> Side effects like dysphagia and neck weakness were the next most common issue, affecting 6-34%. Rigid retreatment schedules and early wearing off of effects also reduced satisfaction. Finally, a lack of experienced injectors capable of proper dosing and muscle targeting was a key barrier.<sup>99</sup> Conversely, the top driver of patient satisfaction with botulinum toxin treatment is relief from dystonia symptoms, with 63-78% reporting improvement. Patients expressed wanting more flexible, personalized injection schedules matched to their symptoms. Satisfaction is higher when given by specialized experts.<sup>99</sup>

## Meta-Analyses

N/A

## Systematic Reviews and Meta-Analyses

Mainka et al. (2019) performed a systematic review and meta-analysis to examine dystonic remission across 31 studies representing 2,551 patients, with over half having CD (n = 1,319).<sup>100</sup> Complete remission of symptoms was experienced by 15.4% of CD patients, while another 7.4% experienced partial improvement. Meta-analysis of 5 CD studies

showed that patients experiencing remission had significantly earlier disease onset by an average of 7 years compared to non-remitters. However, relapse was common, occurring in 58.7% of CD cases with remission.<sup>100</sup> These data suggest that while complete remission occurs in a sizable minority of CD patients, it is more likely in those with earlier onset disease.<sup>100</sup> The long-term durability of symptom remission has not been firmly established even among these select CD subgroups.<sup>100</sup>

A systematic review and meta-analysis conducted by Marsh et al. (2014) focused on determining the duration of clinical effect of onabotulinumtoxinA in treating CD.<sup>101</sup> The analysis included a total of 18 studies involving more than n = 1900 patients. The results revealed that the average duration of effect of onabotulinumtoxinA for CD patients was between 93.2 days (95% CI 91.8-94.6 days) in a fixed effects model and 95.2 days (95% CI 88.9-101.4 days) in a random effects model.<sup>101</sup> It was also found that the dosage of onabotulinumtoxinA significantly influenced the duration of effect. Specifically, doses of 180 units or more resulted in longer effect durations (between 107-109 days) compared to doses less than 180 units (86-88 days). The study also noted that country of origin had an influence on the duration of effect. The main conclusion from this analysis is that on average, patients with CD treated with onabotulinumtoxinA would require approximately 4 treatments per year, with higher doses associated with longer durations of effect.<sup>101</sup>

### Randomized Controlled Trials (RCTs)

Samotus et al. (2018) found that many CD patients discontinue BtA injections early due to insufficient relief.<sup>102</sup> Tailoring injections through kinematic analysis may improve outcomes.<sup>102</sup> In a 38-week study, 28 participants were split into groups receiving expert injector-determined doses by either visual assessment ("vb") or kinematic biomechanics ("kb"). Kinematics utilized sensors capturing severity of static/dynamic neck movements while seated. The "kb" group had a 28.8% TWSTRS score reduction at 6 weeks with sustained improvement. The "vb" group reduced scores by 28.5% at 22 weeks.<sup>102</sup> The "kb" group trended towards better outcomes. Kinematic guidance enabled faster optimal muscle selection and dose minimizing while achieving comparable or superior relief versus standard visual assessment.<sup>102</sup>

Wu et al. (2016) conducted an RCT to compare EMG-guided vs palpation-guided BTIs for CD patients (n = 68).<sup>103</sup> The primary endpoint was the Tsui score at 16 weeks. At 16 weeks, the EMG-guided group had a significantly lower (better) Tsui score than the palpation-guided group. Both groups had significant reductions in pain scores over time, with no between-group differences.<sup>103</sup> There were no significant between-group differences in Hospital Anxiety and Depression Scale scores or patient/clinician global impression scores, though clinician scores were higher than patient scores.<sup>103</sup> The EMG-guided group had significantly more injection site pain but less dysphagia than the palpation-guided group. In

conclusion, EMG-guided injections resulted in better dystonia control at 16 weeks, less dysphagia, and more injection discomfort compared to palpation-guided injections.<sup>103</sup>

A randomized, double-blind, crossover study by Yun et al. (2015) of CD (n = 94) patients that received injections of abobotulinumtoxinA and onabotulinumtoxinA using a 2.5:1 unit conversion ratio.<sup>104</sup> Efficacy was measured using the Tsui scale, with mean improvement from baseline of 4.0 points ( $\pm 3.9$  points) after 4 weeks for abobotulinumtoxinA compared to 4.8 points ( $\pm 4.1$  points) for onabotulinumtoxinA (95% CI -0.13 to 1.70 points;  $P = 0.091$ ). No statistically significant differences were detected between groups in magnitude of symptom improvement per Tsui scale changes. Secondary 10-point TWSTRS scores aligned with the primary outcome. These quantitative clinical efficacy metrics indicate that both formulations reduced CD severity to a similar degree following dose conversion, adding evidence for the non-inferiority of abobotulinumtoxinA to onabotulinumtoxinA in managing this condition when utilized at an optimal 2.5:1 dose ratio.<sup>104</sup>

## Chronic Migraine

### Society Guidelines

We found 6 clinical guidelines that suggest botulinum toxin as an effective treatment for chronic migraine. These recommendations were issued by the AAN, the American Headache Society (AHS), the AAO-HNS, European Headache Federation (EHF) and National Institute for Health and Care Excellence (NICE). Among the evaluated guidelines, 3 were founded on systematic review outcomes (AAN, EHF, NICE). The other 2 (AHS, AAO-HNS, are position statements drafted to reflect clinician consensus.

The AAN issued guidelines in 2016 for treating headaches and migraines, updating their 2008 guidelines. They recommend botulinum toxin as a treatment choice for patients with chronic migraines to help them have more days without headaches.<sup>92</sup>

In 2012, NICE recommended botulinum toxin type A as a preventative therapy option for chronic migraine patients who have failed at least 3 other prophylactic medications.<sup>7</sup>

The EHF issued guidelines in 2018 that states that onabotulinumtoxinA is recommended as an effective and well-tolerated treatment for chronic migraine.<sup>105</sup>

In 2021, the AAO-HNS stated that botulinum toxin can be injected in different regions for the prophylactic treatment of migraine headaches that have failed to improve with other medical management.<sup>94</sup>

The AHS specifically recommends onabotulinumtoxinA (Botox) as an effective treatment for chronic migraine, characterized by headaches on 15 or more days per month, by administering a 155-unit dose without gradual escalation, noted for its rapid efficacy, favorable tolerability, and suitability for patients unresponsive to other treatments, with

ongoing treatment based on patient response and regular assessment of benefits. They state the statement was based on a review of existing guidelines.<sup>42</sup>

Additionally, the American Academy of Pain Medicine (AAPM) 2022 systematic review (and practice guideline)<sup>106</sup> includes a strong recommendation for BtA (onabotulinumtoxinA, specifically) for chronic migraines (and a weak recommendation against BtA for episodic migraine).<sup>106</sup>

## Systematic Reviews

### AAN's Systematic Review

In the AAN's systematic review, a total of 28 RMTs were identified and analyzed to assess the efficacy of botulinum toxin formulations in the treatment of headaches, including chronic migraine and episodic migraine. The systematic review found that onabotulinumtoxinA (onaBoNT-A) is established as safe and effective for reducing the number of headache days in chronic migraine, based on 2 Class I studies. These studies highlighted the effectiveness of onaBoNT-A for reducing total headache days over 28 days, with 1 study showing a mean intergroup difference of 1.4 days. In contrast, the systematic review found that onaBoNT-A is probably ineffective for the treatment of EM, based on 3 Class I studies, including 2 from the 2008 report. One Class I study published after 2008 demonstrated that onaBoNT-A was ineffective in reducing migraine frequency over 180 days. As a result, onaBoNT-A is recommended for chronic migraine to increase headache-free days (Level A) and may be considered for improving health-related quality of life (Level B), but it should not be offered as a treatment for episodic migraine (Level A).

### EHF's Systematic Review

Researchers for EHF reviewed 27 studies, including RCTs like the PREEMPT studies which included almost 1,400 chronic migraine patients. In the pooled analysis of the PREEMPT trials, onabotulinumtoxinA resulted in a modest but statistically significant additional reduction of 2 headache days per month over placebo. Almost 50% of onabotulinumtoxinA patients were responders based on a 50% or greater headache frequency reduction, compared to 35% for placebo. Based on this and additional evidence of improved disability scores and quality of life measures versus placebo, the panel strongly recommends onabotulinumtoxinA for treatment of chronic migraine. It is considered effective and well-tolerated. They recommend patients try at least 2 to 3 other preventive medications first, with certain exceptions. Treatment response should be evaluated after the first one to three treatment cycles and discontinued if patients do not achieve at least a 30% headache frequency reduction.

### NICE's Systematic Review

NICE based its recommendation on a systematic review by the drug manufacturer Allergan, which conducted a systematic review to evaluate the clinical effectiveness of botulinum toxin type A for the prevention of headaches in adults with chronic migraine.<sup>107</sup> The review identified 7 relevant RCTs comparing botulinum toxin type A against placebo. Of these, 4 trials with active comparators were excluded. The remaining 3 placebo-controlled trials focused on people with chronic migraine defined as at least 15 headache days per month, of which at least 8 days met criteria for migraine headaches. The analysis centered on the results from 2 large, identically designed phase III trials (PREEMPT 1 and PREEMPT 2). These enrolled a total of 1,384 participants - 688 in the botulinum toxin type A arms and 696 in the placebo arms.<sup>107</sup> Most patients (approximately 64%) had received at least 1 prior migraine preventive medication, with 35% having failed at least 3 preventatives. The 24-week double-blind treatment phase showed statistically significant improvements with botulinum toxin type A over placebo across several headache symptom outcomes including headache days per month.<sup>107</sup>

### Other Systematic Reviews

A systematic review and meta-analysis by Corasaniti et al. (2023) included 9 randomized clinical trials with a total of  $n = 3,565$  patients to assess the safety of onabotulinumtoxin A compared to placebo or active comparators for treatment of chronic migraine.<sup>108</sup> The meta-analysis included  $n = 1,787$  patients treated with onabotulinumtoxin A and  $n = 1,778$  patients treated with comparators. The use of onabotulinumtoxin A produced more treatment-related adverse events (TRAEs) than placebo, but fewer than oral topiramate. The most common adverse events were neck pain, musculoskeletal pain, muscular weakness, migraine, eyelid ptosis, blurred vision and injection site pain. However, there was high heterogeneity between studies ( $I^2 = 96\%$ ,  $P < 0.00001$ ). The authors concluded that further adequately powered, randomized clinical trials are needed to assess safety of onabotulinumtoxin A, including in combination with newer treatments for chronic migraine.<sup>108</sup>

A systematic review and meta-analysis by Giri et al. (2023) included 3 RCTs ( $n = 1,139$ ) to assess the efficacy of BtA compared to placebo for prevention of chronic migraine with medication overuse headache.<sup>109</sup> The studies showed that treatment with BtA resulted in a reduction of 1.92 headache days per month on average compared to placebo (95% CI -2.68 to -1.16). However, the effect on achieving at least 50% response rate ( $\geq 50\%$  reduction in headache days) was uncertain, with substantial heterogeneity between studies ( $I^2 = 92\%$ ) and an imprecise odds ratio estimate of 1.56 (95% CI 0.42 to 5.76). The dropout rates ranged from 2.9% to 7.4%. No significant differences were found in adverse events between BoNTA and placebo groups. Due to concerns over risk of bias, inconsistency, and imprecision of results, the quality of evidence was very low for response rate and low for change in headache days.<sup>109</sup>

A systematic review by Shaterian et al. (2022) included 24 studies on the efficacy of onabotulinumtoxinA for treating migraine (BtA).<sup>110</sup> The studies included a total of  $n = 4,431$  patients, mostly with chronic migraine. Doses of Botox<sup>®</sup> ranged from 2.5 to 200 units, with most studies using 155 units. BtA was injected at different sites around the head, neck, and shoulders. The review found that BtA treatment decreased the frequency of migraine attacks per month, pain intensity, medication use, emergency visits, and migraine-related disabilities. Botox<sup>®</sup> also improved quality of life. The authors concluded that BtA is an effective and cost-effective option for treating migraine, including chronic, episodic, unilateral, and vestibular types.<sup>110</sup>

A systematic review and meta-analysis by Herd et al. (2019) included 28 RCTs with a total of  $n = 4,190$  participants to assess the efficacy of BtA compared to placebo for prevention of chronic migraine. The analysis found that treatment with BtA resulted in a reduction of 2.0 migraine days per month on average compared to placebo (95% CI -2.8 to -1.1,  $n = 1,384$ ) in patients with chronic migraine. This efficacy was obtained after removing RCTs that were at a high risk of bias due to small study sizes. BtA also showed greater improvements in migraine severity scores and headache days per month versus placebo, though the quality of evidence was low.<sup>111</sup> The risk of TRAEs was twice as high with BtA compared to placebo, but these events (the most common being blepharoptosis, muscle weakness, injection site pain and neck pain) were non-serious and transient.<sup>111</sup>

A systematic review and meta-analysis by Bruloy et al. (2019) included 17 RCTs with a total of  $n = 3,646$  patients to assess the efficacy of BtA compared to placebo for prevention of migraine.<sup>112</sup> The analysis found that treatment with BtA resulted in a significant reduction of 1.56 migraine days per month on average compared to placebo (95% CI -3.05 to -0.07;  $P = 0.04$ ) in patients with chronic migraine ( $n = 1,551$ ). There was also a tendency for reduction in episodic migraine frequency with BtA versus placebo, but this was not statistically significant. The risk of adverse events was higher with BtA compared to placebo, but adverse events were mild. Patient quality of life at 3 months was significantly improved in the botulinum toxin group ( $P < 0.00001$ ).<sup>112</sup>

A systematic review and meta-analysis by Herd et al. (2018) included 28 RCTs with a total of  $n = 4,190$  participants to assess the efficacy of BtA compared to placebo for prevention of chronic migraine.<sup>113</sup> The analysis found that treatment with BtA resulted in a reduction of 2 migraine days per month on average compared to placebo (95% CI -2.8 to -1.1) in patients with chronic migraine, based on data from 2 large trials ( $n = 1,384$ ). BtA also showed greater improvements in number of headache days per month and migraine severity versus placebo, though the quality of evidence was low. The risk of TRAEs was twice as high with BtA compared to placebo, but these events were non-serious and transient.<sup>113</sup>

## Meta-Analyses



A meta-analysis by Lanteri-Minet et al. (2022) included 44 observational studies on the real-world effectiveness of onabotulinumtoxinA for preventing chronic migraine. The meta-analysis included 7 studies assessing change from baseline in number of headache days per month at approximately 24 weeks ( $n = 1,525$ ) and 5 studies at approximately 52 weeks ( $n = 995$ ).<sup>114</sup> onabotulinumtoxinA treatment resulted in a reduction of 10.64 headache days per month (95% CI -12.31, -8.97) at 24 weeks and 10.32 headache days per month (95% CI -14.92, -5.73) at 52 weeks. The meta-analysis also showed reductions in other outcomes like days taking acute headache medications,  $\geq 50\%$  response rates, Headache Impact Test scores, and improvements in Migraine-Specific Quality of Life scores that were consistent with pivotal clinical trials.<sup>114</sup>

In another meta-analysis, conducted by Shen and Wang (2020), BtA was found to significantly reduce headache episodes per month relative to placebo for chronic migraine.<sup>115</sup> Moreover, the therapy improved the impact of chronic migraines after 16 weeks of therapy.

### Randomized Controlled Trials (RCTs)

An RCT by Bono et al. (2023) included 139 patients with chronic migraine who had previously not responded to intramuscular BtA injections. Patients were randomized 2:1 to receive subcutaneous injections of BtA ( $n = 90$ ) or placebo ( $n = 49$ ) targeted to the area of maximum headache pain origin (trigeminal or occipital regions).<sup>116</sup> At 6 months, BtA significantly reduced monthly headache days versus placebo (-13.2 days versus -1.2 days,  $P < 0.0001$ ). Secondary outcomes like migraine disability scores also improved with BtA. The results support using a follow-the-pain targeted subcutaneous injection paradigm with BtA in chronic migraine patients unresponsive to other botulinum toxin regimens.<sup>116</sup>

Overall, there are 12 RCTs comparing BtA to placebo,<sup>116-125</sup> including the 2 PREEMPT trials.<sup>126,127</sup> Additionally, there are 9 RCTs addressing BtA versus an active comparator including, BtA vs. topiramate<sup>128-130</sup>, BtA vs. fixed (muscle)-site and acupoint-site injections<sup>122</sup>, BtA vs. facial nerve blockade<sup>123</sup>, BtA vs. low-level laser therapy<sup>131</sup>, BtA vs. amitriptyline<sup>132</sup>, BtA vs. subcutaneous histamine<sup>133</sup>, and BtA with preserved saline vs. BtA with preserve free saline.<sup>134</sup>

### Focal Hand Dystonia

### Society Guidelines

N/A

### Systematic Reviews

N/A

## Meta-Analyses

N/A

## Randomized Controlled Trials (RCTs)

A Class I trial (strong evidence)<sup>135</sup> randomized 40 study participants with writer's cramp in a double-blind design for botulinum toxin or an equal amount of saline placebo. The primary outcome measured was the patient's indicated request to continue injection therapy. In patients randomized to BTI, 70% requested to maintain treatment in comparison to 32% of those who received a placebo ( $P = 0.03$ ). Patients injected with BTI also had considerable improvement in comparison to patients who had been given a placebo in secondary clinical outcome measures including a visual analog scale, symptoms severity scale, writer's cramp rating scale, and assessment of writing speed, but not in the functional status scale. The only adverse effects reported were temporary weakness and pain at the injection site.

Tsui et al. (1993)<sup>136</sup> conducted a Class II study (weak to moderate evidence) utilizing a placebo-controlled, double-blind, crossover design for 20 individuals with writer's cramp. Clinical assessment was utilized in selecting the muscle to be injected and the dose of BtA was founded on investigator experience. Outcome evaluations included assessment of writing speed, accuracy, writing samples, and patients' subjective report. There was substantial improvement with BTI in the objective measures, but not in patients' own evaluations. The only adverse effect was focal weakness, although this was severe enough to worsen pen control in 1 participant. This study only evaluated the first active therapy session for study participants; therefore, the therapeutic effects achieved were not optimal.

Cole et al. (1995)<sup>137</sup> conducted a Class II trial with a double-blind, placebo-controlled, crossover design with 10 study participants with focal hand dystonia. Muscles and BTI-A (Botox®) doses were selected and optimized during a time of open treatment before the trial. Outcome measures were based on study participant's subjective rankings and observation of videotapes taken during actions relevant to the individual dystonia. Eight participants had improved subjective ratings and 6 showed improvement via videotape with BTI in comparison with placebo. Weakness was noted in the injected muscles of 80% of study participants with active treatment.

Three Class II studies assessed technical issues of BTI administration. In 1 trial, Chen et al. (1999) used a blinded, randomized, crossover design to contrast continuous muscle activation with immobilization immediately following BTI.<sup>138</sup> Blinded assessment of handgrip strength and writing showed a substantial increase in focal weakness with continuous muscle activity, but no subjective or objective improvement in writing.



In a similar study, Geenen et al. (1996)<sup>139</sup> randomized the participants to 1 of 2 muscle localization method groups: EMG recording or electrical stimulation. Injections guided by both techniques were similarly effective in producing weakness in the target muscle.

In a third trial, Molloy et al. (2002) assessed the precision of muscle localization with and without EMG.<sup>140</sup> In needle placements established on surface anatomy, 37% were localized in the targeted muscle.

## Hemifacial Spasm/Facial Dystonia

### Society Guidelines

N/A

### Systematic Reviews

N/A

### Meta-Analyses

N/A

### Randomized Controlled Trials (RCTs)

Despite limited data from high-quality clinical trials, BtA is considered the treatment of choice for HFS patients.<sup>141,142</sup> Overall, 76% to 100% of patients have at least a 75% improvement with a typical duration of response lasting from 3 to 4 months.<sup>143,144</sup> Both primary and secondary HFS patients respond to BtA, and the average efficacy has been reported to be around 20 years, often with the need for a gradual increase in dose.<sup>95,145</sup>

From the literature, the efficacy of botulinum toxin ranged from 73% to 98.4%. The mean duration of the effect was around 12 weeks. There were 3 RCT studies, of which 2 were conducted to investigate the efficacy and safety of botulinum toxin in HFS.<sup>57</sup> The third study compared pretarsal versus preseptal injections of the orbicularis oculi in 31 patients with HFS. The investigators reported that the pretarsal portion of the orbicularis oculi was associated with a significantly high response rate in terms of latency to response, duration of improvement, JRS, self-response scale, and patient satisfaction scale than the preseptal injections.<sup>57</sup>

### Other Studies

A retrospective longitudinal comparative analysis suggested that the duration of relief from symptoms remains unchanged over the long term in patients with HFS.<sup>146</sup> In addition, Tunc

et al. (2008) assessed BTI efficacy in 69 patients with primary HFS (n = 46) and those with HFS due to definite neurovascular compression (n = 23), reported that primary HFS patients presented more improvement.<sup>147</sup>

## Upper Extremity Spasticity

### Society Guidelines

In 2016, the AAN issued guidelines that Botulinum toxin (BoNT) is recommended for the treatment of spasticity in adults in several instances.<sup>92</sup> For upper extremity spasticity, abobotulinumtoxin A (aboBoNT-A), incobotulinumtoxin A (incoBoNT-A), and onabotulinumtoxin A (onaBoNT-A) are established as safe and effective for reducing spasticity and improving passive function (Level A recommendation). Rimabotulinumtoxin B (rimaBoNT-B) is probably safe and effective for reducing upper limb spasticity (Level B recommendation).<sup>92</sup>

### Systematic Reviews

N/A

### Meta-Analyses

N/A

### Randomized Controlled Trials

Four new Class I (strong evidence) trials<sup>148-151</sup> investigating abotulinumtoxinA (aboBoNT-A) demonstrated significant reductions in upper limb tone as measured by the modified Ashworth scale. These studies also measured functional outcomes.

A recent study by Gracies et al. (2015)<sup>151</sup> showed improved response rate (>1 point) on the principal target of treatment of the Disability Assessment Scale (DAS), a measure of self-reported disability, at 4 weeks among participants treated with 1,000 U of aboBoNT-A, but not with 500 U or placebo (62% in 1,000 U,  $p = 0.0018$  vs placebo; 50% in 500 U,  $p = 0.1279$  vs placebo; and 39.2% in placebo). The higher-dose BoNT-A group also demonstrated improved active range of motion in the elbow, wrist, and fingers.

A Class I study conducted by Lam et al. (2012)<sup>152</sup> on patients with upper limb spasticity focused on caregiver burden. This study found that 67% of caregivers of patients receiving aboBoNT-A reported a  $\geq 4$ -point reduction on the caregiver burden scale as compared with 20% of caregivers of patients injected with saline ( $P = 0.001$ ).

Another study, by Rosales et al. (2012)<sup>150</sup>, demonstrated no significant change in functional assessment scores.

Shaw et al. (2011)<sup>149</sup> observed no significant difference between groups for improved active arm function as measured by the Action Research Arm Test at 1 month (risk difference [RD] favoring the BoNT group 5.7%, 95% CI -3.5% to 14.6%). However, participants treated with aboBoNT-A showed improvement in upper limb muscle function at 3 months as measured by the Motricity Index (mean change in index 3.5, 95% CI 0.1 to 6.8, greater number of points in the intervention group).

In another study, McCrory et al. (2009)<sup>148</sup> demonstrated no significant difference in quality of life but observed significantly greater global benefit in patients given BoNT.

## Lower Limb Spasticity

### Society Guidelines

A practice guideline update published by the AAN concluded that AboBoNT-A and onaBoNT-A are safe and effective for the reduction of adult lower limb spasticity (multiple Class I studies).<sup>92</sup>

### Systematic Reviews

N/A

### Meta-Analyses

N/A

### Randomized Controlled Trials (RCTs)

A placebo-controlled Class I study<sup>153</sup> published since the 2008 guideline examined aboBoNT-A use in MS and observed reduced pain in both legs in patients randomized to aboBoNT-A (RD proportion of patients reporting decreased pain at 12 weeks 29.9%, 95% CI 10.9%–46%).

Three Class I studies of onabotulinumtoxinA (onaBoNT-A) in the treatment of adult lower limb spasticity<sup>154-156</sup> demonstrated significant reduction in tone but found inconsistent results regarding functional measures.

## Hyperhidrosis

### Society Guidelines

N/A

### Systematic Reviews

N/A

## Meta-Analyses

N/A

## Randomized Controlled Trials (RCTs)

Only 1 RCT was found, in which 20 patients were randomized to receive toxin injections to 1 axilla and suction-curettage to the contralateral axilla. The primary outcome measure was reduction of sweat rate measured by gravimetry, and the secondary measure was quality of life as measured by a patient-directed questionnaire.<sup>157</sup> At 3 months posttreatment, BTIs decreased baseline resting sweat production by 72.1% versus 60.4% ( $P = .29$ ) for suction-curettage, and baseline exercise-induced sweat production by 73.8% versus 58.8% ( $P = 0.10$ ). When patients were stratified into the categories of light and heavy sweaters, there was a difference among heavy sweaters, with exercise-induced sweat production lower by 10.48 mg/min or 34.3% ( $P = 0.0025$ ) at toxin-treated sites. Compared with suction-curettage, toxin also resulted in greater improvements in quality of life by 0.80 points ( $P = 0.0002$ ) and 0.90 points ( $P = 0.0017$ ) at 3- and 6-months posttreatment, respectively, as measured by the patient questionnaire.<sup>157</sup>

By objective measures 3 months after treatment, neurotoxin injections are nominally more effective than suction-curettage in all cases, and markedly more effective than patients who sweat heavily. Patients have a very significant preference for neurotoxin injections at three months, and this is maintained at 6 months.<sup>157</sup>

At base line, the mean ( $\pm$  SD) rate of sweat production was  $192 \pm 136$  mg per minute. Two weeks after the first injections the mean rate of sweat production in the axilla that received botulinum toxin A was  $24 \pm 27$  mg per minute, as compared with  $144 \pm 113$  mg per minute in the axilla that received placebo ( $P < 0.001$ ). Injection of 100 U into the axilla that had been treated with placebo reduced the mean rate of sweat production in that axilla to  $32 \pm 39$  mg per minute ( $P < 0.001$ ). Twenty-four weeks after the injection of 100 U, the rates of sweat production (in the 136 patients in whom the rates were measured at that time) were still lower than baseline values, at  $67 \pm 66$  mg per minute in the axilla that received 200 U and  $65 \pm 64$  mg per minute in the axilla that received placebo and 100 U of the toxin. Treatment was well tolerated.<sup>158</sup>

## OAB (OAB)/Urinary Incontinence (UI)

## Society Guidelines

American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (AUA/SUFU) guideline<sup>69,159</sup> recommends intradetrusor BtA injection as third-line therapy in select patients who have received adequate counseling and

who have not responded to first- and second-line treatment.<sup>159</sup> They report caution must be used in the elderly, the very frail, and those with pre-existing cognitive deficiencies and further research is needed.

## Systematic Reviews

Truzzi et al. (2022)<sup>160</sup> performed a systematic review on the use of BTIs in the bladder to treat OAB in men to identify clinical trials on efficacy and safety of BtA injections in the detrusor for treatment of OAB. Six studies were conducted exclusively in men. There were only 2 were RCTs. Most were observational studies and case series. Quality-of-life symptom questionnaires and voiding diary parameters were some of the assessment methods. Only 2 studies separately reported urodynamics data for men before and after intravesical BtA injection. For 7 studies, the outcome data could be pooled, with the most common adverse events after BtA injection being UTI (29.8%), urinary retention (20.0%), increased postvoid residual (37.3%), de novo interstitial cystitis (28.3%), and hematuria (12.4%). The authors concluded that information regarding the efficacy and safety of BtA bladder injections for male OAB was limited and derived from relatively low-quality evidence. There is a need for further research to assess the risk-benefit profile of BtA in males. Limitations of the studies include variability in assessment of BtA injections among studies, self-reporting bias, and small sample size.

Abrar et al. (2021)<sup>161</sup> conducted a systematic review to evaluate whether poor response/adverse events to BtA for idiopathic OAB are predictable. This review included 17 cohort studies (primarily level 3 evidence). It was found that the potential predictors for non-responders were male gender, frailty, comorbidity, increasing age, smoking, baseline leakage episodes, and various urodynamic parameters including bladder outlet obstruction index (BOOI), high pretreatment maximum detrusor pressure, and poor bladder compliance. The main predictive factors for clean intermittent self-catheterization (CISC) use were male gender, comorbidity, increasing age, number of vaginal deliveries, hysterectomy, and urodynamic parameters (bladder capacity, postvoid residual volume, projected isovolumetric pressure value, bladder contractility index, and BOOI). Female gender, males with their prostates in situ, and CISC were linked to potential increase UTIs after BtA use. The authors concluded that even though various predictive factors based on the patient, medical conditions, previous surgery, and pretreatment investigations were identified, the quality of included studies was generally poor (poor study design being 1 of the reasons), limiting their conclusions.

Gong and associates (Gong, Xu et al. 2019) conducted a meta-analysis of published RCTs to assess the effect of BTX-A using different dosages for the treatment of UI symptoms. Studies that included patients with neurogenic detrusor overactivity (NLUTD) or idiopathic OAB (IOAB) were included and result in 19 studies which were included for analysis. Authors concluded their analysis suggested BTX-A 200 U and 300 U were more effective as

compared to placebo in the treatment of NLUTD. While BTX-A 300 U and 200 U may improve detrusor compliance of IOAB. Authors further state the need for more robust studies to help determine the effect of BTX-A in NLUTD and OAB management.

Gu et al. (2017)<sup>162</sup> conducted a systematic review to assess the effectiveness and safety of BtA at different dosages for OAB. A total of 11 studies were included. Parameters considered were UI episodes per week as the primary outcomes, urodynamic parameters including maximum cystometric capacity (MCC), and maximum detrusor pressure (MDP) for neurogenic detrusor overactivity (NLUTD) at 6 weeks, and for idiopathic detrusor overactivity (IDO) at 36 weeks as compared to placebo. The authors reported a dose of BtA greater than 50 U is significantly more effective for certain symptoms of OAB compared with placebo, with escalating adverse events with increase dosage. The authors recommend BtA 200 U for management of NLUTD for short-term treatment and stated there was no significant difference from the larger dose of 300 U.

Drake et al. (2017)<sup>163</sup> performed a systematic review/meta-analysis of 56 RCTs to compare the efficacy of BtA, mirabegron, and anticholinergics in adults with idiopathic OAB. They reported 100 U of BtA achieved the greatest reduction in urinary incontinence episodes (UIE), urgency episodes, and micturition frequency. The authors conclude that BtA 100 U provides greater relief of OAB symptoms as compared with other licensed doses of other pharmacotherapies in the network at 12 weeks follow up.

Henriet et al. (2015)<sup>164</sup> conducted a systematic review for the treatment of refractory idiopathic OAB with botulinum toxin. Of the 37 studies included, 8 were placebo controlled RCTs. Intra-detrusor injections were associated with reduced frequency, urgency, nighttime urination, incontinence episodes and improved bladder capacity and quality of life. When considering efficiency and tolerance, they suggest a dose of 100 to 150 U of BtA is optimal as a second-line treatment for refractory non-NLUTD and the effects persist between 3 to 12 months. They state long-term effectiveness data is lacking.

Cui et al. (2013)<sup>165</sup> systematically reviewed the evidence related to the efficacy and safety of BtA in the management of idiopathic OAB. Twelve RCT (n=1020) comparing BtA with placebo, different doses, and injection techniques for management of idiopathic OAB was included. The author concluded that BtA resulted in reduction in urinary frequency, less incontinence episodes, improvement in quality of life, higher post-void residual-related catheterization and higher UTI. They recommended low-dose BtA in effort to balance the benefits with adverse events.

## Meta-Analyses

Gong et al. (2019)<sup>166</sup> conducted a meta-analysis of published RCTs to assess the effect of BtA using different dosages for the treatment of UI symptoms. Studies that included

patients with neurogenic detrusor overactivity (NLUTD) or IOAB were included and result in a total of 19 studies were included for analysis. Authors concluded their analysis suggested BtA 200 U and 300 U were more effective as compared to placebo in the treatment of NLUTD. While BtA 300 U and 200 U may improve detrusor compliance of IOAB. Authors further state the need for more robust studies to help determine the effect of BtA in NLUTD and IOAB management.

### Randomized Controlled Trials (RCTs)

Visco et al. (2012)<sup>167</sup> performed a double-blind, double-placebo-controlled RCT to compare anticholinergic medications and BtA in the treatment of urgency UI. A total of 249 women with idiopathic urgency UI who had 5 or more episodes of incontinence per a 3-day period were assigned to receive anticholinergic plus 1 intradetrusor injection of saline or placebo plus 1 intradetrusor injection of 100 U of BtA and followed for 6 months. Starting was a baseline of 5 episodes of incontinence/day the anticholinergic group had a mean reduction of 3.5 and BtA group reduced to 3.3 ( $P = 0.81$ ). Complete resolution of urgency UI was reported by 13% in the anticholinergic group as compared to 27% in the BtA group ( $P = 0.003$ ). The authors found both treatments effective and different by side effect profiles.

Sherif et al. (2017)<sup>168</sup> conducted a RCT to assess the safety and efficacy of posterior tibial nerve stimulation (PTNS) versus an intradetrusor injection of BtA 100 U in the management of 60 patients with refractory IOAB. Initially both groups showed improvements however the effects diminished for the PTNS group by 9 months. They concluded for refractory IOAB, BtA is more effective than PTNS but had more adverse effects.

Nitti et al. (2013)<sup>169</sup> conducted a phase III, placebo-controlled trial of BtA in 557 patients with OAB and UI inadequately managed with anticholinergics. Patients were randomized to receive either intradetrusor injection of BtA 100 U or placebo. They reported the BtA group experienced a greater decreased frequency of UIE than placebo ( $-2.65$  vs  $-0.87$ ,  $p = 0.001$ ) and 22.9% vs. 6.5% of patients became completely continent. Improvement in treatment benefit scale, all other OAB symptoms, and patient health-related quality of life across multiple measures also improved in the BtA group. Uncomplicated UTI was the most common adverse event and there was a 5.4% rate of urinary retention. They concluded that BtA yielded significant and clinically relevant improvements in all OAB symptoms.

Yokoyama et al. (2020)<sup>170</sup> conducted a phase III RCT to evaluate the efficacy and safety of BtA 100 U in patients with OAB and UI who failed medical management. Enrolled patients were randomized 1:1 to receive a single dose of 100 U of BtA ( $n=124$ ) or placebo ( $n=124$ ) into detrusor muscle. The primary endpoint was the change in the number of daily UIEs at week 12 from baseline and the treatment group yielded a significant decrease in mean number of daily UI occurrences as compared to placebo group from baseline ( $2.16$ ;  $P < 0.001$ ) and improvements were seen in all secondary endpoints. Mild to moderate adverse



events were more common among the treatment group which included UTI, dysuria, urinary retention, and increased volume post-void residual urine. The authors concluded that there were statistically significant improvements in symptoms, suggesting BtA as an option for treatment in patient populations where medications have failed.

Multiple prospective studies investigated the role of repeat injections for management of OAB if initial response was positive. These investigations found the effects of BTI to last 4-10 months with a median of 7.6 months. Improvements were reported in quality of life and OAB symptoms. (Nitti, Ginsberg et al. 2016). An RCT concluded that in patients with dysfunctional voiding and detrusor underactivity repeat injections yielded larger therapeutic effects whereas in non-neurogenic voiding dysfunction patients, success rates of BTX-A was not superior to placebo. (Jiang, Wang et al. 2016)

## **Detrusor Overactivity**

### **Society Guidelines**

N/A

### **Systematic Reviews**

N/A

### **Meta-Analyses**

N/A

### **Randomized Controlled Trials (RCTs)**

In a prospective, long-term (3 years), multicenter, open-label extension study following a 52-week, phase III trial of onabotulinumtoxinA, patients were treated on an 'as needed' basis with intradetrusor onabotulinumtoxinA (200 U or 300 U) for UI due to neurogenic detrusor overactivity. Ninety-four patients received treatment  $\geq 12$  weeks since the previous treatment and a UIE threshold. The primary efficacy endpoint was the change from study baseline in UIEs/day at week 6 after each treatment. Additional efficacy measurements included: percent change in UIEs, the proportions of patients with  $\geq 50\%$  and  $100\%$  reductions from baseline in UIEs/day, changes from baseline in volume/void and Incontinence Quality of Life (I-QOL) total summary scores, IQOL responder rates (proportion of patients achieving a  $\geq 11$ -point increase from baseline in I-QOL total score, which is defined as the minimally important difference for I-QOL in NDO), and duration of treatment effect (time to patient request for retreatment). onabotulinumtoxinA 200U consistently reduced UI episodes/day; reductions from baseline ranged from  $-3.2$  to  $-4.1$  across six treatments. Volume/void consistently increased, nearly doubling after treatment. I-QOL improvements were consistently greater than twice the minimally important difference ( $+11$

points). Overall median duration of effect was 9.0 months (200U). Results were similar for onabotulinumtoxinA 300U. Most common AEs were urinary tract infections and urinary retention. De novo CIC rates were 29.5, 3.4, and 6.0% (200U), and 43.0, 15.0, and 4.8% (300U) for treatments 1–3, respectively; de novo CIC rates were 0% for treatments 4–6. The authors concluded that onabotulinumtoxinA treatments consistently improve UI, volume/void, and quality of life in patients with UI due to neurogenic detrusor overactivity in this 4-year study, with no new safety signals.

The use of intradetrusor injection of botulinum toxin for treatment of IC/BPS may be combined with hydrodistention.<sup>171-180</sup> While botulinum toxin may alleviate symptoms of IC/BPS, there is a risk of urinary retention, which may be particularly problematic for a patient with a painful bladder. Any patient considering this treatment must be willing and able to perform intermittent self-catheterization. Use of botulinum toxin for this indication is not approved by the FDA, and payment may not be covered by health insurance.

In a trial of 67 patients randomized to receive suburothelial injection of BtA (100 or 200 Units) combined with hydrodistention, or hydrodistention alone, BtA treatment resulted in a higher proportion of patients with moderate or marked improvement in symptoms at 3-month follow-up (71% versus 48%), and this difference was maintained through 24 months.<sup>124</sup> Initially, 200 Units of botulinum toxin was used, but adverse reactions occurred in 9 of 15 patients (4 patients had acute or chronic urinary retention, 7 had severe dysuria) and the dose was decreased to 100 Units. With the 100 Units dose, the number of adverse effects decreased but was still more frequent than for hydrodistention alone.

The 100 Units dose was further evaluated in another trial, in which 60 patients were randomized to receive suburothelial injections of BtA 100 Units (n = 40) or normal saline (n = 20). At week 8, a greater reduction of bladder pain was observed in the BtA treatment group, with treatment success rates of 63% in the BtA groups and 15% in the normal saline group.<sup>171</sup>

## Neurogenic Lower Urinary Tract Dysfunction (NLUTD)

### Society Guidelines

The AUA/SUFU guideline amendment<sup>66</sup> strongly recommends (evidence level – grade A) clinicians to use BtA to improve bladder storage parameters, decrease episodes of incontinence, and improve quality of life measures in NLUTD patients with SCI or MS refractory to oral medications. The guideline further makes a conditional recommendation (evidence level – grade C) that clinicians may offer BtA to improve bladder storage parameters, decrease episodes of incontinence, and improve quality of life measures in NLUTD patients, other than those with SCI and MS, who are refractory to oral medications. Finally, the guideline highlights the clinical principle that clinicians must discuss the specific

risks of urinary retention and the potential need for intermittent catheterization prior to selecting BtA for NLUTD patients who spontaneously void.

Additionally, discussions during the Multi-Jurisdiction Contract Advisory Committee (CAC) meeting held on 10/19/23 led to the conclusion that while there is a paucity of high-quality evidence to support the use of BtA as a first-line treatment for neurogenic for overactive bladder (OAB), the use of BtA is supported in patients with severe issues.

## Systematic Reviews

Soljanik et al. (2013)<sup>181</sup> conducted a systematic review on the efficacy and safety of BtA in adults with NLUTD dysfunction. Superior effects were reported in 28 studies of BtA compared with placebo in achieving continence as well as a reduction in incontinence episodes, improvements in urodynamic parameters and quality of life. Commonly reported adverse events included intermittent catheterization, urinary retention, and asymptomatic urinary infection. Review limitations include the inclusion of 22 studies with low levels of evidence, high heterogeneity and short follow up periods. The authors concluded that BtA is safe and effective for NLUTD dysfunction patients. They further call for the need of high-quality studies to define standards such as optimal dose, injection technique, long-term safety, and timing indications for re-injections.

## Systematic Reviews and Meta-Analyses

Cheng et al. (2016)<sup>182</sup> conducted a systematic review and meta-analysis to evaluate the efficacy and safety of BtA for NLUTD. A total of six RCTs comprised of 1,915 patients were included. A significantly decreased mean number of UIEs was observed at 6 weeks in the BtA-treated groups compared to placebo. An increase in adverse events was also reported in the BtA-treated groups (the adverse events included UTIs, urinary retention, hematuria, and muscle weakness). The authors concluded that their meta-analysis suggests BtA is a safe and effective treatment option for patients with NLUTD.

Ni et al.<sup>183</sup> conducted a systematic review and meta-analysis to investigate the value of repeat botulinum toxin A (BtA) injections in patients with NDO. A total of 18 studies (level of evidence ranging from 3-4) comprised of 1533 patients were considered. Stable quality of life improvements were reported in patients that received  $\leq 4$  injections following the first and last injections. However, a significant decrease in quality of life was reported in last injection of patients that received  $\geq 5$  injections. Repeat BtA injections intervals yielded no significant differences.

## Randomized Controlled Trials (RCTs)

Fu et al. (2015)<sup>184</sup> conducted an RCT to evaluate the efficacy of BtA injection in 60 patients with SCI. Mean UI frequencies were reduced from 15.2 $\pm$ 3.2 episodes/day to 2.9  $\pm$  1.2 in the

200 U group and  $16.2 \pm 2.9$  episodes/day to  $2.5 \pm 1.4$ ) in the 300 U groups at 4 weeks. There were no significant differences in continence rates between the dose group (63% [19/30] vs 70% [21/30], respectively;  $P > 0.05$ ). No adverse effects were reported in either group during the follow-up period. The authors concluded that BtA injections may serve as a safe and effective treatment in SCI patients.

Cruz et al. (2011)<sup>185</sup> conducted a multicenter RCT to evaluate the effects of BtA on UI, urodynamic variables, and quality of life in incontinent patients with neurogenic detrusor overactivity. Enrollment consisted of patients with MS ( $n = 154$ ) or SCI ( $n = 121$ ) with UI ( $\geq 14$  UIEs per week). Patients were randomized to receive intradetrusor injections of BtA 200 U ( $n = 92$ ), 300 U ( $n = 91$ ), or placebo ( $n = 92$ ). At 6 weeks a significantly reduced incontinence episodes were observed in the BtA 200 U and 300 U groups ( $-21.8$  and  $-19.4$ , respectively) compared with placebo ( $-13.2$ ;  $P < 0.01$ ). Authors concluded both doses of BtA significantly reduced UI while improving urodynamics and quality of life in both MS and SCI patients with NLUTD.

## Interstitial Cystitis (IC) / Bladder Pain Syndrome (BPS)

### Society Guidelines

The American Urology Association (AUA) produced a systematic review and guideline for the diagnosis and treatment of IC/BPS in 2022.<sup>71,72</sup> They give an 'optional' recommendation (evidence level – grade C) that patients, who are willing to accept the possibility that post-treatment intermittent self-catheterization may be required, may be subjected to a trial of neuromodulation if other treatments are not able to adequately improve their symptoms and quality of life.

European Association of Urology Guidelines on Chronic Pelvic Pain<sup>186</sup> state that BtA may result in urodynamic improvements.

During a Multi-Jurisdictional CAC meeting held on 10/19/23, subject matter experts (SMEs) reported that BtA for IC/BPS is included in algorithms for treatment and particularly at the time of operative intervention to reduce urgency and frequency symptoms.

### Systematic Reviews

Vargas et al. (2018)<sup>187</sup> conducted a systematic review to assess the efficacy and safety of BtA, compared with other interventions for the treatment of BPS to improve quality of life. A total of 4 studies were included for analysis. The authors concluded that there was a lack of evidence because of which they were unable to determine the efficacy of BtA for the treatment of IC to improve quality of life.

Wang et al. (2016)<sup>188</sup> conducted a systematic review including 7 RCTs and 1 retrospective study to evaluate efficacy and safety of BtA injections for IC/PBS. Results showed a

significant improvement in daytime frequency of urination (WMD  $-2.36$ ; 95% CI  $-4.23$  to  $-0.49$ ;  $p=0.01$ ) and maximum cystometric capacity (MCC) (WMD  $50.49$  mL; 95% CI  $25.27$  to  $75.71$ ;  $p<0.00001$ ). This study was limited by quality of included studies which included small sample sizes, lack of randomization methods reported, lack of standardization in BtA dosage and short duration of follow up.

Tirumuru et al. (2010)<sup>189</sup> conducted a systematic review to assess the effectiveness and adverse effects of intravesical BtA in IC. RCTs and prospective cohort studies comprised of 260 adult subjects with a clinical diagnosis of IC/BPS were included. Variations in urodynamic parameters across studies were noted. Adverse events included dysuria ( $n=35$ ), temporary hematuria ( $n=4$ ), UTI ( $n=4$ ), voiding difficulty resulting in clean intermittent self-catheterization ( $n=19$ ), impaired detrusor contractility ( $n=18$ ) and decrease in force of urinary stream ( $n=18$ ). The authors concluded that BtA may be beneficial to patients with refractory BPS but there is a need for further research.

## Systematic Reviews and Meta-Analyses

Shim et al. (2016)<sup>190</sup> conducted a systematic review and meta-analysis to evaluate the efficacy and safety of BtA injections as compared to placebo in patients with IC/BPS. Five studies comprised of 252 subjects with 8-12week follow-up duration. Dosage administered ranged from 50-200 U of BtA. Pooled mean change of visual analog score for the treatment group versus control group was  $-0.49$  (95 % CI  $-0.74$ ,  $-0.23$ ). Improvements were also reported in ICPI, frequency, and the occurrence of dysuria in the treatment group. Authors conclude their study resulted in significant differences in pain control and efficacy of treatment as compared to placebo.

Imamura et al. (2020)<sup>191</sup> conducted a Cochrane systematic review and meta-analysis to evaluate the impact of treatments for BPS patients. Articles considered for inclusion were 13 RCTs which evaluated any treatments for adults with BPS. Neuromuscular blockade was achieved by injection of BtA. The authors concluded that while neuromuscular blockade may be helpful to patients with BPS, the lack of evidence makes it difficult to determine if this treatment is superior to other modalities with very low level of certainty.

## Randomized Controlled Trials (RCTs)

Li et al. (2020)<sup>192</sup> conducted a randomized study to compare the therapeutic effect of intravesical instillation hyaluronic acid with intradetrusor botulinum toxin A (BtA) injection or cystoscopic hydrodistention (HD) for ketamine-associated cystitis in 36 patients. A significantly more favorable outcome was seen in the BtA group when considering IC Symptom Index, maximal capacity, and maximal cystometric capacity as compared to the HD group. The authors concluded that BtA injection may provide long-term effectiveness

compared with cystoscopic HD. However, the study was limited by a small sample size, and unclear effect of the addition of hyaluronic acid.

Kuo et al. (2016)<sup>172</sup> conducted a prospective, multicenter, double-blind, placebo controlled RCT to assess the benefit of intravesical BtA for treatment of IC/BPS. The treatment group (n=40) received hydrodistention plus suburothelial injections of BtA 100 while the control group (n=20) received an equivalent amount of normal saline. The primary endpoint was considered successful if a decrease of pain was evaluated at 8 weeks following treatment which was assessed by utilization of the visual analog scale (VAS). A significant decrease in pain at the 8-week follow up was reported in the Botox<sup>®</sup> group as compared to the saline group  $-2.6 \pm 2.8$  vs.  $-0.9 \pm 2.2$ ,  $P = 0.021$ ) with an overall rate of 63% (26/40) in the Botox<sup>®</sup> group and 15% (3/20) in the saline group ( $P = 0.028$ ). Authors concluded that Botox<sup>®</sup> successfully reduced bladder pain in IC/BPS patients. This study was limited by small sample size, short-term follow-up, baseline imbalance and combining procedures.

Kuo and Chancellor (2009)<sup>171</sup> conducted a prospective, randomized study comprised of 67 patients with IC/PBS who failed conventional treatments. Patients were randomized to either receive a suburothelial injection with 200 U (15) or 100 U (29) of BtA followed by cystoscopic HD 2 weeks later (BtA groups) (n=44) or to receive HD with no BtA injection (n=23). A significant decrease in IC/PBS symptoms were seen in all 3 patient outcomes with greater improvement in bladder capacity, reduced VAS score in the BtA group, however they also had more adverse effects with gross hematuria (n=2), urination difficulties (n=12), urinary retention (n=12) and UTI (n=3) only in the BtA groups. The authors concluded that injections of BoNT-A and HD yielded significantly favorable results as compared to HD alone.

## Other Studies

Kuo (2013)<sup>180</sup> conducted a prospective study comprised of 81 patients with IC/BPS who received intravesical BtA 100 U injections every 6 months for up to 4 times or until the symptoms improved. Active controls were comprised of patients who received a single injection (n=20). Other patients received either 2 (n=19), 3 (n=12), or 4 injections (n=30). Significant improvements following repeat injections were reported in symptom scores, VAS, functional bladder capacity, and daytime frequency with markedly improved rates in patients that received 4 repeated injections ( $P = 0.0242$ ) and 3 injections ( $P = 0.050$ ), as compared to single injection patients. The authors concluded that repeat injections were safe and effective in decreasing pain and increasing bladder capacity as compared to single injections.

## Laryngeal Dystonia Spasmodic Dysphonia (SD)

## Society Guidelines



The AAN published a guideline on botulinum toxin in 2008 at which time a “level B” recommendation was made for botulinum toxin use in adductor LD (insufficient evidence was determined for abductor LD).<sup>92</sup> This guideline was retired following the publication of an updated version in 2016. This latest version did not include LD.<sup>92</sup>

In 2013, an international consortium of neurologists based in the U.S. and Europe published an evidence-based assessment of botulinum toxin for movement disorders that included a “level C” recommendation specifically for BtA as a treatment option for ADSD only. The evidence for ABSD and other botulinum toxin products was found to be insufficient or inexistent for some products.<sup>95</sup>

The 2018 AAO-HNS guideline for the treatment of dysphonia<sup>63</sup> is the only active medical society practice guideline identified. The working group recommended the use of BTIs in patients with LD without restricting the recommendation to any LD subgroups or botulinum toxin products. Specifically, it stated that clinicians should offer, or refer to someone who can offer, BTIs for the treatment of dysphonia caused by spasmodic dysphonia and other types of LD. These injections offer the benefits of improved voice quality and voice related quality of life. However, there are inadequate peer-reviewed published studies to support the effectiveness for botulinum neurotoxin in ABSD.

## Systematic Reviews

The 2006 Cochrane systematic review on botulinum toxin for treating LD restricted eligibility to RCTs.<sup>193</sup> Only 1 RCT was found.<sup>194</sup> Its quality was downgraded 1 level due to unclear concealment. The authors concluded, however, that the trial demonstrated benefits in objective and subjective outcomes, but that there is insufficient data to generalize. The other 2 systematic reviews mostly included prospective single-arm NRSIs. The primary studies consistently reported clinically and statistically significant post-injection improvements in objective and subjective voice quality scores, as well as in quality-of-life measures.

Faham et al. (2019)<sup>195</sup> included 17 nonrandomized studies of interventions (NRSIs) (16 prospective) (n=474 patients) in their systematic review that aimed to evaluate the efficacy of bilateral (most common) or unilateral injections in the thyroarytenoid muscles. The quality-of-life measures included in this work were voice handicap index (VHI) and voice-related quality of life (V-RQOL). The minimum post-injection assessment interval was 1 week. Meta-analysis revealed statistically and clinically significant post-injection improvements in VHI and V-RQOL (VHI SMD, -0.36 [-0.58, -0.14]; V-RQOL SMD, [-3.27, -1.32]).

van Esch et al. (2017)<sup>196</sup> included 2 RCTs and 11 NRSIs in their work that assessed bilateral or unilateral injections of Botox<sup>®</sup>, 1 to 7.5 U per thyroarytenoid muscle under EMG guidance. A total of 927 patients with adductor LD were included. Objective (5 studies) and subjective (13 studies) voice quality scores as well as quality of life measures (VHI, V-RQOL) were



evaluated. The time points ranged from 4 days to 5 years. Consistent and significant improvements were observed in the objective outcome, subjective outcome, and quality of life measurements across studies. The duration of effect, reported by 6 studies, ranged between 14.7 and 18.0 weeks.

Watts et al. (2006)<sup>193</sup> conducted a Cochrane systematic review that included only 1 RCT evaluating the effect of Botox® injections in thyroarytenoid muscle. The RCT included 13 patients with adductor LD. The outcomes measured were fundamental frequencies, phonation time, spectrographic ratios, and perceptual voice quality measures. This single RCT demonstrated a benefit regarding subjective and objective measures of voice (improved fundamental frequency range, perturbation, spectrographic analysis in and perceptual voice quality) in active group. However, due to the small number of available RCTs, generalizations regarding the degree of effectiveness must be withheld at this point.

### **Randomized Controlled Trials (RCTs)**

Only RCTs comparing botulinum toxin to another active control or to placebo were eligible for this review. Two were found. The first one, which is reported above under the Watts 2006 Cochrane SR, was a double blind cross-over, placebo-controlled trial conducted at the University of California Medical Center in 1991.<sup>194</sup> The second RCT, also a double-blind placebo controlled study, was recently conducted at the Kyoto University in Japan.<sup>197</sup> In the former, the primary endpoint was the change in the number of aberrant morae (phonemes) at 4 weeks after drug injection. The number of aberrant morae at 4 weeks was reduced by  $7.0 \pm 2.30$  in the botulinum toxin adductor group compared to  $0.2 \pm 0.46$  in the adductor placebo group. The improvement persisted for 12 weeks following injections. Adverse events included breathy hoarseness (77.3%) and aspiration when drinking (40.9%) but were mild and resolved in 4 weeks. As 2 patients only presented with abductor LD, no conclusion could be drawn for that group.

### **Neurogenic Bladder**

#### **Society Guidelines**

N/A

#### **Systematic Reviews**

Soljanik et al. (2013)<sup>181</sup> conducted a systematic review to compare and critically discuss the reported efficacy and safety of BtA in adults with neurogenic bladder dysfunction. Studies published between January 1985 and July 2012 were identified. Analysis of the 28 included studies showed that onabotulinumtoxinA had superior efficacy, compared with placebo, in achieving continence, reducing incontinence episodes, improving urodynamic parameters and health-related quality of life. The most frequently reported adverse events were de novo

intermittent catheterization, urinary retention and asymptomatic urinary infection.

Limitations of this review are the inclusion of studies with level 3 evidence (22/28 studies), the heterogeneity of outcome parameters and time points chosen for follow-up reported in the reviewed studies. The authors concluded that onabotulinumtoxinA therapy is effective, safe and well tolerated in adults with neurogenic bladder dysfunction. However, further high-quality prospective trials are necessary to determine important parameters such as optimal dose, injection technique, favorable timing, indications for re-injections, long-term safety, and the impact of concomitant antimuscarinics on onabotulinumtoxinA therapy.

## Meta-Analyses

N/A

## Randomized Controlled Trials (RCTs)

N/A

## Strabismus

### Society Guidelines

N/A

## Systematic Reviews

Binenbaum et al. (2021)<sup>198</sup> conducted a systematic review that included 2 RCTs, 3 non-randomized comparative studies and 9 case series. It was found that successful motor outcomes were consistent across 4/5 of the comparative studies. In the 5<sup>th</sup> level 2 study, success was significantly higher with BtA compared to surgery. The level III case series demonstrated higher motor success rates when children were treated in 2 muscles at a time. In comparison, successful alignment was lower in adults treated with a single muscle BtA injection. BtA injections into the extraocular muscle showed a high rate of successful motor alignment compared to that achieved after eye muscle surgery for nonparalytic, nonrestrictive horizontal strabismus. In this systematic review, the limitations included that the studies were restricted to level II and level III studies.

Bort et al. (2023)<sup>199</sup> conducted a systematic review that included a total of 242 patients in 4 RCTs. This systematic review found that patients treated with botulinum toxin may need to re-treat with surgery more frequently. Low certainty evidence suggests that surgery may be more likely to improve or correct strabismus versus BtA. The limitations of this systematic review included that there were not enough studies to be confident of the results and the studies were small.

A systematic review conducted by Escuder et al. (2019)<sup>200</sup> revealed that BtA can have positive effects on ocular alignment, promote binocularity, and reduce diplopia. The successful treatment of strabismus with BtA is increasing, with the concomitant reduction in length of anesthesia and increase in muscle preservation. Further studies should be conducted so that we can learn more about the long-term outcomes of BtA in strabismus.

### Meta-Analyses

N/A

### Randomized Controlled Trials (RCTs)

N/A

### Other Studies

A pilot study was conducted that compared the effectiveness of a BTI in combination with surgery compared to surgery alone for the correction of large angle horizontal deviations. The sample size for this study was small with 23 patients, 12 in the Botox<sup>®</sup> group and 11 in the control group. It was found that there was significant increase in the percent net change in deviations between 7 and 30 days but this did not last longer than 3 months.<sup>201</sup> Limitations for this study included a small sample size of patients. In another study, it was found that there was a trend toward fewer injections and that treatment of strabismus with BtA is practical for patients with poor binocular potential, complicated strabismus, or multiple previous strabismus operations.<sup>202</sup>

### Sialorrhea

#### Society Guidelines

N/A

### Systematic Reviews

Most systematic reviews targeting sialorrhea in Parkinson's disease concluded that botulinum toxin is an effective therapeutic option with a certainty of evidence generally rated as moderate.

The authors of a 2022 Cochrane review on treatments for sialorrhea in amyotrophic lateral sclerosis (ALS) concluded that there is some low to moderate certainty evidence for the use of the Myobloc in that indication.<sup>203</sup>

In a recent well-conducted systematic review with meta-analyses, it was found that either Myobloc<sup>®</sup> or Xeomin<sup>®</sup> were likely to alleviate drooling frequency and severity in adults with neurological diseases at 4- and 12-week follow-ups without significantly increasing

dysphagia. The review included 17 RCTs involving a total of 981 patients. Results were statistically and clinically significant with a certainty of evidence rated as moderate [Drooling Frequency and Severity Scale (DFSS), Xeomin<sup>®</sup>: MD = -1.20 [-1.89 to -0.51], Myobloc<sup>®</sup>: MD = -1.62 (-2.07 to -1.17)]. Xeomin<sup>®</sup>, but not Myobloc<sup>®</sup>, remained effective after 12 weeks.<sup>204</sup>

## Meta-Analyses

N/A

## Randomized Controlled Trials (RCTs)

Note: BtA: incobotulinumtoxinA (Xeomin<sup>®</sup>); BtB: rimabotulinumtoxinB (Myobloc<sup>®</sup>/RIMA)

Three RCTs specifically targeting Parkinson's disease.<sup>205-207</sup> These included trials testing BtB 1,500 U vs. BtB 2,500 U vs. BtB 3,500 U vs. placebo<sup>205</sup>, BtB 4000 vs placebo<sup>206</sup>, and BtB 100 U vs placebo<sup>207</sup>.

Two RCTs in ALS, which includes trials testing BoNT-A vs. radiotherapy<sup>208</sup>, and BtB vs. placebo<sup>209</sup>.

One trial in neuroleptic-induced sialorrhea testing BtB vs. placebo.<sup>210</sup>

Four trials targeting varied etiologies, including Parkinson's disease, atypical Parkinson's disease, ALS, stroke, or traumatic brain injury (TBI). These include trials testing BtA vs. BtB<sup>211</sup>, BtA 50 U vs. BtA 100 U vs. BtA 200 U<sup>212</sup>, BtA 75 U vs. BtA 100 U<sup>213,214</sup> ("SIAXI" trial), and BtB 2500 U vs. BtB 3500 U vs placebo<sup>215</sup> (RIMA).

The 2 RCTs cited below were the largest and most consequential trials as they supported FDA approvals of Xeomin<sup>®</sup> (incobotulinumtoxinA) and Myobloc<sup>®</sup> (rimabotulinumtoxinB) for sialorrhea. Unstimulated salivary flow rate (uSFR) from baseline to 4 weeks was the primary endpoint in both.

The SIAXI trial was a European multicenter double-blind placebo-controlled trial testing Xeomin<sup>®</sup> 75 U vs. 100 U vs. placebo in 184 adults with Parkinson's disease, stroke, atypical Parkinson's disease, or TBI.<sup>213</sup> Mean age was 65.2 years. Compared to placebo, both treatment groups showed reduced uSFR, but only the 100-unit treatment group showed a statistically significant reduction (-0.09 g/min least squares mean difference versus placebo,  $P = 0.004$ ).<sup>213</sup> The reduction in uSFR was maintained in both groups in the 64-week follow-up study, which included 173 patients.<sup>214</sup> Treatment-related adverse events included dry mouth and dysphagia.

The second large RCT (RIMA) was an international double-blind trial that evaluated Myobloc<sup>®</sup> (rimabotulinumtoxinB) 2500 U vs. 3500 U vs. placebo in 249 adult patients with a

wide range of pathologies, including but not limited to Parkinson's disease, ALS, stroke, adult cerebral palsy, TBI, oral cancer, and drug-induced sialorrhea. Sixty-three percent were 65 years of age or older. The trial showed a statistically significant reduction in uSFR in both treatment groups compared to placebo (mean uSFR reduction of 0.3 g/min in both,  $p < 0.001$ ). The most common treatment-related adverse event was dry mouth (38% and 45% of patients in the 2500 U and 3500 U treatment groups respectively).<sup>215</sup> There were no overall differences between older and younger patients regarding safety or effectiveness.

## Analysis of Evidence (Rationale for Determination)

### Achalasia

PubMed® and Google Scholar databases were queried for studies published in the past 10 years with terms including “achalasia” and “botulinum toxin”. A Multi-Jurisdictional CAC meeting was conducted in October 2023 with SMEs, after which additional articles selected by them were included. Four society guidelines and 5 additional systematic reviews in were also included in the analysis.

Intra-sphincteric BTI is effective for esophageal achalasia treatment but is limited by its short-term efficacy. Four society guidelines (from the ASGE<sup>13</sup>, the ISDE<sup>2</sup>, the ACG<sup>8</sup>, and the ESGE<sup>1</sup>) support the use of BTI as a treatment for achalasia only in patients who are not candidates for alternative invasive therapies, such as PD, LHM, or POEM. Four additional systematic reviews and meta-analyses comparing the use of BTI with alternative treatments have found that, while BTI may provide short-term symptom relief and is associated with low complication rates, its results are inferior in both medium- and long-term compared to all other non-pharmacological alternatives.<sup>10,12,14,78</sup>

The 2018 ISDE achalasia guidelines also recommend the use of BTI as a bridge for those awaiting more effective treatments such as Heller myotomy, PD or POEM.<sup>2</sup> Further, the use of BTI during work-up and treatment planning of definitive treatments for achalasia is recommended by SMEs and Katzka et al. (1999).<sup>17</sup>

In non-achalasia esophageal motility disorders the evidence of BTI is inconsistent but SME supports its use. Few non-randomized small trials favor this approach, such as the one by Vanuytsel et al. (2013)<sup>216</sup>, while others do not.<sup>217</sup> ESGE guidelines specifically do not recommend BTI in those cases, but state that if in individual patients the decision of BTI is chosen, they recommend performing additional BTI in the lower third of the esophagus.<sup>1</sup>

BTIs in achalasia are overall safe.<sup>2,12,14</sup> Patients with contraindications for upper endoscopy should not undergo BTI. Those contraindications have been well established and include perforated bowel, peritonitis, toxic megacolon in an unstable patient.<sup>18</sup>

BTI in the esophageal body is not recommended in the treatment of idiopathic achalasia. Only in patients exhibiting dysphagia in “achalasia-like” conditions such as opioid induced esophageal dysmotility, could these be considered (ISDE guidelines<sup>2</sup> and SME recommendation).

Escalating doses above 100 U of onabotulinumtoxinA (or equivalent) have not been shown to have superior treatment benefit.<sup>1,2</sup> ESGE recommends that BTI should be performed using 100 units of the toxin diluted in preservative-free saline that is injected in aliquots of 0.5mL–1mL using an injection needle in at least 4 quadrants of the LES.<sup>1</sup>

Frequency intervals of BTI vary. ESGE guidelines found that although the initial (1 month) response rate is high (> 75%), the therapeutic effect of repeated treatments substantially reduces over time and about half of patients required further injections at intervals of 6-24 months with some reporting long lasting effects when repeating BTI after a month.<sup>1</sup> Thus, we recommend performing less or equal to 3 BTIs per year for the treatment of achalasia.

### **Anal Fissure**

A literature search was conducted by querying PubMed® and Google Scholar. Literature published over a 10-year period ending December 31, 2023, was identified using search terms "anal fissures" and "botulinum toxin". Society guidelines, systematic reviews, meta-analyses, RCTs were included.

The ASCRS' systematic review provides strong, moderate quality evidence supporting the use of BTIs as an effective first and second-line treatment option for patients with chronic anal fissures.<sup>79</sup> With healing rates approaching 70% and fewer side effects than topical agents, these guidelines have significant practical implications for physicians aiming to use Botox® therapy for fissure patients who have failed conservative measures. Specifically, the guidelines endorse offering Botox® as an alternative to lateral internal sphincterotomy surgery in this situation. This evidence-based recommendation strongly supports coverage for Botox® on both clinical efficacy and cost-saving grounds compared to surgical intervention.

Across systematic reviews included in the present document, BTI consistently demonstrates around 65-70% healing rates for chronic anal fissures, similar to topical nitrates but with fewer side effects like headaches.<sup>218-221</sup> Risk of transient anal incontinence is the main downside, reported around 10-15%.<sup>219,220</sup> Multiple reviews found no definitive optimal dose, with similar efficacy but dose-dependent side effects, suggesting the lowest effective dose is appropriate.<sup>218-221</sup> Various injection locations are described with no consensus on optimal site. Compared to surgery, healing rates are lower but surgery carries permanent incontinence risks in just over 2% of patients.<sup>219</sup> These systematic analyses appear to confirm that Botox® is an effective medical therapy for chronic fissures prior to considering surgery. The data supports starting with lower doses (10-20 units) injected off midline into

the internal sphincter.<sup>218,221</sup> Patients should be counseled on risks of temporary incontinence but lower recurrence and side effects than topical agents.

The synthesis of key findings from 4 RCTs and meta-analyses presents moderate-grade evidence for the efficacy of botulinum toxin in the treatment of chronic anal fissures.<sup>80,84-86</sup> The 2016 meta-analysis reviewed here demonstrated no correlation between botulinum toxin dose and healing rates, complication rates, or risks of transient incontinence.<sup>83</sup>

### **Blepharospasm**

The FDA has approved onabotulinum toxin A and incobotulinum toxin A for the treatment of blepharospasm with comparable dosing. The literature supports the use of these botulinum toxins at approximately 12 week intervals as effective symptomatic treatment for blepharospasm. While the majority of the studies supported onabotulinum toxinA for the treatment of blepharospasm, a class I trial showed identical effectiveness between onabotulinum toxinA and incobotulinum toxinA in efficacy of treatment and side effect profile. (see additional reference to include in bibliography Simpson DM, Blitzler A, Brashear A, et al. Assessment: Botulinum neurotoxin for the treatment of movement disorders (an evidence-based review): report of the Therapeutics and Technology Assessment Subcommittee of the American Academy of Neurology. *Neurology*. May 6 2008;70(19):1699-706. doi:10.1212/01.wnl.0000311389.26145.95

Therefore, given the FDA approval for Xeomin, both botulinum toxins have sufficient literature to support the use in this condition.

### **Cervical Dystonia**

Relevant literature was collected through a systematic search strategy implemented on PubMed® and Google Scholar. Briefly, literature published over a 10-year period ending December 31, 2023 was identified using search terms "cervical dystonia" and "botulinum toxin". Society guidelines, systematic reviews, meta-analyses, RCTs were included.

BTI has become a mainstream first-line intervention for managing moderate to severe cases of CD.<sup>63,95,194-197,222-226</sup> There is strong consensus across systematic reviews and clinical guidelines that botulinum toxin is efficacious for alleviating symptoms and reducing disability in CD patients.<sup>63,95,194-197,222-226</sup> The AAN and the EFNS have highlighted the high efficacy of aboBoNT-A and rimaBoNT-B, with OnaBoNT-A and incoBoNT-A also being effective treatment options for CD.<sup>63,222</sup>

At least 7 systematic reviews published since 2013 examined botulinum toxin for CD. Sample sizes ranged from 441 to over 1,900 patients.<sup>63,193-197,225</sup> The reviews demonstrate botulinum toxin type A and B reduce dystonia severity by 15-30% and pain levels by over 2 points on the TWSTRS scale versus placebo.<sup>63,193-197,225</sup> While adverse events are more



common, including a 3-7 times higher risk of dysphagia and dry mouth, they are generally transient.<sup>197</sup> Patient satisfaction is driven by symptom improvement but reduced by perceived ineffectiveness in up to 66% of patients.<sup>195</sup> Expert administration and tailored dosing help maximize relief.<sup>195</sup>

Recent randomized trials also show that botulinum toxin reduces CD severity by 4-5 points on scales like TWSTRS and Tsui with patient-reported improvements versus placebo.<sup>224,225,227</sup> Trials suggest techniques like EMG-guidance and kinematic sensors enable more precise muscle selection and dosing compared to standard visual methods, resulting in faster optimal doses with similar or better efficacy.<sup>224,225,227</sup> Studies validating a 2.5:1 dosing ratio provide evidence for the non-inferiority of Dysport® and Botox® in managing CD.<sup>227</sup>

In summary, given the breadth and quality of evidence demonstrating botulinum toxin effectiveness along with its favorable safety profile relative to alternative treatments, expanding coverage to support its use as standard first-line therapy for CD patients would be well supported by the literature. The evidence also has meaningful practice implications regarding the importance of administration by highly trained experts and individualized dosing strategies to ensure patients have access to optimal treatment regimens tailored to provide the greatest symptom relief.

### Chronic Migraine

Relevant literature was identified by querying online databases such as PubMed® and Google Scholar. Literature published over a 10-year period ending December 31, 2023 was collected using search terms "migraine" and "botulinum toxin". Society guidelines, systematic reviews, meta-analyses, RCTs were included.

Multiple clinical guidelines based on systematic reviews recommend botulinum toxins as an effective option for preventing chronic migraine headaches in patients who have failed other treatments.<sup>116-118</sup> Guidelines from the AAN, NICE, and EHF support the statement that Botox® reduces headache days compared to placebos.<sup>116-118</sup>

Across the systematic analyses reviewed, Botox® treatment consistently showed a reduction in the number of monthly headache or migraine days compared to placebo.<sup>122-124,126,127</sup> The meta-analyses found that Botox® reduced headache days per month by 1.9 to 2.7 days on average. Botox® also reduced migraine severity, the number of days needing acute headache medications, and improved quality of life scales related to migraine.<sup>122-124,126,127</sup> The reviews concluded that Botox® is an effective option for preventing chronic migraine.<sup>122-124,126,127</sup>

Several RCTs including the pivotal 24-week Phase III Research Evaluating Migraine Prophylaxis Therapy (PREEMPT) trials have demonstrated Botox®'s ability to reduce

headache days per month versus placebo in chronic migraine patients.<sup>129</sup> The PREEMPT pooled data (n=1,384) showed a significant benefit for Botox<sup>®</sup> over placebo on headache days and 50% responder rates.<sup>129</sup>

An RCT by Bono et al. (2023) evaluated Botox<sup>®</sup> injected subcutaneously in trigeminal and occipital regions in 139 chronic migraine patients unresponsive to other Botox<sup>®</sup> regimens.<sup>116</sup> At 6 months, Botox<sup>®</sup> significantly lowered headache days per month by 13 days versus only 1 day for placebo. This approach could benefit non-responders.<sup>116</sup>

Regarding safety, Botox<sup>®</sup> does have a higher risk of transient adverse events like neck pain, injection site reactions, and muscle weakness compared to placebo. However, serious adverse events are rare, and the drug has an acceptable safety profile overall.<sup>122-124,126,127</sup> Discontinuation rates due to side effects were quite low at 2-3%.<sup>122-124,126,127</sup>

The meta-analyses were limited by heterogeneity between studies in terms of doses, injection sites/techniques, study durations, and populations included. Additionally, the longest study follow-up was 9 months, so long-term efficacy and safety remains unclear.<sup>128</sup>

In summary, evidence from multiple clinical practice guidelines, systematic reviews, meta-analyses, and RCTs supports the efficacy of BTIs for preventing chronic migraine headaches, especially in patients who have failed other treatments. Guidelines from the AAN, NICE, and EHF recommend botulinum toxin based on analyses finding around a 2 day per month reduction in headache days compared to placebo. Larger meta-analyses and the pivotal regulatory PREEMPT trials corroborate this with significant differences in responder rates and quality of life outcomes versus placebo as well. The treatment also demonstrates better tolerability than common oral options like topiramate, albeit at increased cost. The totality of evidence provides a rationale for considering botulinum toxin in appropriate patients with chronic migraine refractory to other therapies.

### **Focal Hand Dystonia**

Because FHD is a rare and heterogenous disease, studies had small sample sizes and different BTI dosing schemes. Most studies used subjective scales involving either patient or clinician perceptions of efficacy. None of the scales have been rigorously evaluated for clinical utility or validity. Whether studies involved treatments or pathophysiologic assays, there was a heterogeneous choice of rating scales used with no clear standard. As a result, the collective interpretive value of these studies is limited because the results are confounded by measurement effects.

### **Hemifacial Spasm/Facial Dystonia**

Articles published between 1991 and 2021 support the effectiveness and safety of BoNT in HFS as well as some differences among BoNT formulations with most published studies

citing BoNT-A as the preferred treatment. The efficacy of BoNT for HFS treatment ranged from 73% to 98.4%. The mean duration of the effect was around 12 weeks. Effectiveness did not decrease over time. Adverse effects were usually mild and transient. The efficacy and tolerability of the different preparations appeared to be similar. Among the studies, dosage, injected muscles, intervals of treatment, and rating scales were variable, thus leading to some challenges in comparing the results. In general, BoNT-A is considered the treatment of choice for HFS due to its efficacy and safety profile. Further studies are needed to investigate the factors that influence the outcome, including the optimal timing of treatment, injection techniques, dosage, and the best selection criteria for formulations.<sup>57</sup>

### Upper and Lower Limb Spasticity

AboBoNT-A, incoBoNT-A, and onaBoNT-A are established as effective and should be offered (Level A), and rimaBoNT-B is probably effective and should be considered (Level B), for upper limb spasticity. AboBoNT-A and onaBoNT-A are established as effective and should be offered (Level A) for lower limb spasticity.<sup>92</sup>

AboBoNT-A, incoBoNT-A, and onaBoNT-A are established as safe and effective for the reduction of adult upper limb spasticity and improvement of passive function (multiple Class I studies for all preparations). RimaBoNT-B is probably safe and effective for the reduction of adult upper limb spasticity (1 Class I study). Data are inadequate to determine the efficacy of aboBoNT-A, onaBoNT-A, incoBoNT-A, or rimaBoNT-B for improvement of active function associated with adult upper limb spasticity (Class I studies, inconsistent results dependent on active functional outcome).<sup>92</sup>

Therefore, for focal manifestations of adult spasticity involving the upper limb, aboBoNT-A, incoBoNT-A, and onaBoNT-A should be offered (Level A), and rimaBoNT-B should be considered (Level B), as treatment options.

For focal manifestations of adult spasticity involving the lower limb that warrant treatment, onaBoNT-A and aboBoNT-A should be offered (Level A) as treatment options. There is insufficient evidence to support or refute a benefit of incoBoNT-A or rimaBoNT-B for treatment of adult lower limb spasticity.<sup>92</sup>

Although botulinum toxin can reduce increased tone in spasticity, the impact of BTIs on functional outcomes is mixed, suggesting that potential functional gains are highly patient-specific. Because of the lack of comparative trials, there is insufficient evidence to indicate that any 1 of the botulinum toxin formulations is superior to the others.

AboBoNT-A and onaBoNT-A are established as safe and effective for the reduction of adult lower limb spasticity (multiple Class I studies).<sup>148,149,151,152</sup>

### Hyperhidrosis

According to published data, there is little difference in the treatment effect seen with 200 U (81.4% reduction in sweat production) and that seen with 100 U (76.5% reduction). Comparison of the results 2 weeks after the initial injections must be interpreted cautiously, because of the effect of the placebo injections. The reductions in the rate of sweat production at 4 weeks (2 weeks after the injection of 100 U of botulinum toxin A into the axillae that had received placebo), however, were similar in the 2 axillae. Follow-up measurements of the rates of sweat production showed no advantage to the higher dose.<sup>158</sup>

For sustained relief from symptoms of hyperhidrosis, additional injections of botulinum toxin A at varying intervals are usually required. There are no explicit criteria for the dose and frequency of the subsequent treatments; according to previous reports, patients have requested additional injections 4 to 17 months after the first treatment. We found that although the mean rate of sweat production gradually increased after injection of botulinum toxin A, after 6 months it was still well below half the initial mean rate.<sup>228</sup>

### **Hemifacial Spasm/Facial Dystonia/Limb Spasticity**

Botulinum toxin has revolutionized the treatment of spasticity, facial dystonia, and hyperhidrosis and is now administered worldwide. There are currently 3 leading botulinum toxin type A products available in the Western Hemisphere for this purpose: onabotulinum toxin-A (ONA) Botox<sup>®</sup>, abobotulinum toxin-A (ABO) Dysport<sup>®</sup>, and incobotulinum toxin-A (INCO) Xeomin<sup>®</sup>. Several others have found their way to the marketplace including Jeuveau<sup>®</sup> (prabotulinumtoxin-A) and more recently, Daxxify<sup>®</sup> (daxibotulinumtoxin-A).

Although the efficacies are similar, there is an intense debate regarding the comparability of various preparations. The clinical issues of potency and conversion ratios, as well as safety issues such as toxin spread and immunogenicity, to provide guidance for BtA use on clinical practice need to be considered during treatment. INCO was shown to be as effective as ONA with a comparable adverse event profile when a clinical conversion ratio of 1:1 was used. The available clinical and preclinical data suggest that a conversion ratio ABO:ONA of 3:1 – or even lower – could be appropriate for treating spasticity and HFS. A higher conversion ratio may lead to an overdosing of ABO. While uncommon, distant spread may occur; however, several factors other than the pharmaceutical preparation are thought to affect spread. Finally, whereas the 3 products have similar efficacy when properly dosed, ABO has a better cost-efficiency profile.<sup>229</sup>

In summary, published peer-reviewed literature has shown that BTIs are supported by level 1 evidence in the treatment of facial dystonias, either primary or secondary, hyperhidrosis and spasticity. The majority of the literature supports the use of onabotulinumtoxinA, however, there is also supporting evidence for the other serotypes in certain situations. The exact dosing for each serotype is typically compared to the number of onabotulinumtoxinA

units. In addition, the dosing may vary in terms of clinical responses from patient to patient. It seems important to reiterate that comparability between the various BtA preparations was determined with indirect methods and since there is no standardized potency test among all 3 products, further clinical trials are needed to establish the exact conversion ratio.

### Laryngeal Dystonia (spasmodic dysphonia)

Various databases such as PubMed®, the Cochrane Library, and the Guideline International Network (GIN) were queried for primary clinical studies, systematic reviews, and practice guidelines regarding the efficacy and risks of BTIs in LD. Overall, 488 citations were screened and 13 were selected based on the following eligibility criteria:

- Practice guidelines published in the last 10 years.
- Systematic reviews of RCTs or NRSIs, published in the last 10 years.
- RCTs comparing botulinum toxin to another active control or placebo, with no time limit.
- NRSIs or case series with more than 200 patients; no time limit.

Two practice guidelines<sup>63,95</sup>, 3 systematic reviews<sup>193,195,196</sup>, 2 small randomized trials (13 and 24 participants)<sup>194,197</sup>, and 8 large (> 200 patients) case series were identified.<sup>223-225,227,230-233</sup> It appears that Botox® is the most commonly used botulinum toxin product, but other formulations, such as Myobloc® and Dysport®, have been reported in case series and single-arm trials. The systematic reviews mostly included prospective single-arm NRSIs. Clinical and statistically significant improvements in objective and subjective voice measures were consistently reported in both RCTs and systematic reviews. These results are consistent with the outcomes of the identified cases series, with optimal response rates of at least 70% and duration of effect ranging from 10 to 18 weeks. Based upon the available evidence, treatment with botulinum toxin A for ADSD is currently recommended for off-label coverage. In addition, there is insufficient evidence to support the off-label usage of botulinum neurotoxin for ABSD at this time.

### IC/BPS

The rationale for non-coverage of IC/BPS is that the quality of evidence to support a role of BTX-A in IC/BPS is low quality (Grade C). The literature is challenged by lack of selection criteria for patients, standardized protocols for administration including dosage, administration timing and technique, comparison studies to other treatment modalities, short-term follow-up, small sample sizes and risk of bias in the limited literature. The meta-analyses are limited by these factors, high heterogeneity and they call for more robust evidence to guide practice management.

### Sialorrhea

Eleven RCTs and 11 systematic reviews addressing the use of botulinum toxin in adults with sialorrhea were found using a specific search strategy in the PubMed® database.

Note that there are 2 formulations of botulinum toxins with FDA-approval for chronic sialorrhea in adults currently, namely, incobotulinumtoxinA (BoNT-A, Xeomin®) and rimabotulinumtoxinB (BoNT-B, Myobloc®/RIMA). Both have been equally investigated in sialorrhea clinical research.

Xeomin® (incobotulinumtoxinA) injection, powder, lyophilized, for solution (Merz Pharmaceuticals, LLC) is approved by the FDA for the treatment of chronic sialorrhea in patients 2 years of age and older. In adults: the recommended total dose is 100 Units per treatment session consisting of 30 Units per parotid gland and 20 Units per submandibular gland, no sooner than every 16 weeks.

Myobloc® (rimabotulinumtoxinB) injection, solution (Solstice Neurosciences, LLC) is approved by the FDA for the treatment of chronic sialorrhea in adults. The recommended dosage is 1,500 Units to 3,500 Units; 500 Units to 1,500 Units per parotid gland and 250 Units per submandibular gland; no more frequent than every 12 weeks.

### **Strabismus**

The concept of a minimally invasive procedure for strabismus has long been an ambition of strabismus surgeons. There is vast experience regarding the use of botulinum toxin, and, in the setting of strabismus, it is a treatment modality with favorable efficacy and safety profiles. Botulinum toxin presents major advantages over surgical modalities such as shorter procedure duration, lower costs, and potentially lower exposure to general anesthesia. It can be planned before surgical procedures as it is less invasive. However, more extensive trials are needed to further clarify the efficacy of the exotoxin across strabismic entities and different patient populations.<sup>234</sup>

### **Overactive Bladder (OAB)/Urinary Incontinence (UI)**

The AUA/SUFU guideline recommends intradetrusor botulinum toxin A (BtA) injection as a third-line therapy for select OAB and UI patients who have not responded to first- and second-line treatments, although they caution its use in certain populations pending further research.<sup>69,159</sup> Several systematic reviews and meta-analyses have evaluated the efficacy and safety of BtA injections for OAB/UI.<sup>160-166</sup> These studies generally suggest that BtA doses greater than 50 U are more effective than placebo for certain OAB symptoms, with 100-200 U providing optimal balance of efficacy and safety as a second- or third-line option for refractory OAB/UI.<sup>160-166</sup> However, higher doses are associated with increased adverse events, and long-term efficacy data is limited. Potential predictors of poor response or adverse events include male gender, frailty, comorbidities, increasing age, and certain urodynamic parameters, although study quality was variable.<sup>160-166</sup>



RCTs have also demonstrated the efficacy of BtA 100 U for refractory OAB/UI, with significant improvements in UIE, other OAB symptoms, and quality of life measures compared to placebo or active comparators like anticholinergics and posterior tibial nerve stimulation.<sup>167-170</sup> These benefits appear to be consistent across different patient populations, including those who have failed prior pharmacologic therapies. However, adverse events such as UTIs, urinary retention, dysuria, and increased post-void residual volumes were more common with BtA compared to placebo.<sup>167-170</sup> Overall, while BtA appears to be an effective later-line treatment option for carefully selected patients with refractory OAB/UI, its use should be balanced against the increased risk of adverse events, particularly in higher-risk populations. More research is needed to fully characterize its long-term efficacy, safety, and comparative effectiveness. Appropriate patient selection and counseling remain important.<sup>167-170</sup>

---

## General Information

### Associated Information

N/A

### Sources of Information

N/A

### Bibliography

1. Weusten B, Barret M, Bredenoord AJ, et al. Endoscopic management of gastrointestinal motility disorders - part 1: European Society of Gastrointestinal Endoscopy (ESGE) Guideline. *Endoscopy*. 2020;52(6):498-515.
2. Zaninotto G, Bennett C, Boeckxstaens G, et al. The 2018 ISDE achalasia guidelines. *Dis Esophagus*. 2018;31(9):1-29.
3. Awaiz A, Yunus RM, Khan S, Memon B, Memon MA. Systematic review and meta-analysis of perioperative outcomes of peroral endoscopic myotomy (POEM) and laparoscopic heller myotomy (LHM) for achalasia. *Surg Laparosc Endosc Percutan Tech*. 2017;27(3):123-131.
4. Defazio G, Jinnah HA, Berardelli A, et al. Diagnostic criteria for blepharospasm: A multicenter international study. *Parkinsonism & Related Disorders*. 2021;91:109-114.
5. Kilic-Berkmen G, Defazio G, Hallett M, et al. Diagnosis and classification of blepharospasm: Recommendations based on empirical evidence. *Journal of the Neurological Sciences*. 2022;439:120319.
6. Pandolfino JE, Kwiatek MA, Nealis T, Bulsiewicz W, Post J, Kahrilas PJ. Achalasia: A new clinically relevant classification by high-resolution manometry. *Gastroenterology*.



- 2008;135(5):1526-1533.
7. Ramos VF, Karp BI, Hallett M. Tricks in dystonia: Ordering the complexity. *J Neurol Neurosurg Psychiatry*. 2014;85(9):987-993.
  8. Vaezi MF, Pandolfino JE, Yadlapati RH, Greer KB, Kavitt RT. ACG clinical guidelines: Diagnosis and management of achalasia. *Am J Gastroenterol*. 2020;115(9):1393-1411.
  9. Vaezi MF, Richter JE. Current therapies for achalasia: Comparison and efficacy. *J Clin Gastroenterol*. 1998;27(1):21-35.
  10. Andolfi C, Fisichella PM. Meta-analysis of clinical outcome after treatment for achalasia based on manometric subtypes. *Br J Surg*. 2019;106(4):332-341.
  11. Annese V, Bassotti G, Coccia G, et al. A multicentre randomised study of intrasphincteric botulinum toxin in patients with oesophageal achalasia. GISMAD Achalasia Study Group. *Gut*. 2000;46(5):597-600.
  12. Gong F, Li Y, Ye S. Effectiveness and complication of achalasia treatment: A systematic review and network meta-analysis of randomized controlled trials. *Asian J Surg*. 2023;46(1):24-34.
  13. Khashab MA, Vela MF, Thosani N, et al. ASGE guideline on the management of achalasia. *Gastrointest Endosc*. 2020;91(2):213-227.
  14. Leyden JE, Moss AC, MacMathuna P. Endoscopic pneumatic dilation versus botulinum toxin injection in the management of primary achalasia. *Cochrane Database Syst Rev*. 2014;2014(12):CD005046.
  15. Oude Nijhuis RAB, Zaninotto G, Roman S, et al. European guidelines on achalasia: United European Gastroenterology and European Society of Neurogastroenterology and Motility recommendations. *United European Gastroenterol J*. 2020;8(1):13-33.
  16. Kahrilas PJ, Bredenoord AJ, Fox M, et al. Expert consensus document: advances in the management of oesophageal motility disorders in the era of high-resolution manometry: A focus on achalasia syndromes. *Nat Rev Gastroenterol Hepatol*. 2017;14(11):677-688.
  17. Katzka DA, Castell DO. Use of botulinum toxin as a diagnostic/therapeutic trial to help clarify an indication for definitive therapy in patients with achalasia. *Am J Gastroenterol*. 1999;94(3):637-642.
  18. Ahlawat R, Hoilat GJ, Ross AB. *Esophagogastroduodenoscopy*. StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; Jan 2024.
  19. Boland PA, Kelly ME, Donlon NE, et al. Management options for chronic anal fissure: A systematic review of randomised controlled trials. *Int J Colorectal Dis*. 2020;35(10):1807-1815.
  20. Davids JS, Hawkins AT, Bhama AR, et al. The American Society of Colon and Rectal Surgeons clinical practice guidelines for the management of anal fissures. *Diseases of the Colon & Rectum*. 2023;66(2):190-199.
  21. Brisinda G, Chiarello M, Crocco A, Bentivoglio A, Cariatì M, Vanella S. Botulinum toxin injection for the treatment of chronic anal fissure: Uni- and multivariate analysis of the

- factors that promote healing. *International Journal of Colorectal Disease*. 2022;37(3):693-700.
22. Yiannakopoulou E. Botulinum toxin and anal fissure: Efficacy and safety systematic review. *Int J Colorectal Dis*. 2012;27(1):1-9.
23. Freeze M, Khine K, Ford J. Oculofacial diagnosis and management of benign essential blepharospasm. *Eyenet Magazine*. 2022:31-32.
24. Ramos VFML, Karp BI, Hallett M. Tricks in dystonia: Ordering the complexity. *Journal of Neurology, Neurosurgery & Psychiatry*. 2014;85(9):987-993.
25. Ma H, Qu J, Ye L, Shu Y, Qu Q. Blepharospasm, oromandibular dystonia, and Meige syndrome: Clinical and genetic update. *Frontiers in Neurology*. 2021;12:630221.
26. Wabbels B, Jost WH, Roggenkämper P. Difficulties with differentiating botulinum toxin treatment effects in essential blepharospasm. *J Neural Transm (Vienna)*. 2011;118(6):925-943.
27. S. Food and Drug Administration (FDA). [BOTOX® \(onabotulinumtoxinA\)](#) . Irvine, CA: Allergan Pharmaceuticals Ireland; 2017. Accessed 4/2/24.
28. S. Food and Drug Administration (FDA). [XEOMIN® \(incobotulinumtoxinA\)](#) . Frankfurt, Germany: Merz Pharmaceuticals; 2018. Accessed 4/2/24.
29. Dresner SC, Gauthier S, Codere F. Essential blepharospasm and the Meige syndrome. *Archives of Ophthalmology*. 1984;102(9):1268.
30. Weiss EM, Hershey T, Karimi M, et al. Relative risk of spread of symptoms among the focal onset primary dystonias. *Movement Disorders: Official Journal of the Movement Disorder Society*. 2006;21(8):1175-1181.
31. Ameer MA, Bhatti D. Chemodenervation for oromandibular dystonia utilizing Botulinum toxins. *Cureus*. 2021;13(10):e18425.
32. National Organization for Rare Disorders. [Cervical dystonia](#) . NORD®. Updated 7/19/19. Accessed 4/2/24.
33. Garcia-Ruiz PJ, Slawek J, Sitek EJ, Castrillo JCM. Art and dystonia. *Journal of the Neurological Sciences*. 2015;356(1-2):49-54.
34. Ortiz R, Scheperjans F, Mertsalmi T, Pekkonen E. The prevalence of adult-onset isolated dystonia in Finland 2007-2016. *PLoS One*. 2018;13(11):e0207729.
35. Lew MF, Brashear A, Factor S. The safety and efficacy of Botulinum toxin type B in the treatment of patients with cervical dystonia: Summary of three controlled clinical trials. *Neurology*. 2000;55(12 Suppl 5):S29-S35.
36. Velickovic M, Benabou R, Brin MF. Cervical dystonia: Pathophysiology and treatment options. *Drugs*. 2001;61:1921-1943.
37. Defazio G, Fabbrini G, Erro R, et al. Does acute peripheral trauma contribute to idiopathic adult-onset dystonia? *Parkinsonism & Related Disorders*. 2020;71:40-43.
38. Simpson DM, Hallett M, Ashman EJ, et al. Practice guideline update summary: Botulinum neurotoxin for the treatment of blepharospasm, cervical dystonia, adult spasticity, and headache: Report of the Guideline Development Subcommittee of the American Academy of Neurology. *Neurology*. 2016;86(19):1818-1826.

39. Albanese A, Sorbo FD, Comella C, et al. Dystonia rating scales: Critique and recommendations. *Mov Disord*. 2013;28(7):874-883.
40. US Food and Drug Administration (FDA). [DYSPORT® \(abobotulinumtoxinA\)](#) . Fort Worth, TX: Ipsen Biopharm Ltd; 2019. Accessed 4/2/24.
41. US Food and Drug Administration (FDA). [MYOBLOC® \(rimabotulinumtoxinB\)](#) . Louisville, KY: Solstice Neurosciences LLC; 2019. Accessed 4/2/24.
42. American Headache Society. The American Headache Society position statement on integrating new migraine treatments into clinical practice. *Headache*. 2019;59(1):1-18.
43. [Botulinum toxin type A for the prevention of headaches in adults with chronic migraine](#) . National Institute for Health and Care Excellence. Published 6/27/12. Accessed 4/2/24.
44. Niere K, Jerak A. Measurement of headache frequency, intensity and duration: Comparison of patient report by questionnaire and headache diary. *Physiother Res Int*. 2004;9(4):149-156.
45. Miller VE, Faurot KR, Palsson OS, et al. Comparing prospective headache diary and retrospective four-week headache questionnaire over 20 weeks: Secondary data analysis from a randomized controlled trial. *Cephalalgia*. 2020;40(13):1523-1531.
46. Nappi G, Jensen R, Nappi RE, Sances G, Torelli P, Olesen J. Diaries and calendars for migraine: A review. *Cephalalgia*. 2006;26(8):905-916.
47. van der Meer HA, Visscher CM, Vredevelde T, Nijhuis van der Sanden MW, Engelbert RH, Speksnijder CM. The diagnostic accuracy of headache measurement instruments: A systematic review and meta-analysis focusing on headaches associated with musculoskeletal symptoms. *Cephalalgia*. 2019;39(10):1313-1332.
48. Murase N, Rothwell JC, Kaji R, et al. Subthreshold low-frequency repetitive transcranial magnetic stimulation over the premotor cortex modulates writer's cramp. *Brain*. 2005;128(1):104-115.
49. Priori A, Pesenti A, Cappellari A, Scarlato G, Barbieri S. Limb immobilization for the treatment of focal occupational dystonia. *Neurology*. 2001;57(3):405-409.
50. Siebner HR, Tormos J, Ceballos-Baumann A, et al. Low-frequency repetitive transcranial magnetic stimulation of the motor cortex in writer's cramp. *Neurology*. 1999;52(3):529-537.
51. Taira T, Hori T. Stereotactic ventrooralis thalamotomy for task-specific focal hand dystonia (writer's cramp). *Stereotactic and functional neurosurgery*. 2004;80(1-4):88-91.
52. Zeuner KE, Bara-Jimenez W, Noguchi PS, Goldstein SR, Dambrosia JM, Hallett M. Sensory training for patients with focal hand dystonia. *Ann Neurol*. 2002;51(5):593-598.
53. Pandey S. A practical approach to management of focal hand dystonia. *Ann Indian Acad Neurol*. 2015;18(2):146-153.
54. Batla A, Goyal C, Shukla G, Goyal V, Srivastava A, Behari M. Hemifacial spasm: Clinical characteristics of 321 Indian patients. *J Neurol*. 2012;259(8):1561-1565.

55. Yaltho TC, Jankovic J. The many faces of hemifacial spasm: Differential diagnosis of unilateral facial spasms. *Mov Disord*. 2011;26(9):1582-1592.
56. Mocan BO. Imaging anatomy and pathology of the intracranial and intratemporal facial nerve. *Neuroimaging Clin N Am*. 2021;31(4):553-570.
57. Tambasco N, Filidei M, Nigro P, Parnetti L, Simoni S. Botulinum toxin for the treatment of hemifacial spasm: An update on clinical studies. *Toxins (Basel)*. 2021;13(12):881.
58. American Academy of Family Physicians. Dysphonia (hoarseness): AAO-HNSF releases updated clinical guideline for treatment. *American Family Physician*. 2018;98(10):606-607.
59. International Hyperhidrosis Society®. [Hyperhidrosis disease severity scale](#) . Sweat Help Website. Accessed 4/2/24.
60. Lyra Rde M, Campos JR, Kang DW, et al. Guidelines for the prevention, diagnosis and treatment of compensatory hyperhidrosis. *J Bras Pneumol*. 2008;34(11):967-977.
61. Doft MA, Hardy KL, Ascherman JA. Treatment of hyperhidrosis with botulinum toxin. *Aesthet Surg J*. 2012;32(2):238-244.
62. Lowe N, Naumann M, Eadie N. Treatment of hyperhidrosis with Botox (onabotulinumtoxinA): Development, insights, and impact. *Medicine*. 2023;102(Suppl 1):S51-S58.
63. Stachler RJ, Francis DO, Schwartz SR, et al. Clinical practice guideline: Hoarseness (dysphonia) (update). *Otolaryngol Head Neck Surg*. 2018;158(Suppl 1):S1-S42.
64. Kotby MN, Baraka M, El Sady SR, Ghanem M, Shoeib R. Psychogenic stress as a possible etiological factor in non-organic dysphonia. *International Congress Series*. 2003;1240:1251-1256.
65. French G, Bosch JD, Randall DR. Retrospective review of dosing trends in botulinum toxin injections for the treatment of adductor spasmodic dysphonia in a long-term cohort. *Journal of Otolaryngology - Head & Neck Surgery*. 2020;49(1):4.
66. Ginsberg DA, Boone TB, Cameron AP, et al. The AUA/SUFU guideline on adult neurogenic lower urinary tract dysfunction: Diagnosis and evaluation. *J Urol*. 2021;206(5):1097-1105.
67. gov. [Botulinum toxin a vs anticholinergic treatment of neurogenic overactive bladder in patients with multiple sclerosis \(SEPTOX\)](#) . Published 3/26/21. Updated 2/21/22. Accessed 4/2/24.
68. Kuo HC. Botulinun A toxin urethral sphincter injection for neurogenic or nonneurogenic voiding dysfunction. *Ci Ji Yi Xue Za Zhi*. 2016;28(3):89-93.
69. Lightner DJ, Gomelsky A, Souter L, Vasavada SP. Diagnosis and treatment of overactive bladder (non-neurogenic) in adults: AUA/SUFU guideline amendment 2019. *J Urol*. 2019;202(3):558-563.
70. Yamaguchi D, Tsuruoka N, Sakata Y, Shimoda R, Fujimoto K, Iwakiri R. Safety and efficacy of botulinum toxin injection therapy for esophageal achalasia in Japan. *J Clin Biochem Nutr*. 2015;57(3):239-243.

71. Clemens JQ, Erickson DR, Varela NP, Lai HH. Diagnosis and treatment of interstitial cystitis/bladder pain syndrome. *J Urol*. 2022;208(1):34-42.
72. Hanno PM, Erickson D, Moldwin R, Faraday MM. Diagnosis and treatment of interstitial cystitis/bladder pain syndrome: AUA guideline amendment. *J Urol*. 2015;193(5):1545-1553.
73. Lance JW. Symposium synopsis. Spasticity: Disordered motor control. Chicago: Yearbook Medical; Published 1980. p. 485–494.
74. Dagi L, Velez F, Archer S, et al. Adult Strabismus Preferred Practice Pattern®. *Ophthalmology*. 2020;127(1):P182-P298.
75. American Optometric Association. *Optometric clinical practice guideline: Care of the patient with strabismus: Esotropia and exotropia*. [↗](#) Louis, MO: American Optometric Association; 2011. Accessed 4/2/24.
76. Social Security Administration (SSA). [1861 Part E—Miscellaneous Provisions](#) [↗](#). Accessed 4/2/24.
77. Zhao X, Pasricha PJ. Botulinum toxin for spastic GI disorders: A systematic review. *Gastrointest Endosc*. 2003;57(2):219-235.
78. Shiu SI, Chang CH, Tu YK, Ko CW. The comparisons of different therapeutic modalities for idiopathic achalasia: A systematic review and network meta-analysis. *Medicine*. 2022;101(24):e29441.
79. Akalin Ç, Yavuzarslan AB, Akyol C. Efficacy and safety of endoanal ultrasound-guided botulinum toxin in chronic anal fissure. *Am Surg*. 2023;89(5):2125-2128.
80. Vitoopinyoparb K, Insin P, Thadanipon K, et al. Comparison of doses and injection sites of botulinum toxin for chronic anal fissure: A systematic review and network meta-analysis of randomized controlled trials. *Int J Surg*. 2022;104:106798.
81. Sahebally SM, Meshkat B, Walsh SR, Beddy D. Botulinum toxin injection vs topical nitrates for chronic anal fissure: An updated systematic review and meta-analysis of randomized controlled trials. *Colorectal Dis*. 2018;20(1):6-15.
82. Lin JX, Krishna S, Su'a B, Hill AG. Optimal dosing of botulinum toxin for treatment of chronic anal fissure: A systematic review and meta-analysis. *Dis Colon Rectum*. 2016;59(9):886-894.
83. Bobkiewicz A, Francuzik W, Krokowicz L, et al. Botulinum toxin injection for treatment of chronic anal fissure: Is there any dose-dependent efficiency? A meta-analysis. *World J Surg*. 2016;40(12):3064-3072.
84. Pilkington SA, Bhome R, Welch RE, et al. Bilateral versus unilateral botulinum toxin injections for chronic anal fissure: A randomised trial. *Tech Coloproctol*. 2018;22(7):545-551.
85. Gandomkar H, Zeinoddini A, Heidari R, Amoli HA. Partial lateral internal sphincterotomy versus combined botulinum toxin A injection and topical diltiazem in the treatment of chronic anal fissure: A randomized clinical trial. *Dis Colon Rectum*. 2015;58(2):228-234.
86. Berkel AE, Rosman C, Koop R, van Duijvendijk P, van der Palen J, Klaase JM. Isosorbide dinitrate ointment vs botulinum toxin A (Dysport) as the primary treatment for chronic



- anal fissure: A randomized multicentre study. *Colorectal Dis.* 2014;16(10):O360-O366.
87. Duarte GS, Rodrigues FB, Marques RE, et al. Botulinum toxin type A therapy for blepharospasm. *Cochrane Database Syst Rev.* 2020;11(11):CD004900.
88. Calace P, Cortese G, Piscopo R, et al. Treatment of blepharospasm with botulinum neurotoxin type A: Long-term results. *Eur J Ophthalmol.* 2003;13(4):331-336.
89. Engstrom PF, Arnoult JB, Mazow ML, et al. Effectiveness of botulinum toxin therapy for essential blepharospasm. *Ophthalmology.* 1987;94(8):971-975.
90. Shorr N, Seiff SR, Kopelman J. The use of botulinum toxin in blepharospasm. *Am J Ophthalmol.* 1985;99(5):542-546.
91. Scott AB, Kennedy RA, Stubbs HA. Botulinum A toxin injection as a treatment for blepharospasm. *Arch Ophthalmol.* 1985;103(3):347-350.
92. Simpson DM, Hallett M, Ashman EJ, et al. Practice guideline update summary: Botulinum neurotoxin for the treatment of blepharospasm, cervical dystonia, adult spasticity, and headache: Report of the Guideline Development Subcommittee of the American Academy of Neurology. *Neurology.* 2016;86(19):1818-1826.
93. Spiegel LL, Ostrem JL, Bledsoe IO. FDA approvals and consensus guidelines for botulinum toxins in the treatment of dystonia. *Toxins (Basel).* 2020;12(5):332.
94. AAO-HNS. [Position statement: Botulinum toxin treatment](#). American Academy of Ophthalmology-Head and Neck Surgery. Updated 4/21/21. Accessed 4/2/24.
95. Hallett M, Albanese A, Dressler D, et al. Evidence-based review and assessment of botulinum neurotoxin for the treatment of movement disorders. *Toxicon.* 2013;67:94-114.
96. Albanese A, Asmus F, Bhatia K, et al. EFNS guidelines on diagnosis and treatment of primary dystonias. *Eur J Neurol.* 2011;18(1):5-18.
97. Rodrigues FB, Duarte GS, Marques RE, et al. Botulinum toxin type A therapy for cervical dystonia. *Cochrane Database Syst Rev.* 2020;11(11):CD003633.
98. Marques RE, Duarte GS, Rodrigues FB, et al. Botulinum toxin type B for cervical dystonia. *Cochrane Database Syst Rev.* 2016;2016(5):CD004315.
99. Boyce MJ, McCambridge AB, Bradnam LV, Canning CG, Verhagen AP. The barriers and facilitators to satisfaction with botulinum neurotoxin treatment in people with cervical dystonia: A systematic review. *Neurol Sci.* 2022;43(8):4663-4670.
100. Mainka T, Erro R, Rothwell J, Kühn AA, Bhatia KP, Ganos C. Remission in dystonia - systematic review of the literature and meta-analysis. *Parkinsonism Relat Disord.* 2019;66:9-15.
101. Marsh WA, Monroe DM, Brin MF, Gallagher CJ. Systematic review and meta-analysis of the duration of clinical effect of onabotulinumtoxinA in cervical dystonia. *BMC Neurol.* 2014;14:91.
102. Samotus O, Lee J, Jog M. Personalized botulinum toxin type A therapy for cervical dystonia based on kinematic guidance. *J Neurol.* 2018;265(6):1269-1278.
103. Wu C, Xue F, Chang W, et al. Botulinum toxin type A with or without needle electromyographic guidance in patients with cervical dystonia. *Springerplus.*

2016;5(1):1292.

104. Yun JY, Kim JW, Kim HT, et al. Dysport and Botox at a ratio of 2.5:1 units in cervical dystonia: A double-blind, randomized study. *Mov Disord*. 2015;30(2):206-213.
105. Bendtsen L, Sacco S, Ashina M, et al. Guideline on the use of onabotulinumtoxinA in chronic migraine: A consensus statement from the European Headache Federation. *J Headache Pain*. 2018;19(1):91.
106. Barad M, Ailani J, Hakim SM, Kissoon NR, Schuster NM. Percutaneous interventional strategies for migraine prevention: A systematic review and practice guideline. *Pain Med*. 2022;23(1):164-188.
107. National Institute for Health and Care Excellence. [Botulinum toxin type A for the prevention of headaches in adults with chronic migraine](#) . NICE. Published 6/27/12. Accessed 4/2/24.
108. Corasaniti MT, Bagetta G, Nicotera P, et al. Safety of onabotulinumtoxin A in chronic migraine: A systematic review and meta-analysis of randomized clinical trials. *Toxins (Basel)*. 2023;15(5):332.
109. Giri S, Tronvik E, Linde M, Pedersen SA, Hagen K. Randomized controlled studies evaluating Topiramate, Botulinum toxin type A, and mABs targeting CGRP in patients with chronic migraine and medication overuse headache: A systematic review and meta-analysis. *Cephalalgia*. 2023;43(4):3331024231156922.
110. Shaterian N, Shaterian N, Ghanaatpisheh A, et al. Botox (onabotulinumtoxinA) for treatment of migraine symptoms: A systematic review. *Pain Res Manag*. 2022;2022:3284446.
111. Herd CP, Tomlinson CL, Rick C, et al. Cochrane systematic review and meta-analysis of botulinum toxin for the prevention of migraine. *BMJ Open*. 2019;9(7):e027953.
112. Bruloy E, Sinna R, Grolleau JL, Bout-Roumazielles A, Berard E, Chaput B. Botulinum toxin versus placebo: A meta-analysis of prophylactic treatment for migraine. *Plast Reconstr Surg*. 2019;143(1):239-250.
113. Herd CP, Tomlinson CL, Rick C, et al. Botulinum toxins for the prevention of migraine in adults. *Cochrane Database Syst Rev*. 2018;6(6):CD011616.
114. Lanteri-Minet M, Ducros A, Francois C, Olewinska E, Nikodem M, Dupont-Benjamin L. Effectiveness of onabotulinumtoxinA (BOTOX®) for the preventive treatment of chronic migraine: A meta-analysis on 10 years of real-world data. *Cephalalgia*. 2022;42(14):1543-1564.
115. Shen B, Wang L. Impact of the botulinum-A toxin on prevention of adult migraine disorders. *J Integr Neurosci*. 2020;19(1):201-208.
116. Bono F, Mazza MR, Magro G, et al. Regional targeted subcutaneous injection of botulinum neurotoxin type A in refractory chronic migraine: A randomized, double-blind, placebo-controlled study. *Toxins (Basel)*. 2023;15(5):324.
117. Cady R, Turner I, Dexter K, Beach ME, Cady R, Durham P. An exploratory study of salivary calcitonin gene-related peptide levels relative to acute interventions and



- preventative treatment with onabotulinumtoxinA in chronic migraine. *Headache*. 2014;54(2):269-277.
118. Chankrachang S, Arayawichanont A, Pongvarin N, et al. Prophylactic botulinum type A toxin complex (Dysport®) for migraine without aura. *Headache*. 2011;51(1):52-63.
119. Evers S, Vollmer-Haase J, Schwaag S, Rahmann A, Husstedt IW, Frese A. Botulinum toxin A in the prophylactic treatment of migraine—A randomized, double-blind, placebo-controlled study. *Cephalalgia*. 2004;24(10):838-843.
120. Freitag FG, Diamond S, Diamond M, Urban G. Botulinum toxin type A in the treatment of chronic migraine without medication overuse. *Headache*. 2008;48(2):201-209.
121. Hollanda L, Monteiro L, Melo A. Botulinum toxin type a for cephalic cutaneous allodynia in chronic migraine: a randomized, double-blinded, placebo-controlled trial. *Neurol Int*. 2014;6(4):5133.
122. Hou M, Xie JF, Kong XP, et al. Acupoint injection of onabotulinumtoxin A for migraines. *Toxins (Basel)*. 2015;7(11):4442-4454.
123. Lauretti GR, Rosa CP, Kitayama A, Lopes BC. Comparison of Botox® or Prosigne® and facial nerve blockade as adjuvant in chronic migraine. *J Biomed Sci Eng*. 2014;7:446-452.
124. Petri S, Tölle T, Straube A, Pfaffenrath V, Stefenelli U, Ceballos-Baumann A. Botulinum toxin as preventive treatment for migraine: A randomized double-blind study. *Eur Neurol*. 2009;62(4):204-211.
125. Pijpers JA, Kies DA, Louter MA, van Zwet EW, Ferrari MD, Terwindt GM. Acute withdrawal and botulinum toxin A in chronic migraine with medication overuse: A double-blind randomized controlled trial. *Brain*. 2019;142(5):1203-1214.
126. Aurora SK, Dodick DW, Diener HC, et al. OnabotulinumtoxinA for chronic migraine: Efficacy, safety, and tolerability in patients who received all five treatment cycles in the PREEMPT clinical program. *Acta Neurol Scand*. 2014;129(1):61-70.
127. Diener HC, Dodick DW, Aurora SK, et al. OnabotulinumtoxinA for treatment of chronic migraine: Results from the double-blind, randomized, placebo-controlled phase of the PREEMPT 2 trial. *Cephalalgia*. 2010;30(7):804-814.
128. Cady RK, Schreiber CP, Porter JA, Blumenfeld AM, Farmer KU. A multi-center double-blind pilot comparison of onabotulinumtoxinA and topiramate for the prophylactic treatment of chronic migraine. *Headache*. 2011;51(1):21-32.
129. Mathew NT, Jaffri SF. A double-blind comparison of onabotulinumtoxinA (BOTOX®) and topiramate (TOPAMAX®) for the prophylactic treatment of chronic migraine: A pilot study. *Headache*. 2009;49(10):1466-1478.
130. Rothrock JF, Adams AM, Lipton RB, et al. FORWARD study: Evaluating the comparative effectiveness of onabotulinumtoxinA and topiramate for headache prevention in adults with chronic migraine. *Headache*. 2019;59(10):1700-1713.
131. Loeb LM, Amorim RP, Mazzacoratti M, Scorza FA, Peres MFP. Botulinum toxin A (BT-A) versus low-level laser therapy (LLLT) in chronic migraine treatment: A comparison. *Arg Neuropsiquiatr*. 2018;76(10):663-667.

132. Magalhães E, Menezes C, Cardeal M, Melo A. Botulinum toxin type A versus amitriptyline for the treatment of chronic daily migraine. *Clin Neurol Neurosurg.* 2010;112(6):463-466.
133. Millán-Guerrero RO, Isais-Millán S, Barreto-Vizcaíno S, Rivera-Castaño L, Rios-Madariaga C. Subcutaneous histamine versus botulinum toxin type A in migraine prophylaxis: A randomized, double-blind study. *Eur J Neurol.* 2009;16(1):88-94.
134. Zidan A, Hussaini S, Gibson S, Brooks G, Mejico L. Onabotulinumtoxin type A reconstitution with preserved versus preservative-free saline in chronic migraine (B-RECON): A randomised, double-blind trial. *Int J Clin Pract.* 2020;74(9):e13522.
135. Zakin E, Simpson DM. Botulinum toxin therapy in writer's cramp and musician's dystonia. *Toxins (Basel).* 2021;13(12):899.
136. Tsui JK, Bhatt M, Calne S, Calne DB. Botulinum toxin in the treatment of writer's cramp: A double-blind study. *Neurology.* 1993;43(1):183-185.
137. Cole R, Hallett M, Cohen LG. Double-blind trial of botulinum toxin for treatment of focal hand dystonia. *Mov Disord.* 1995;10(4):466-471.
138. Chen R, Karp BI, Goldstein SR, Bara-Jimenez W, Yaseen Z, Hallett M. Effect of muscle activity immediately after botulinum toxin injection for writer's cramp. *Mov Disord.* 1999;14(2):307-312.
139. Geenen C, Consky E, Ashby P. Localizing muscles for botulinum toxin treatment of focal hand dystonia. *Can J Neurol Sci.* 1996;23(3):194-197.
140. Molloy FM, Shill HA, Kaelin-Lang A, Karp BI. Accuracy of muscle localization without EMG: Implications for treatment of limb dystonia. *Neurology.* 2002;58(5):805-807.
141. Colakoglu BD, Cakmur R, Uzunel F. Is it always necessary to apply botulinum toxin into the lower facial muscles in hemifacial spasm? A randomized, single-blind, crossover trial. *Eur Neurol.* 2011;65(5):286-290.
142. Choe WJ, Kim J. Increasing the area and varying the dosage of Botulinum toxin a injections for effective treatment of hemifacial spasm. *Acta Otolaryngol.* 2016;136(9):952-955.
143. Pellizzari R, Rossetto O, Schiavo G, Montecucco C. Tetanus and botulinum neurotoxins: Mechanism of action and therapeutic uses. *Philos Trans R Soc Lond B Biol Sci.* 1999;354(1381):259-268.
144. Ababneh OH, Cetinkaya A, Kulwin DR. Long-term efficacy and safety of botulinum toxin A injections to treat blepharospasm and hemifacial spasm. *Clin Exp Ophthalmol.* 2014;42(3):254-261.
145. Streitová H, Bareš M. Long-term therapy of benign essential blepharospasm and facial hemispasm with botulinum toxin A: Retrospective assessment of the clinical and quality of life impact in patients treated for more than 15 years. *Acta Neurol Belg.* 2014;114(4):285-291.
146. Gill HS, Kraft SP. Long-term efficacy of botulinum a toxin for blepharospasm and hemifacial spasm. *Can J Neurol Sci.* 2010;37(5):631-636.

147. Tunç T, Cavdar L, Karadag YS, Okuyucu E, Coskun O, Inan LE. Differences in improvement between patients with idiopathic versus neurovascular hemifacial spasm after botulinum toxin treatment. *J Clin Neurosci*. 2008;15(3):253-256.
148. McCrory P, Turner-Stokes L, Baguley IJ, et al. Botulinum toxin A for treatment of upper limb spasticity following stroke: A multi-centre randomized placebo-controlled study of the effects on quality of life and other person-centred outcomes. *J Rehabil Med*. 2009;41(7):536-544.
149. Shaw LC, Price CI, van Wijck FM, et al. Botulinum toxin for the upper limb after stroke (BoTULS) trial: Effect on impairment, activity limitation, and pain. *Stroke*. 2011;42(5):1371-1379.
150. Rosales RL, Kong KH, Goh KJ, et al. Botulinum toxin injection for hypertonicity of the upper extremity within 12 weeks after stroke: A randomized controlled trial. *Neurorehabil Neural Repair*. 2012;26(7):812-821.
151. Gracies JM, Brashear A, Jech R, et al. Safety and efficacy of abobotulinumtoxinA for hemiparesis in adults with upper limb spasticity after stroke or traumatic brain injury: A double-blind randomised controlled trial. *Lancet Neurol*. 2015;14(10):992-1001.
152. Lam K, Lau KK, So KK, et al. Can botulinum toxin decrease carer burden in long term care residents with upper limb spasticity? A randomized controlled study. *J Am Med Dir Assoc*. 2012;13(5):477-484.
153. Gusev YI, Banach M, Simonow A, et al. Efficacy and safety of botulinum type A toxin in adductor spasticity due to multiple sclerosis. *J Musculoskelet Pain*. 2008;16(3):175-188.
154. Kaji R, Osako Y, Suyama K, Maeda T, Uechi Y, Iwasaki M. Botulinum toxin type A in post-stroke lower limb spasticity: A multicenter, double-blind, placebo-controlled trial. *J Neurol*. 2010;257(8):1330-1337.
155. Maanum G, Jahnsen R, Stanghelle JK, Sandvik L, Keller A. Effects of botulinum toxin A in ambulant adults with spastic cerebral palsy: A randomized double-blind placebo controlled-trial. *J Rehabil Med*. 2011;43(4):338-347.
156. Dunne JW, Gracies JM, Hayes M, Zeman B, Singer BJ. A prospective, multicentre, randomized, double-blind, placebo-controlled trial of onabotulinumtoxinA to treat plantarflexor/invertor overactivity after stroke. *Clin Rehabil*. 2012;26(9):787-797.
157. Ibrahim O, Kakar R, Bolotin D, et al. The comparative effectiveness of suction-curettage and onabotulinumtoxin-A injections for the treatment of primary focal axillary hyperhidrosis: A randomized control trial. *J Am Acad Dermatol*. 2013;69(1):88-95.
158. Heckmann M, Ceballos-Baumann AO, Plewig G. Botulinum toxin A for axillary hyperhidrosis (excessive sweating). *N Engl J Med*. 2001;344(7):488-493.
159. Gormley EA, Lightner DJ, Faraday M, Vasavada SP. Diagnosis and treatment of overactive bladder (non-neurogenic) in adults: AUA/SUFU guideline amendment. *J Urol*. 2015;193(5):1572-1580.
160. Truzzi JC, Lapitan MC, Truzzi NC, Iacovelli V, Averbek MA. Botulinum toxin for treating overactive bladder in men: A systematic review. *Neurol Urodyn*. 2022;41(3):710-723.

161. Abrar M, Pindoria N, Malde S, Chancellor M, DeRidder D, Sahai A. Predictors of poor response and adverse events following botulinum toxin A for refractory idiopathic overactive bladder: A systematic review. *Eur Urol Focus*. 2021;7(6):1448-1467.
162. Gu HY, Song JK, Zhang WJ, et al. A systematic review and meta-analysis of effectiveness and safety of therapy for overactive bladder using botulinum toxin A at different dosages. *Oncotarget*. 2017;8(52):90338-90350.
163. Drake MJ, Nitti VW, Ginsberg DA, et al. Comparative assessment of the efficacy of onabotulinumtoxinA and oral therapies (anticholinergics and mirabegron) for overactive bladder: A systematic review and network meta-analysis. *BJU Int*. 2017;120(5):611-622.
164. Henriët B, Roumeguère T. [Botulinum toxin injection for refractory non-neurogenic overactive bladder. Systematic review]. *Rev Med Brux*. 2015;36(1):29-37.
165. Cui Y, Wang L, Liu L, et al. Botulinum toxin-A injections for idiopathic overactive bladder: A systematic review and meta-analysis. *Urol Int*. 2013;91(4):429-438.
166. Gong QQ, Xu YQ, Xu J, Ding XY, Guo C. Meta-analysis of randomized controlled trials using botulinum toxin A at different dosages for urinary incontinence in patients with overactive bladder. *Front Pharmacol*. 2019;10:1618.
167. Visco AG, Brubaker L, Richter HE, et al. Anticholinergic therapy vs. onabotulinumtoxinA for urgency urinary incontinence. *N Engl J Med*. 2012;367(19):1803-1813.
168. Sherif H, Khalil M, Omar R. Management of refractory idiopathic overactive bladder: Intradetrusor injection of botulinum toxin type A versus posterior tibial nerve stimulation. *Can J Urol*. 2017;24(3):8838-8846.
169. Nitti VW, Dmochowski R, Herschorn S, et al. OnabotulinumtoxinA for the treatment of patients with overactive bladder and urinary incontinence: Results of a phase 3, randomized, placebo controlled trial. *J Urol*. 2013;189(6):2186-2193.
170. Yokoyama O, Honda M, Yamanishi T, et al. OnabotulinumtoxinA (botulinum toxin type A) for the treatment of Japanese patients with overactive bladder and urinary incontinence: Results of single-dose treatment from a phase III, randomized, double-blind, placebo-controlled trial (interim analysis). *Int J Urol*. 2020;27(3):227-234.
171. Kuo HC, Chancellor MB. Comparison of intravesical botulinum toxin type A injections plus hydrodistention with hydrodistention alone for the treatment of refractory interstitial cystitis/painful bladder syndrome. *BJU Int*. 2009;104(5):657-661.
172. Kuo HC, Jiang YH, Tsai YC, Kuo YC. Intravesical botulinum toxin-A injections reduce bladder pain of interstitial cystitis/bladder pain syndrome refractory to conventional treatment - A prospective, multicenter, randomized, double-blind, placebo-controlled clinical trial. *Neurourol Urodyn*. 2016;35(5):609-614.
173. Pinto R, Lopes T, Frias B, et al. Trigonal injection of botulinum toxin A in patients with refractory bladder pain syndrome/interstitial cystitis. *Eur Urol*. 2010;58(3):360-365.
174. Liu HT, Kuo HC. Intravesical botulinum toxin A injections plus hydrodistension can reduce nerve growth factor production and control bladder pain in interstitial cystitis. *Urology*. 2007;70(3):463-468.

175. Giannantoni A, Porena M, Costantini E, Zucchi A, Mearini L, Mearini E. Botulinum A toxin intravesical injection in patients with painful bladder syndrome: 1-year followup. *J Urol*. 2008;179(3):1031-1034.
176. Giannantoni A, Costantini E, Di Stasi SM, Tascini MC, Bini V, Porena M. Botulinum A toxin intravesical injections in the treatment of painful bladder syndrome: A pilot study. *Eur Urol*. 2006;49(4):704-709.
177. Kuo HC. Preliminary results of suburothelial injection of botulinum a toxin in the treatment of chronic interstitial cystitis. *Urol Int*. 2005;75(2):170-174.
178. Pinto R, Lopes T, Silva J, Silva C, Dinis P, Cruz F. Persistent therapeutic effect of repeated injections of onabotulinum toxin a in refractory bladder pain syndrome/interstitial cystitis. *J Urol*. 2013;189(2):548-553.
179. Shie JH, Liu HT, Wang YS, Kuo HC. Immunohistochemical evidence suggests repeated intravesical application of botulinum toxin A injections may improve treatment efficacy of interstitial cystitis/bladder pain syndrome. *BJU Int*. 2013;111(4):638-646.
180. Kuo HC. Repeated onabotulinumtoxin-a injections provide better results than single injection in treatment of painful bladder syndrome. *Pain Physician*. 2013;16(1):E15-E23.
181. Soljanik I. Efficacy and safety of botulinum toxin A intradetrusor injections in adults with neurogenic detrusor overactivity/neurogenic overactive bladder: A systematic review. *Drugs*. 2013;73(10):1055-1066.
182. Cheng T, Shuang WB, Jia DD, et al. Efficacy and safety of onabotulinumtoxinA in patients with neurogenic detrusor overactivity: A systematic review and meta-analysis of randomized controlled trials. *PLoS One*. 2016;11(7):e0159307.
183. Ni J, Wang X, Cao N, Si J, Gu B. Is repeat Botulinum Toxin A injection valuable for neurogenic detrusor overactivity—A systematic review and meta-analysis. *Neurology and Urodynamics*. 2018;37(2):542-553.
184. Fu G, Wu J, Cong H, et al. [Efficacy of Botulinum-A toxin injection into bladder to treat neurogenic incontinence in patients with spinal cord injury: comparison of two doses]. *Zhonghua Yi Xue Za Zhi*. 2015;95(48):3920-3923.
185. Cruz F, Herschorn S, Aliotta P, et al. Efficacy and safety of onabotulinumtoxinA in patients with urinary incontinence due to neurogenic detrusor overactivity: A randomised, double-blind, placebo-controlled trial. *Eur Urol*. 2011;60(4):742-750.
186. Engeler D, Baranowski AP, Berghmans B, et al. [EAU Guidelines on Chronic Pelvic Pain](#)<sup>↗</sup>. EAU Guidelines Office, Arnhem, The Netherlands; Limited Update March 2022. Accessed 4/2/24.
187. Ochoa Vargas DC, García Perdómo HA. Efficacy of botulinum toxin A for the treatment of bladder pain syndrome: A systematic review. *Actas Urol Esp (Engl Ed)*. 2018;42(3):152-162.
188. Wang J, Wang Q, Wu Q, Chen Y, Wu P. Intravesical botulinum toxin A injections for bladder pain syndrome/interstitial cystitis: A systematic review and meta-analysis of controlled studies. *Med Sci Monit*. 2016;22:3257-3267.



189. Tirumuru S, Al-Kurdi D, Latthe P. Intravesical botulinum toxin A injections in the treatment of painful bladder syndrome/interstitial cystitis: A systematic review. *Int Urogynecol J*. 2010;21(10):1285-1300.
190. Shim SR, Cho YJ, Shin IS, Kim JH. Efficacy and safety of botulinum toxin injection for interstitial cystitis/bladder pain syndrome: A systematic review and meta-analysis. *Int Urol Nephrol*. 2016;48(8):1215-1227.
191. Imamura M, Scott NW, Wallace SA, et al. Interventions for treating people with symptoms of bladder pain syndrome: A network meta-analysis. *Cochrane Database Syst Rev*. 2020;7(7):CD013325.
192. Li B, Leng Q, Li C, Tan X, Su W, Li C. Comparison of intravesical instillation of hyaluronic acid with intradetrusor botulinum toxin A injection or cystoscopic hydrodistention for ketamine-associated cystitis. *J Int Med Res*. 2020;48(11):300060520973100.
193. Watts C, Nye C, Whurr R. Botulinum toxin for treating spasmodic dysphonia (laryngeal dystonia): A systematic Cochrane review. *Clin Rehabil*. 2006;20(2):112-122.
194. Troung DD, Rontal M, Rolnick M, Aronson AE, Mistura K. Double-blind controlled study of botulinum toxin in adductor spasmodic dysphonia. *Laryngoscope*. 1991;101(6 Pt 1):630-634.
195. Faham M, Ahmadi A, Silverman E, Harouni GG, Dabirmoghaddam P. Quality of life after botulinum toxin injection in patients with adductor spasmodic dysphonia; A systematic review and meta-analysis. *J Voice*. 2021;35(2):271-283.
196. van Esch BF, Wegner I, Stegeman I, Grolman W. Effect of botulinum toxin and surgery among spasmodic dysphonia patients. *Otolaryngol Head Neck Surg*. 2017;156(2):238-254.
197. Hyodo M, Nagao A, Asano K, et al. Botulinum toxin injection into the intrinsic laryngeal muscles to treat spasmodic dysphonia: a multicenter, placebo-controlled, randomized, double-blinded, parallel-group comparison/open-label clinical trial. *Eur J Neurol*. 2021;28(5):1548-1556.
198. Binenbaum G, Chang MY, Heidary G, et al. Botulinum toxin injection for the treatment of strabismus: A report by the American Academy of Ophthalmology. *Ophthalmology*. 2021;128(12):1766-1776.
199. Bort-Martí AR, Rowe FJ, Ruiz Sifre L, Ng SM, Bort-Martí S, Ruiz Garcia V. Botulinum toxin for the treatment of strabismus. *Cochrane Database Syst Rev*. 2023;3(3):CD006499.
200. Escuder AG, Hunter DG. The role of botulinum toxin in the treatment of strabismus. *Semin Ophthalmol*. 2019;34(4):198-204.
201. Minguini N, de Carvalho KM, Bosso FL, Hirata FE, Kara-José N. Surgery with intraoperative botulinum toxin-A injection for the treatment of large-angle horizontal strabismus: A pilot study. *Clinics (Sao Paulo)*. 2012;67(3):279-282.
202. Gardner R, Dawson EL, Adams GG, Lee JP. Long-term management of strabismus with multiple repeated injections of botulinum toxin. *J AAPOS*. 2008;12(6):569-575.

203. James E, Ellis C, Brassington R, Sathasivam S, Young CA. Treatment for sialorrhea (excessive saliva) in people with motor neuron disease/amyotrophic lateral sclerosis. *Cochrane Database Syst Rev*. 2022;5(5):CD006981.
204. Yu YC, Chung CC, Tu YK, et al. Efficacy and safety of botulinum toxin for treating sialorrhea: A systematic review and meta-analysis. *Eur J Neurol*. 2022;29(1):69-80.
205. Chinnapongse R, Gullo K, Nemeth P, Zhang Y, Griggs L. Safety and efficacy of botulinum toxin type B for treatment of sialorrhea in Parkinson's disease: A prospective double-blind trial. *Mov Disord*. 2012;27(2):219-226.
206. Lagalla G, Millevolte M, Capecci M, Provinciali L, Ceravolo MG. Long-lasting benefits of botulinum toxin type B in Parkinson's disease-related drooling. *J Neurol*. 2009;256(4):563-567.
207. Narayanaswami P, Geisbush T, Tarulli A, et al. Drooling in parkinson's disease: A randomized controlled trial of incobotulinum toxin A and meta-analysis of botulinum toxins. *Parkinsonism Relat Disord*. 2016;30:73-77.
208. Weikamp JG, Schinagl DA, Verstappen CC, Schelhaas HJ, de Swart BJ, Kalf JG. Botulinum toxin-A injections vs radiotherapy for drooling in ALS. *Acta Neurol Scand*. 2016;134(3):224-231.
209. Jackson CE, Gronseth G, Rosenfeld J, et al. Randomized double-blind study of botulinum toxin type B for sialorrhea in ALS patients. *Muscle Nerve*. 2009;39(2):137-143.
210. Steinlechner S, Klein C, Moser A, Lencer R, Hagenah J. Botulinum toxin B as an effective and safe treatment for neuroleptic-induced sialorrhea. *Psychopharmacology (Berl)*. 2010;207(4):593-597.
211. Guidubaldi A, Fasano A, Ialongo T, et al. Botulinum toxin A versus B in sialorrhea: A prospective, randomized, double-blind, crossover pilot study in patients with amyotrophic lateral sclerosis or Parkinson's disease. *Mov Disord*. 2011;26(2):313-319.
212. Mazlan M, Rajasegaran S, Engkasan JP, Nawawi O, Goh KJ, Freddy SJ. A double-blind randomized controlled trial investigating the most efficacious dose of botulinum toxin-A for sialorrhea treatment in Asian adults with neurological diseases. *Toxins (Basel)*. 2015;7(9):3758-3770.
213. Jost WH, Friedman A, Michel O, et al. SIAXI: Placebo-controlled, randomized, double-blind study of incobotulinumtoxinA for sialorrhea. *Neurology*. 2019;92(17):e1982-e1991.
214. Jost WH, Friedman A, Michel O, et al. Long-term incobotulinumtoxinA treatment for chronic sialorrhea: Efficacy and safety over 64 weeks. *Parkinsonism Relat Disord*. 2020;70:23-30.
215. Isaacson SH, Ondo W, Jackson CE, et al. Safety and efficacy of rimabotulinumtoxinB for treatment of sialorrhea in adults: A randomized clinical trial. *JAMA Neurol*. 2020;77(4):461-469.
216. Vanuytsel T, Bisschops R, Farré R, et al. Botulinum toxin reduces Dysphagia in patients with nonachalasia primary esophageal motility disorders. *Clin Gastroenterol Hepatol*. 2013;11(9):1115-1121.



217. Mion F, Marjoux S, Subtil F, et al. Botulinum toxin for the treatment of hypercontractile esophagus: Results of a double-blind randomized sham-controlled study. *Neurogastroenterol Motil.* 2019;31(5):e13587.
218. Alsaleh N, Aljunaydil AI, Aljamili GA. Clinical trial of combining botulinum toxin injection and fissurectomy for chronic anal fissure: A dose-dependent study. *Ann Coloproctol.* 2021.
219. Alyanak A, Gulen M, Ege B. Comparison of botulinum toxin (BoNT) injection and lateral internal sphincterotomy (redo-LIS) for recurrent anal fissure treatment. *Front Surg.* 2022;9:988082.
220. Amorim H, Santoalha J, Cadilha R, Festas MJ, Barbosa P, Gomes A. Botulinum toxin improves pain in chronic anal fissure. *Porto Biomed J.* 2017;2(6):273-276.
221. Barbeiro S, Atalaia-Martins C, Marcos P, et al. Long-term outcomes of Botulinum toxin in the treatment of chronic anal fissure: 5 years of follow-up. *United European Gastroenterol J.* 2017;5(2):293-297.
222. Simonyan K, Barkmeier-Kraemer J, Blitzer A, et al. Laryngeal dystonia: Multidisciplinary update on terminology, pathophysiology, and research priorities. *Neurology.* 2021;96(21):989-1001.
223. Patel PN, Kabagambe EK, Starkweather JC, et al. Outcomes of onabotulinum toxin A treatment for adductor spasmodic dysphonia and laryngeal tremor. *JAMA Otolaryngol Head Neck Surg.* 2018;144(4):293-299.
224. Blitzer A. Spasmodic dysphonia and botulinum toxin: Experience from the largest treatment series. *Eur J Neurol.* 2010;17(Suppl 1):28-30.
225. Lerner MZ, Lerner BA, Patel AA, Blitzer A. Gender differences in onabotulinum toxin A dosing for adductor spasmodic dysphonia. *Laryngoscope.* 2017;127(5):1131-1134.
226. Patel AB, Bansberg SF, Adler CH, Lott DG, Crujido L. The mayo clinic Arizona spasmodic dysphonia experience: A demographic analysis of 718 patients. *Ann Otol Rhinol Laryngol.* 2015;124(11):859-863.
227. Mor N, Tang C, Blitzer A. Botulinum toxin in secondarily nonresponsive patients with spasmodic dysphonia. *Otolaryngol Head Neck Surg.* 2016;155(3):458-461.
228. Karamfilov T, Konrad H, Karte K, Wollina U. Lower relapse rate of botulinum toxin A therapy for axillary hyperhidrosis by dose increase. *Arch Dermatol.* 2000;136(4):487-490.
229. Scaglione F. Conversion ratio between Botox<sup>®</sup>, Dysport<sup>®</sup>, and Xeomin<sup>®</sup> in clinical practice. *Toxins (Basel).* 2016;8(3):65.
230. Rosow DE, Parikh P, Vivero RJ, Casiano RR, Lundy DS. Considerations for initial dosing of botulinum toxin in treatment of adductor spasmodic dysphonia. *Otolaryngol Head Neck Surg.* 2013;148(6):1003-1006.
231. Rosow DE, Pechman A, Saint-Victor S, Lo K, Lundy DS, Casiano RR. Factors influencing botulinum toxin dose instability in spasmodic dysphonia patients. *J Voice.* 2015;29(3):352-355.

232. Stone A, Powell ME, Hamers K, et al. Optimizing botox regimens in patients with adductor spasmodic dysphonia and essential tremor of voice: a 31-year experience. *Laryngoscope Investig Otolaryngol*. 2022;7(5):1499-1505.
233. Venkatesan NN, Johns MM, Hapner ER, DelGaudio JM. Abductor paralysis after botox injection for adductor spasmodic dysphonia. *Laryngoscope*. 2010;120(6):1177-1180.
234. Miguel AI, Almeida C, Mahjoub A, et al. [Botulinum toxin use in strabismus](#) . American Academy of Ophthalmology EyeWiki® Updated 11/20/23. Accessed 4/2/24.
235. Abbruzzese G, Berardelli A, Girlanda P, et al. Long-term assessment of the risk of spread in primary late-onset focal dystonia. *J Neurol Neurosurg Psychiatry*. 2008;79(4):392-396.
236. Abrams P, Andersson KE, Apostolidis A, et al. 6th International Consultation on Incontinence. Recommendations of the International Scientific Committee: Evaluation and treatment of urinary incontinence, pelvic organ prolapse and faecal incontinence. *Neurourol Urodyn*. 2018;37(7):2271-2272.
237. Albanese A, Bhatia K, Bressman SB, et al. Phenomenology and classification of dystonia: a consensus update. *Movement disorders*. 2013;28(7):863-873.
238. Zeuner K, Hallett M. Sensory training as treatment for focal hand dystonia: A 1-year follow-up. *Movement Disorders: Official Journal of the Movement Disorder Society*. 2003;18(9):1044-1047.
239. American College of Obstetricians and Gynecologists. Evaluation of uncomplicated stress urinary incontinence in women before surgical treatment. *AUGS and ACOG Committee Opinion No. 603*. Updated June 2014.
240. American College of Obstetricians and Gynecologists. ACOG Practice bulletin no. 155: Urinary incontinence in women. *Obstet Gynecol*. 2015;126(5):e66-e81.
241. American Headache Society. The American Headache Society position statement on integrating new migraine treatments into clinical practice. *Headache: The Journal of Head and Face Pain*. 2019;59(1):1-18.
242. Jacobson BH, Johnson A, Grywalski C, et al. The voice handicap index (VHI): Development and validation. *American Journal of Speech-Language Pathology*. 1997;6(3):66-70.
243. Aromataris E, Stern C, Lockwood C, et al. JBI series paper 2: Tailored evidence synthesis approaches are required to answer diverse questions: A pragmatic evidence synthesis toolkit from JBI. *Journal of Clinical Epidemiology*. 2022;150:196-202.
244. Aurora S, Dodick D, Turkel C, et al. OnabotulinumtoxinA for treatment of chronic migraine: Results from the double-blind, randomized, placebo-controlled phase of the PREEMPT 1 trial. *Cephalalgia*. 2010;30(7):793-803.
245. Aurora S, Gawel M, Brandes J, Pokta S, VanDenburgh A, BOTOX North American Episodic Migraine Study Group. Botulinum toxin type a prophylactic treatment of episodic migraine: A randomized, double-blind, placebo-controlled exploratory study. *Headache: The Journal of Head and Face Pain*. 2007;47(4):486-499.

246. Aurora SK, Winner P, Freeman MC, et al. OnabotulinumtoxinA for treatment of chronic migraine: Pooled analyses of the 56-week PREEMPT clinical program. *Headache: The Journal of Head and Face Pain*. 2011;51(9):1358-1373.
247. Bastola P, Chaudhary M, Agrawal J, Shah D. The role of the injection botulinum toxin A in cases of essential blepharospasm syndrome, hemifacial spasm and Meige's syndrome. *Kathmandu University Medical Journal*. 2010;8(3):305-310.
248. Batla A, Goyal C, Shukla G, Goyal V, Srivastava A, Behari M. Hemifacial spasm: Clinical characteristics of 321 Indian patients. *J Neurol*. 2012;259(8):1561-1565.
249. Becker W. Botulinum toxin in the treatment of headache. *Toxins*. 2020;12(12):803.
250. Berman BD, Groth CL, Sillau SH, et al. Risk of spread in adult-onset isolated focal dystonia: A prospective international cohort study. *J Neurol Neurosurg Psychiatry*. 2020;91(3):314-320.
251. Bo K, Frawley HC, Haylen BT, et al. An International Urogynecological Association (IUGA)/International Continence Society (ICS) joint report on the terminology for the conservative and nonpharmacological management of female pelvic floor dysfunction. *Neurourol Urodyn*. 2017;36(2):221-244.
252. Borsuk DJ, Studniarek A, Park JJ, Marecik SJ, Mellgren A, Kochar K. Use of botulinum toxin injections for the treatment of chronic anal fissure: Results from an American Society of Colon and Rectal Surgeons Survey. *The American Surgeon*. 2023;89(3):346-354.
253. Botox® [Cervical dystonia](#) . AbbVie. Updated July 2022. Accessed 4/2/24.
254. Brin M, Lew M, Adler C, et al. Safety and efficacy of NeuroBloc (botulinum toxin type B) in type A-resistant cervical dystonia. *Neurology*. 1999;53(7):1431.
255. Charles D, Brashear A, Hauser RA, Li HI, Boo LM, Brin MF. Efficacy, tolerability, and immunogenicity of onabotulinumtoxinA in a randomized, double-blind, placebo-controlled trial for cervical dystonia. *Clin Neuropharmacol*. 2012;35(5):208-214.
256. Chen H, Xie K, Jiang C. A single-blind randomized control trial of trigonal versus nontrigonal Botulinum toxin-A injections for patients with urinary incontinence and poor bladder compliance secondary to spinal cord injury. *J Spinal Cord Med*. 2021;44(5):757-764.
257. Cho H, Kilic-Berkmen G, McKay J, et al. Clinical features and evolution of Blepharospasm: a multicenter international cohort and systematic literature review. *Dystonia*. 2022;1:10359.
258. Cillino S, Raimondi G, Guépratte N, et al. Long-term efficacy of botulinum toxin A for treatment of blepharospasm, hemifacial spasm, and spastic entropion: A multicentre study using two drug-dose escalation indexes. *Eye (Lond)*. 2010;24(4):600-607.
259. Clemens JQ. [Interstitial cystitis/bladder pain syndrome: Management](#) . UpToDate. Updated 12/12/23. Accessed 4/2/24.
260. Yaltho TC, Jankovic J. The many faces of hemifacial spasm: Differential diagnosis of unilateral facial spasms. *Movement Disorders*. 2011;26(9):1582-1592.

261. Comella CL, Jankovic J, Truong DD, Hanschmann A, Grafe S. Efficacy and safety of incobotulinumtoxinA (NT 201, XEOMIN<sup>®</sup>, botulinum neurotoxin type A, without accessory proteins) in patients with cervical dystonia. *J Neurol Sci*. 2011;308(1-2):103-109.
262. Cook TA, Brading AF, Mortensen NJ. The pharmacology of the internal anal sphincter and new treatments of ano-rectal disorders. *Aliment Pharmacol Ther*. 2001;15(7):887-898.
263. Costanzo M, Belvisi D, Berardelli I, et al. Effect of botulinum toxin on non-motor symptoms in cervical dystonia. *Toxins*. 2021;13(9):647.
264. Winters JC, Dmochowski RR, Goldman HB, et al. Urodynamic studies in adults: AUA/SUFU guideline. *J Urol*. 2012;188(6 Suppl):2464-2472.
265. Dagostino S, Ercoli T, Gigante AF, Pellicciari R, Fadda L, Defazio G. Sensory trick in upper limb dystonia. *Parkinsonism Relat Disord*. 2019;63:221-223.
266. Defazio G, Abbruzzese G, Girlanda P, et al. Botulinum toxin A treatment for primary hemifacial spasm: A 10-year multicenter study. *Archives of Neurology*. 2002;59(3):418-420.
267. Defazio G, Fabbrini G, Erro R, et al. Does acute peripheral trauma contribute to idiopathic adult-onset dystonia? *Parkinsonism Relat Disord*. 2020;71:40-43.
268. Diener H, Dodick D, Lipton R, Manack Adams A, DeGryse R, Silberstein S. Benefits beyond headache days with onabotulinumtoxinA treatment: a pooled PREEMPT analysis. *Pain and Therapy*. 2020;9:683-694.
269. Diener HC, Dodick D, Turkel C, et al. Pooled analysis of the safety and tolerability of onabotulinumtoxinA in the treatment of chronic migraine. *European journal of neurology*. 2014;21(6):851-859.
270. Dodick D, Silberstein S, Lipton R, DeGryse R, Adams A, Diener H. Early onset of effect of onabotulinumtoxinA for chronic migraine treatment: analysis of PREEMPT data. *Cephalalgia*. 2019;39(8):945-956.
271. Dodick DW, Turkel CC, DeGryse RE, et al. OnabotulinumtoxinA for treatment of chronic migraine: Pooled results from the double-blind, randomized, placebo-controlled phases of the PREEMPT clinical program. *Headache: The Journal of Head and Face Pain*. 2010;50(6):921-936.
272. Dresner SC, Gauthier S, Codere F. Essential blepharospasm and the Meige syndrome. *Arch Ophthalmol*. 1984;102(9):1268.
273. Dressler D, Altavista M, Altenmueller E, et al. Consensus guidelines for botulinum toxin therapy: general algorithms and dosing tables for dystonia and spasticity. *Journal of Neural Transmission*. 2021;128:321-335.
274. Duarte G, Rodrigues F, Marques R, et al. Botulinum toxin type A therapy for blepharospasm. *The Cochrane Database of Systematic Reviews*. 2020;11(11):CD004900.
275. Ferrazzano G, Conte A, Gigante A, Defazio G, Berardelli A, Fabbrini G. Disease progression in blepharospasm: A 5-year longitudinal study. *European Journal of*

- Neurology*. 2019;26(2):268-273.
276. Frank F, Ulmer H, Sidoroff V, Broessner G. CGRP-antibodies, topiramate and botulinum toxin type A in episodic and chronic migraine: A systematic review and meta-analysis. *Cephalalgia*. 2021;41(11-12):1222-1239.
277. Garcia-Ruiz PJ, Slawek J, Sitek EJ, Martinez Castrillo JC. Art and dystonia. *J Neurol Sci*. 2015;356(1-2):49-54.
278. Goorah R, Ahmed F. OnabotulinumtoxinA for chronic migraine: A critical appraisal. *Therapeutics and Clinical Risk Management*. 2015;11:1003-1013.
279. Gray SL, Anderson ML, Dublin S, et al. Cumulative use of strong anticholinergics and incident dementia: A prospective cohort study. *JAMA Intern Med*. 2015;175(3):401-407.
280. Gronseth G, Cox J, Gloss D, et al. *2017 Edition Clinical Practice Guideline Process Manual*. Minneapolis, MN: American Academy of Neurology®; 2017.
281. Heddle R, Cock C. Role of botulinum toxin injection in treatment of achalasia. *Annals of Esophagus*. 2020;3:26.
282. Hogikyan ND. The voice-related quality of life (V-RQOL) measure: History and ongoing utility of a validated voice outcomes instrument. *Perspectives on Voice and Voice Disorders*. 2004;14(1):3-5.
283. Hok P, Hvizdošová L, Otruba P, et al. Botulinum toxin injection changes resting state cerebellar connectivity in cervical dystonia. *Sci Rep*. 2021;11(1):8322.
284. Jiang YH, Liao CH, Tang DL, Kuo HC. Efficacy and safety of intravesical onabotulinumtoxinA injection on elderly patients with chronic central nervous system lesions and overactive bladder. *PLoS One*. 2014;9(8):e105989.
285. Jitpimolmard S, Tiamkao S, Laopaiboon M. Long term results of botulinum toxin type A (Dysport) in the treatment of hemifacial spasm: a report of 175 cases. *Journal of Neurology, Neurosurgery & Psychiatry*. 1998;64(6):751-757.
286. Jost WH, Hefter H, Stenner A, Reichel G. Rating scales for cervical dystonia: A critical evaluation of tools for outcome assessment of botulinum toxin therapy. *Journal of neural transmission*. 2013;120:487-496.
287. Jung HK, Hong SJ, Lee OY, et al. 2019 Seoul Consensus on Esophageal Achalasia Guidelines. *J Neurogastroenterol Motil*. 2020;26(2):180-203.
288. Kahrilas P, Bredenoord A, Fox M, et al. Advances in the management of oesophageal motility disorders in the era of high-resolution manometry: A focus on achalasia syndromes. *Nature reviews Gastroenterology & hepatology*. 2017;14(11):677-688.
289. Kilic-Berkmen G, Defazio G, Hallett M, et al. Diagnosis and classification of blepharospasm: Recommendations based on empirical evidence. *J Neurol Sci*. 2022;439:120319.
290. Kobashi KC, Albo ME, Dmochowski RR, et al. Surgical treatment of female stress urinary incontinence: AUA/SUFU guideline. *J Urol*. 2017;198(4):875-883.
291. Kollewe K, Mohammadi B, Dengler R, Dressler D. Hemifacial spasm and reinnervation synkinesias: Long-term treatment with either Botox® or Dysport®. *Journal of Neural Transmission*. 2010;117:759-763.



292. Kruisdijk J, Koelman J, De Visser BO, De Haan R, Speelman J. Botulinum toxin for writer's cramp: A randomised, placebo-controlled trial and 1-year follow-up. *Journal of Neurology, Neurosurgery & Psychiatry*. 2007;78(3):264-270.
293. gov. [Exercise program for patients with cervical dystonia who are treated with botulinum toxin type A](#) . Published 8/16/22. Updated 2/29/24. Accessed 4/2/24.
294. Lew MF, Adornato BT, Duane DD, et al. Botulinum toxin type B: A double-blind, placebo-controlled, safety and efficacy study in cervical dystonia. *Neurology*. 1997;49(3):701-707.
295. Lipton R, Varon S, Grosberg B, et al. OnabotulinumtoxinA improves quality of life and reduces impact of chronic migraine. *Neurology*. 2011;77(15):1465-1472.
296. Lipton RB, Rosen NL, Ailani J, DeGryse RE, Gillard PJ, Varon SF. OnabotulinumtoxinA improves quality of life and reduces impact of chronic migraine over one year of treatment: pooled results from the PREEMPT randomized clinical trial program. *Cephalalgia*. 2016;36(9):899-908.
297. Loudovici-Krug D, Derlien S, Best N, Günther A. Physiotherapy for cervical yystonia: A systematic review of randomised controlled trials. *Toxins*. 2022;14(11):784.
298. Lu J, Zhang Q, Guo X, et al. Calcitonin gene-related peptide monoclonal antibody versus Botulinum toxin for the preventive treatment of chronic migraine: Evidence from indirect treatment comparison. *Frontiers in Pharmacology*. 2021;12:631204.
299. Ma H, Qu J, Ye L, Shu Y, Qu Q. Blepharospasm, oromandibular dystonia, and Meige syndrome: Clinical and genetic update. *Front Neurol*. 2021;12:630221.
300. Martino D, Berardelli A, Abbruzzese G, et al. Age at onset and symptom spread in primary adult-onset blepharospasm and cervical dystonia. *Movement disorders*. 2012;27(11):1447-1450.
301. Weiss EM, Hershey T, Karimi M, et al. Relative risk of spread of symptoms among the focal onset primary dystonias. *Mov Disord*. 2006;21(8):1175-1181.
302. Matharu M, Halker R, Pozo-Rosich P, DeGryse R, Manack Adams A, Aurora SK. The impact of onabotulinumtoxinA on severe headache days: PREEMPT 56-week pooled analysis. *The Journal of Headache and Pain*. 2017;18(1):1-8.
303. Williams L, McGovern E, Kimmich O, et al. Epidemiological, clinical and genetic aspects of adult onset isolated focal dystonia in Ireland. *Eur J Neurol*. 2017;24(1):73-81.
304. Naumann M, So Y, Argoff C, et al. Assessment: Botulinum neurotoxin in the treatment of autonomic disorders and pain (an evidence-based review): Report of the Therapeutics and Technology Assessment Subcommittee of the American Academy of Neurology. *Neurology*. 2008;70(19):1707-1714.
305. Nijhuis RO, Zaninotto G, Roman S, et al. European Guideline on Achalasia-UEG and ESNM recommendations. *United European Gastroenterology Journal*. 2020;8(1):13-33.
306. Nullens S, Fockens P, Bredenoord AJ. Long-term outcomes of treatments for achalasia. *Current Opinion in Gastroenterology*. 2021;37(4):408-413.
307. Park Y, Lim J, Lee D, Yi S. Botulinum a toxin treatment of hemifacial spasm and blepharospasm. *Journal of Korean Medical Science*. 1993;8(5):334-340.

308. Patel N, Hanfelt J, Marsh L, Jankovic J. Alleviating manoeuvres (sensory tricks) in cervical dystonia. *J Neurol Neurosurg Psychiatry*. 2014;85(8):882-884.
309. Ramzan Z, Nassri AB. The role of Botulinum toxin injection in the management of achalasia. *Curr Opin Gastroenterol*. 2013;29(4):468-473.
310. Relja M, Poole A, Schoenen J, Pascual J, Lei X, Thompson C. A multicentre, double-blind, randomized, placebo-controlled, parallel group study of multiple treatments of botulinum toxin type A (BoNTA) for the prophylaxis of episodic migraine headaches. *Cephalalgia*. 2007;27(6):492-503.
311. Rogers RG, Pauls RN, Thakar R, et al. An International Urogynecological Association (IUGA)/International Continence Society (ICS) joint report on the terminology for the assessment of sexual health of women with pelvic floor dysfunction. *International urogynecology journal*. 2018;29:647-666.
312. Rogers RG, Pauls RN, Thakar R, et al. An International Urogynecological Association (IUGA)/International Continence Society (ICS) joint report on the terminology for the assessment of sexual health of women with pelvic floor dysfunction. *Neurol Urodyn*. 2018;37(4):1220-1240.
313. Saper JR, Mathew NT, Loder EW, DeGryse R, VanDenburgh AM. A double-blind, randomized, placebo-controlled comparison of botulinum toxin type a injection sites and doses in the prevention of episodic migraine. *Pain Medicine*. 2007;8(6):478-485.
314. Scuteri D, Tonin P, Nicotera P, et al. Pooled analysis of real-world evidence supports anti-CGRP mAbs and OnabotulinumtoxinA combined trial in chronic migraine. *Toxins*. 2022;14(8):529.
315. Siddiqui N, Reynolds W. [Botulinum toxin for treatment of lower urinary tract conditions: Indications and clinical evaluation](#)  UpToDate. Updated 11/30/22. Accessed 4/2/24.
316. Silberstein SD, Blumenfeld AM, Cady RK, et al. OnabotulinumtoxinA for treatment of chronic migraine: PREEMPT 24-week pooled subgroup analysis of patients who had acute headache medication overuse at baseline. *Journal of the Neurological Sciences*. 2013;331(1-2):48-56.
317. Silberstein SD, Diener H-C, Dodick DW, Manack Adams A, DeGryse RE, Lipton RB. The impact of onabotulinumtoxinA vs. placebo on efficacy outcomes in headache day responder and nonresponder patients with chronic migraine. *Pain and Therapy*. 2020;9:695-707.
318. Silberstein SD, Dodick DW, Aurora SK, et al. Per cent of patients with chronic migraine who responded per onabotulinumtoxinA treatment cycle: PREEMPT. *Journal of Neurology, Neurosurgery & Psychiatry*. 2015;86(9):996-1001.
319. Simpson D, Blitzer A, Brashear A, et al. Assessment: Botulinum neurotoxin for the treatment of movement disorders (an evidence-based review): Report of the Therapeutics and Technology Assessment Subcommittee of the American Academy of Neurology. *Neurology*. 2008;70(19):1699-1706.
320. Stewart WF, Lipton RB, Kolodner KB, Sawyer J, Lee C, Liberman JN. Validity of the migraine disability assessment (midas) score in comparison to a diary-based measure



- in a population sample of migraine sufferers. *Pain*. 2000;88(1):41-52.
321. Sung Y, Nam S, Lew H. Clinical outcomes of individualized botulinum neurotoxin type A injection techniques in patients with essential blepharospasm. *Korean Journal of Ophthalmology*. 2015;29(2):115-120.
  322. Svetel M, Pekmezovic T, Jovic J, et al. Spread of primary dystonia in relation to initially affected region. *J Neurol*. 2007;254(7):879-883.
  323. Svetel M, Pekmezovic T, Tomic A, Kresojevic N, Kostic VS. The spread of primary late-onset focal dystonia in a long-term follow up study. *Clin Neurol Neurosurg*. 2015;132:41-43.
  324. Truong D, Brodsky M, Lew M, et al. Long-term efficacy and safety of botulinum toxin type A (Dysport) in cervical dystonia. *Parkinsonism Relat Disord*. 2010;16(5):316-323.
  325. Velickovic M, Benabou R, Brin MF. Cervical dystonia pathophysiology and treatment options. *Drugs*. 2001;61(13):1921-1943.
  326. Truong D, Duane DD, Jankovic J, et al. Efficacy and safety of botulinum type A toxin (Dysport) in cervical dystonia: Results of the first US randomized, double-blind, placebo-controlled study. *Mov Disord*. Jul 2005;20(7):783-791.
  327. Tucker H, Osei-Poku F, Ashton D, et al. Management of secondary poor response to botulinum toxin in cervical dystonia: A multicenter audit. *Mov Disord Clin Pract*. 2021;8(4):541-545.
  328. Turkel CC, Aurora S, Diener H-C, et al. Treatment of chronic migraine with Botox (onabotulinumtoxinA): Development, insights, and impact. *Medicine*. 2023;102(S1):e32600.
  329. Comella CL, Jankovic J, Hauser RA, et al. Efficacy and safety of DaxibotulinumtoxinA for injection in cervical dystonia: ASPEN-1 phase 3 randomized controlled trial. *Neurology*. 2024;102(4):e208091.
  330. Food and Drug Administration (FDA). DAXXIFY® (daxibotulinumtoxinA-lanm). Newark, CA: Revance Therapeutics, Inc; 2023. Accessed 5/17/24.  
[https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2023/761127s002lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/761127s002lbl.pdf)
  331. Averbeck MA. Editorial Comment: Intradetrusor botulinum toxin injections (300 units) for the treatment of poorly compliant bladders in patients with adult neurogenic lower urinary tract dysfunction. *Int Braz J Urol*. 2022 Jan-Feb;48(1):182-183. doi: 10.1590/S1677-5538.IBJU.2022.01.05. PMID: 34735089; PMCID: PMC8691242.
  332. Cooley LF, Kielb S. A Review of Botulinum Toxin A for the Treatment of Neurogenic Bladder. *PM R*. 2019 Feb;11(2):192-200. doi: 10.1016/j.pmrj.2018.07.016. Epub 2019 Feb 13. PMID: 30138720.

Additional references reviewed: 3,4,6,7,9,21,34,35,39,44-52,57,109,235-329.

---

## Revision History Information

| Revision History Date | Revision History Number | Revision History Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Reasons for Change                                                            |
|-----------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 02/22/2026            | R17                     | Proposed LCD released to Final.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Provider Education/Guidance</li> </ul> |
| 10/01/2019            | R16                     | <p><b>Under Coverage Indications, Limitations and/or Medical Necessity:</b> Deleted 17. <i>Due to the short life of Botulinum toxin, Medicare will reimburse the unused portion of these drugs only when vials are not split between patients. Use modifier JW to code for drug wastage on a separate line of the claim form. The documentation must show in the patient's medical record the exact dosage of the drug given, exact amount and reason for unavoidable wastage, and the exact amount of the discarded portion of the drug.</i> 18. <i>Scheduling of more than one patient is encouraged to prevent wastage of Botulinum toxins. If a vial is split between two patients, the billing in these instances must be for the exact amount of Botulinum toxin used on each individual patient. Medicare would not expect to see billing for the</i></p> | <ul style="list-style-type: none"> <li>Provider Education/Guidance</li> </ul> |

| Revision History Date | Revision History Number | Revision History Explanation | Reasons for Change |
|-----------------------|-------------------------|------------------------------|--------------------|
|-----------------------|-------------------------|------------------------------|--------------------|

*full fee amount for Botulinum toxin on each beneficiary when the vial is split between two or more patients. Corrected grammatical and typographical.*

|            |     |                                                                                                                                                                                                                                                     |                                                                                 |
|------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 10/01/2019 | R15 | 10/01/2019: At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage.<br><br>LCD was converted to the "no-codes" format. | <ul style="list-style-type: none"> <li>Revisions Due To Code Removal</li> </ul> |
|------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|

|            |     |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                           |
|------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 10/01/2019 | R14 | <p>10/01/2019: At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage.</p> <p>The following codes were added for 2019 ICD10 Coding Update:</p> <p>Group I Additions:<br/>H02.042;H02.045;H02.141;H02.142;H02.144;H02.145;G44.221;G44.229</p> <p>Group II Additions:<br/>G51.31*;G51.32*;G51.33*;G</p> | <ul style="list-style-type: none"> <li>Revisions Due To ICD-10-CM Code Changes</li> </ul> |
|------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

| Revision History Date | Revision History Number | Revision History Explanation                                                                                                                                                                                                                                                                                                                                                                                                        | Reasons for Change                                                                          |
|-----------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|                       |                         | <p>82.51*;G36.1*;G80.3*;G82.52*;G82.53*;G83.82*;G83.89*</p> <p>Group III Additions:<br/> I63.343*;<br/> I69.932*;I69.942*;I69.943*;<br/> I69.944*;I69.961*;I69.963*;<br/> I69.964*;I69.965*</p>                                                                                                                                                                                                                                     |                                                                                             |
| 10/01/2018            | R13                     | <p>10/08/2018: At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</p> <p>Typographical Error corrected - Asterisk Explanation in Group I coding to include N31.0; N31.1 and N31.9</p> | <ul style="list-style-type: none"> <li>• Typographical Error</li> </ul>                     |
| 10/01/2018            | R12                     | <p>08/30/2018: At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage</p>                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Revisions Due To ICD-10-CM Code Changes</li> </ul> |

| Revision History Date | Revision History Number | Revision History Explanation                                                                                                                                                                                                                                              | Reasons for Change |
|-----------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|                       |                         | determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.<br><br>Effective 10/1/2018, LCD is revised per the annual ICD-10-CM code update to:<br><br>Add ICD-10-CM codes:<br>I63.81; I63.89; G51.31;<br>G51.32; G51.33 |                    |

| Revision History Date | Revision History Number | Revision History Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Reasons for Change                                                                        |
|-----------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 10/01/2017            | R11                     | <p>DATE (08/24/2017): At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</p> <p>Effective 10/1/2017, LCD is revised per the annual ICD-10-CM code update to:</p> <p>Add ICD-10-CM codes:<br/> I63.211*; I63.212*; I63.323*;<br/> I63.333*; I63.513*; I63.523*;<br/> I63.533* to Group 3</p> | <ul style="list-style-type: none"> <li>Revisions Due To ICD-10-CM Code Changes</li> </ul> |
| 02/13/2017            | R10                     | <p>Revision to correct 49499 to 43499 and to clarify K22.0 may be billed with other appropriate CPT codes in Group I asterisk in Medical Necessity ICD-10 Codes Asterisk Explanation and to indicate the G93.4 in the R5 Revision History Table was to be G83.4</p>                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Typographical Error</li> </ul>                     |

| Revision History Date | Revision History Number | Revision History Explanation                                                                                                                                                                                                                                                                               | Reasons for Change                                                                                     |
|-----------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 02/13/2017            | R9                      | Effective for dates of service 6/1/2016 - code 92265 is removed from the LCD. Please see the Nerve Conduction Studies and Electromyography LCD for coverage criteria. Typographical errors in Group 1 Paragraph - corrected CPT codes in the ICD-10 Section: 95875 was changed to 95873 and 92265 removed. | <ul style="list-style-type: none"> <li>• Typographical Error</li> </ul>                                |
| 02/13/2017            | R8                      | This LCD version was created as a result of DL35172 being released to a Final LCD and Coding Guidelines removed due to information included in LCD.                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Creation of Uniform LCDs Within a MAC Jurisdiction</li> </ul> |
| 10/01/2016            | R7                      | N31.0, N31.1 had addition of * and N31.9* added to Group One codes. G93.4* and G95.89* added to secondary diagnosis.                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• Revisions Due To ICD-10-CM Code Changes</li> </ul>            |
| 10/01/2016            | R6                      | 2016-2017 ICD 10 Update: To Add I63.013*, I63.033*, I63.113*, I63.133*, I63.213, I63.233, I63.313, I60.2* to Group III Codes                                                                                                                                                                               | <ul style="list-style-type: none"> <li>• Revisions Due To ICD-10-CM Code Changes</li> </ul>            |



| Revision History Date | Revision History Number | Revision History Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Reasons for Change                                                                                   |
|-----------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 06/01/2016            | R5                      | Effective for dates of service 6/1/2016 92265 is removed from this LCD. Please see the Nerve Conduction Studies and Electromyography LCD for coverage criteria.                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Revisions Due To CPT/HCPCS Code Changes</li> </ul>            |
| 10/01/2015            | R4                      | The LCD revised to add the ICD-10-CM subsequent (D) and sequela (S) codes for S14.0XX - S24.154.                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Reconsideration Request</li> </ul>                            |
| 10/01/2015            | R3                      | <p>The LCD is revised to add the sentence "Recently, abotulinumtoxinA (Dysport) has been approved by the FDA for upper limb spasticity" in the Coverage Indications, Limitations and/or Medical Necessity section.</p> <p>Also added ICD10 codes G82.53, G83.21, G83.22, G83.23, G83.24, I69.931, I69.932, I69.933, I69.934, I69.951, I69.952, I69.953, I69.954 and J38.3. ICD10 code J38.7 was removed as this was added in error and the correct ICD10 code should have been J38.3.</p> | <ul style="list-style-type: none"> <li>Creation of Uniform LCDs Within a MAC Jurisdiction</li> </ul> |

| Revision History Date | Revision History Number | Revision History Explanation                                                                                                                                                                    | Reasons for Change                                                                          |
|-----------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 10/01/2015            | R2                      | The "ICD-10 Codes that Support Medical Necessity" was revised to correct the information in this section. Some of the CPT codes were not transferred when it was converted to the ICD10 policy. | <ul style="list-style-type: none"> <li>• Typographical Error</li> </ul>                     |
| 10/01/2015            | R1                      | The LCD is revised to add ICD-10-CM G43.109 to the Group 1 list codes with an asterisk explanation that the chart must document the patient has chronic migraine.                               | <ul style="list-style-type: none"> <li>• Revisions Due To ICD-10-CM Code Changes</li> </ul> |

## Associated Documents

### Attachments

N/A

### Related Local Coverage Documents

#### Articles

[A57186 - Billing and Coding: Botulinum Toxin Injections](#) 

[A60366 - Response to Comments: Botulinum Toxin Injections](#) 

### Related National Coverage Documents

#### NCDs

N/A

### Public Versions

| Updated On | Effective Dates         | Status              |  |
|------------|-------------------------|---------------------|-------------------------------------------------------------------------------------|
| 12/29/2025 | 02/22/2026 - N/A        | Currently in Effect | You are here                                                                        |
| 04/12/2022 | 10/01/2019 - 02/21/2026 | Superseded          | <a href="#">View</a>                                                                |
| 09/19/2019 | 10/01/2019 - N/A        | Superseded          | <a href="#">View</a>                                                                |

Some older versions have been archived. Please visit the [MCD Archive Site](#) to retrieve them.

## Keywords

N/A